

Anatomy of Male & Female Reproductive System



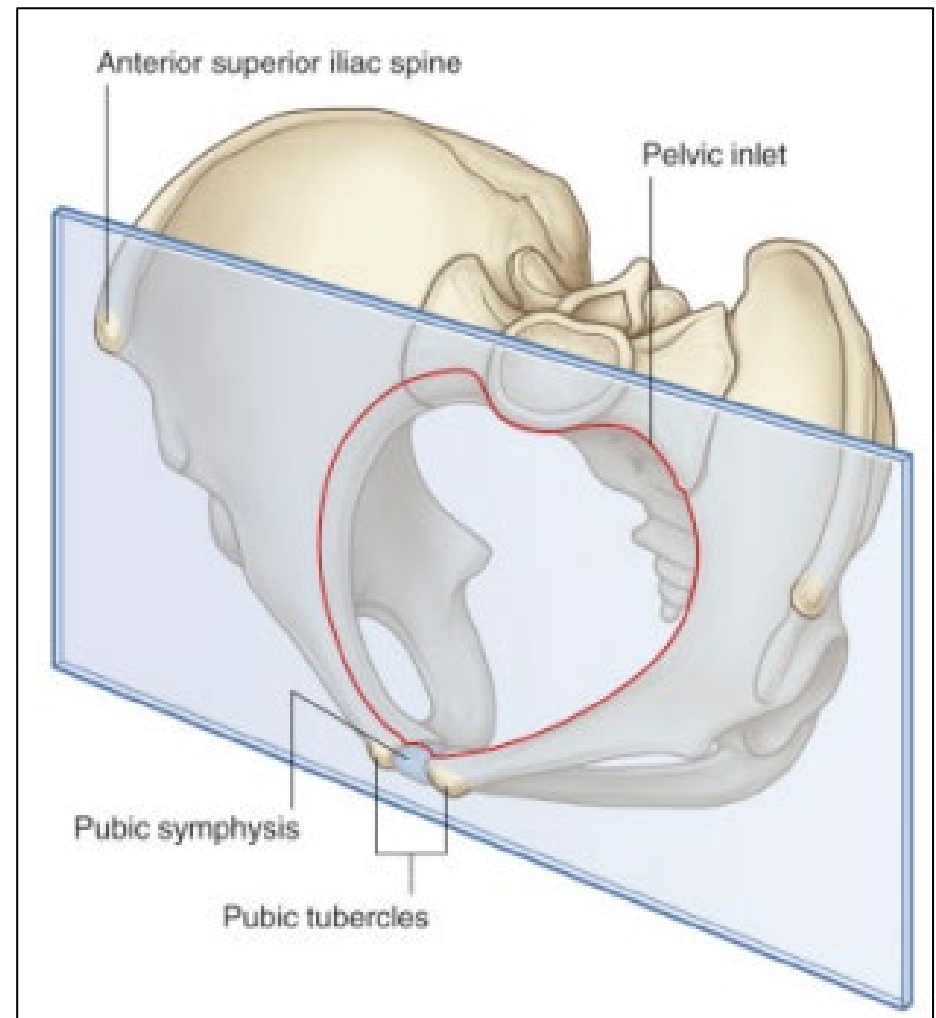
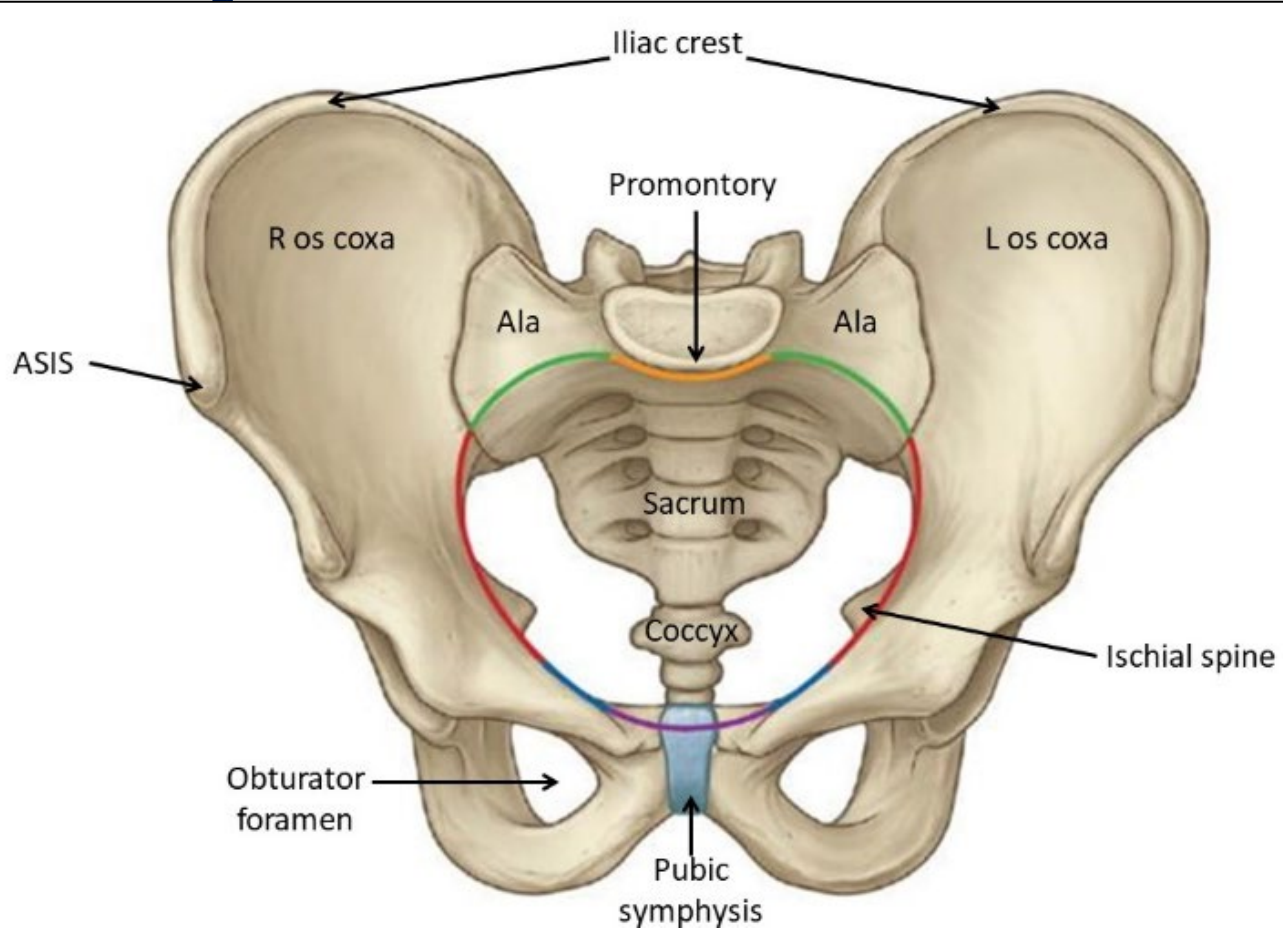
Wessam Ibrahim, MD, PhD, MEHP

(For use only by CIMED students)

Learning Objectives

- Describe the locations and relationships between the true pelvic cavity, the false pelvis, and the perineum.
- Identify the features of the bony pelvis that are important for adequacy of the pelvis for childbirth (pelvic inlet via diagonal conjugate, interspinous distance, intertuberous distance, width of subpubic arch).
- Explain the function of the pelvic floor, its innervation, and the two major muscles that comprise it.
- Describe the peritoneal relationships of the pelvic viscera, including pouches and spaces.
- Describe the anatomical features and general functions of the pelvic viscera in females (bladder, uterus, uterine tube, ovary, cervix, vagina, and rectum).
- Describe the anatomical features and general functions of the pelvic viscera in males (bladder, ductus deferens, seminal vesicle, ejaculatory duct, prostate, prostatic urethra, and rectum).
- Trace the arterial blood supply and venous drainage of internal pelvic structures to their vascular origins and terminations.
- Relate the pelvic vasculature to neighboring organs, fascial planes, and clinically significant anatomical landmarks.
- Apply anatomical knowledge explaining methods of achieving hemostasis during common gynecologic, urologic, and colorectal surgical procedures.

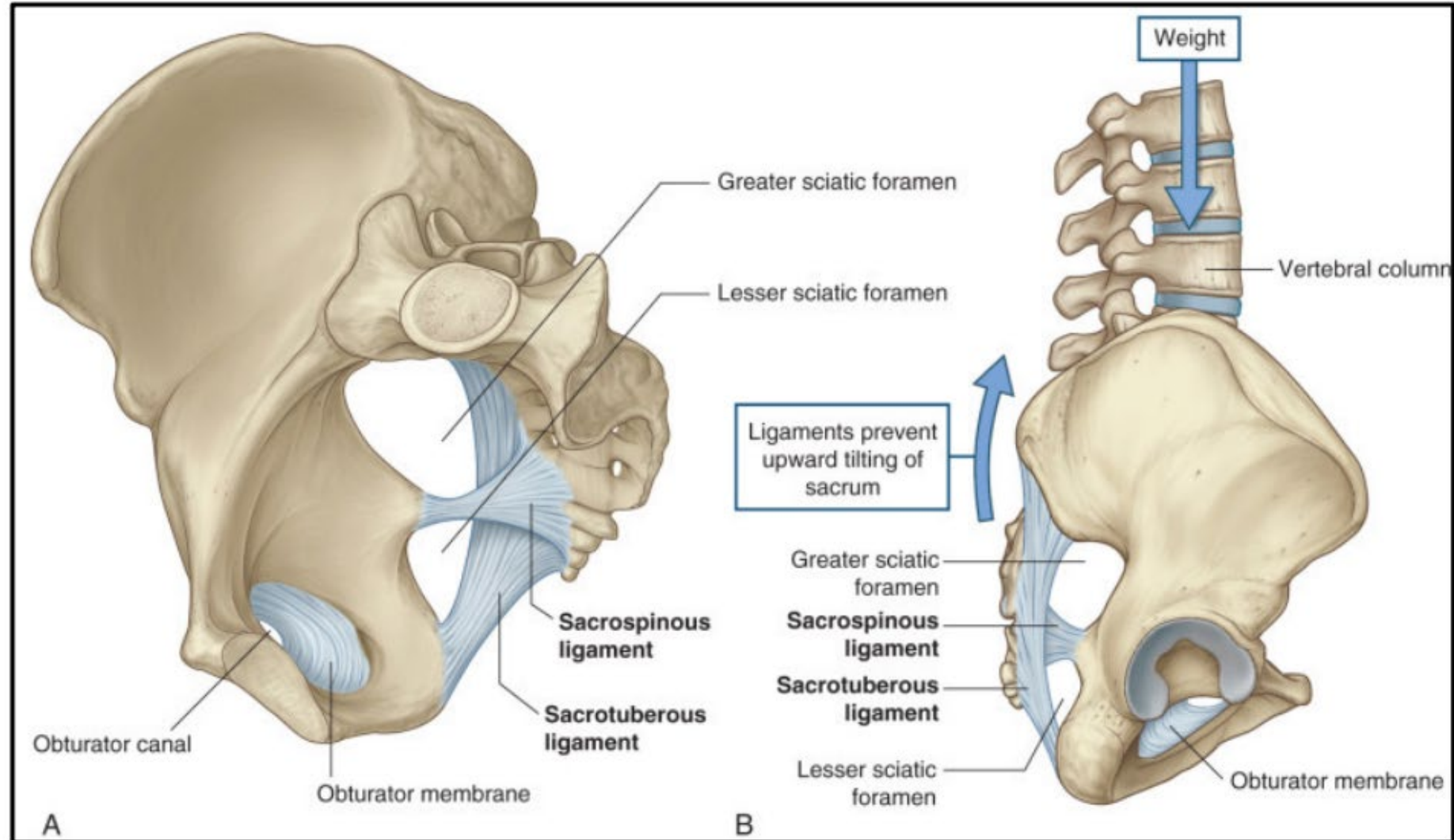
Bony Pelvis



Functions:

- Bears & transfers the weight of axial to the appendicular skeleton
- Provides attachment for muscles
- Protects pelvic viscera and provides passage of fetus (in females) via the perineum
- Provides attachment for the erectile bodies of the external genitalia

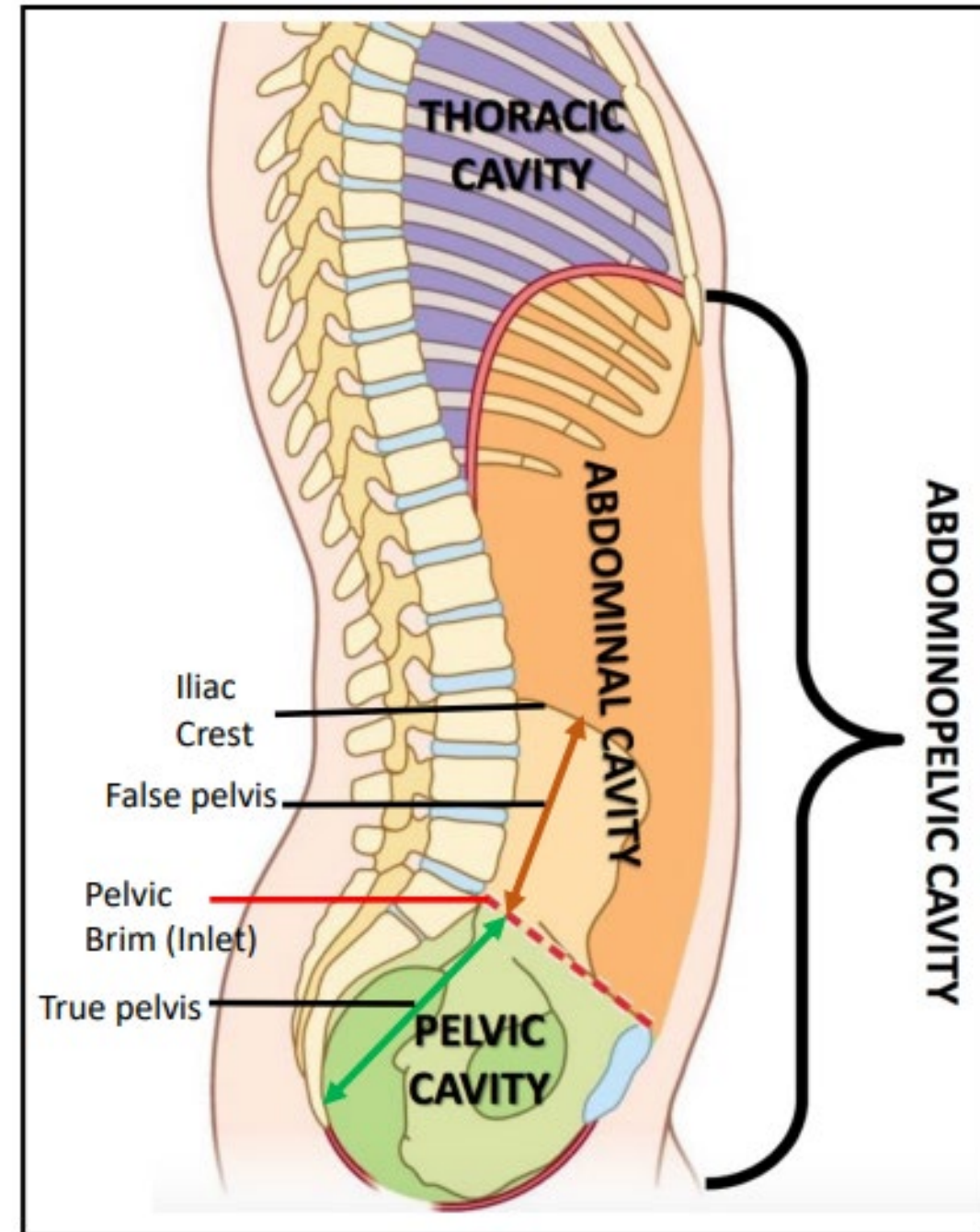
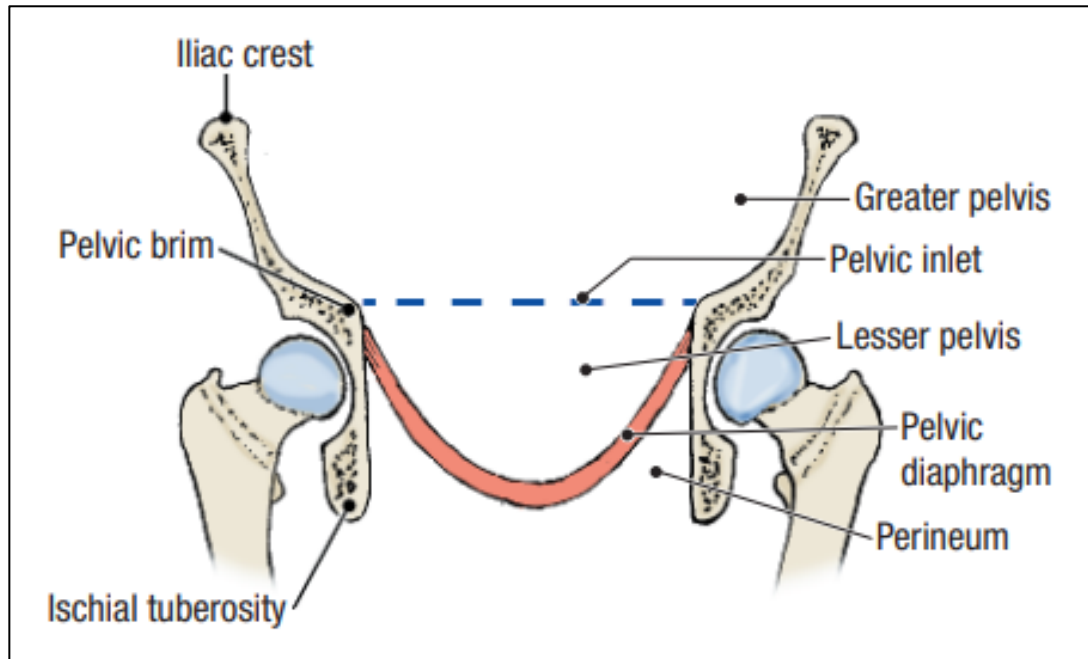
Pelvic Ligaments and Foramina



Pelvic Region

Pelvic brim separates the false (or greater) pelvis from the true (or lesser) pelvis.

- **False (greater) pelvis** mostly bounded by the iliac blades; occupied by abdominal viscera.
- **True (lesser) pelvis** is a bony canal through which the fetus passes during normal childbirth.



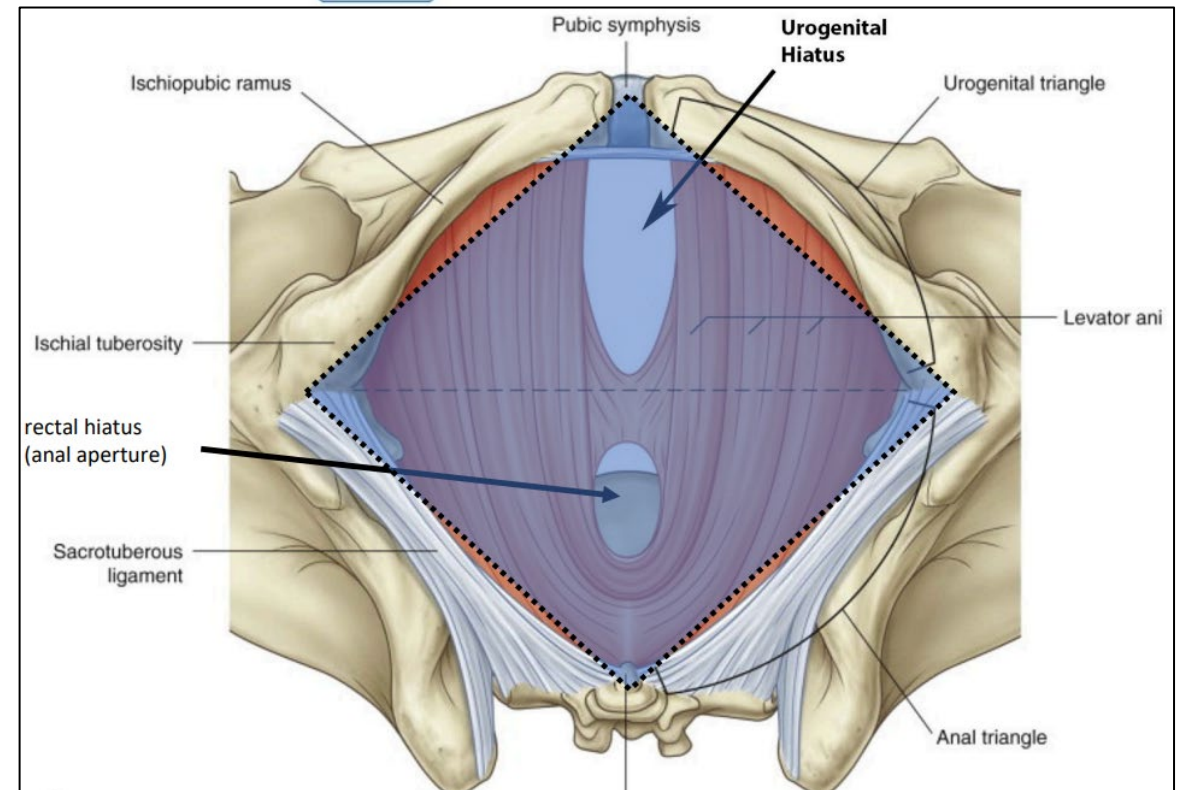
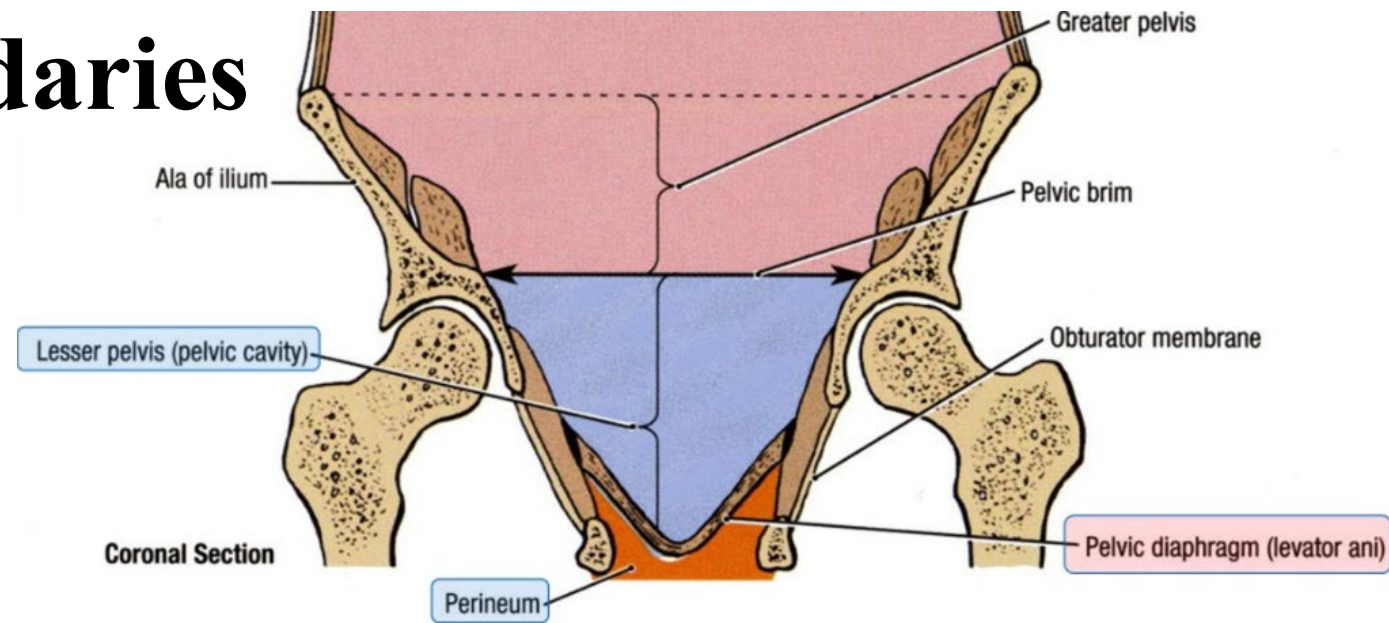
Pelvis & Perineum Boundaries

The pelvic floor (diaphragm) forms a physical boundary between the pelvic cavity and the perineum.

The perineum is a diamond-shaped region of the body wall inferior to the pelvic diaphragm between the buttocks and thighs.

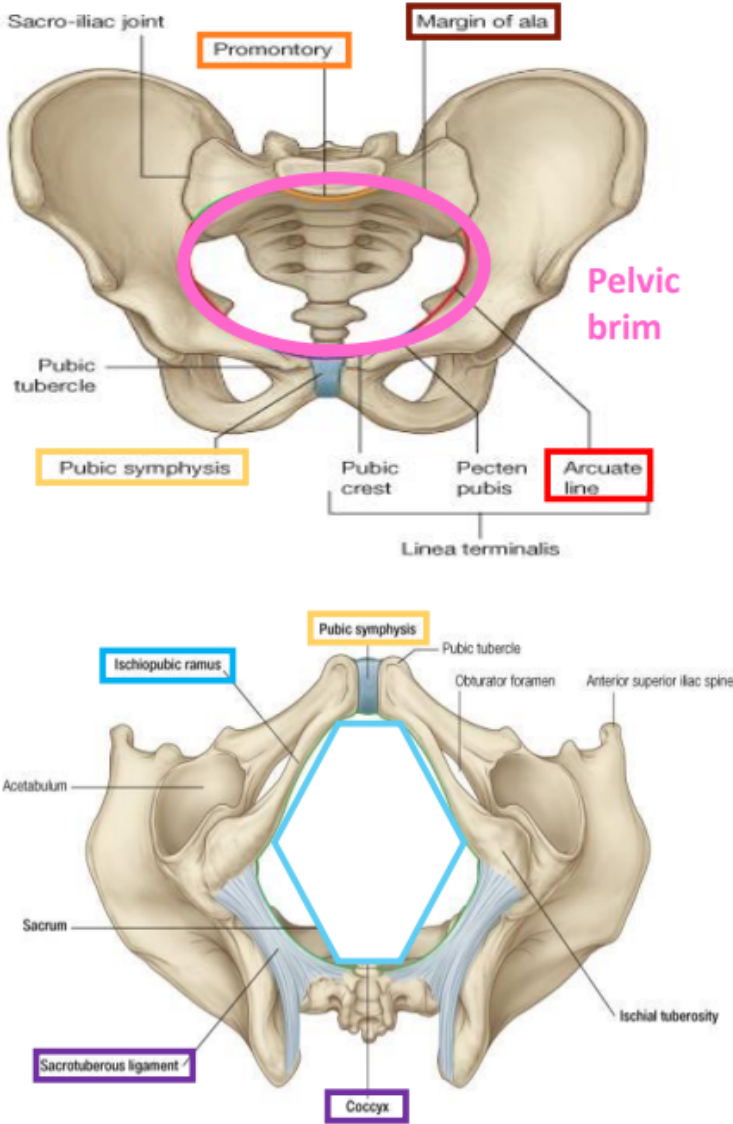
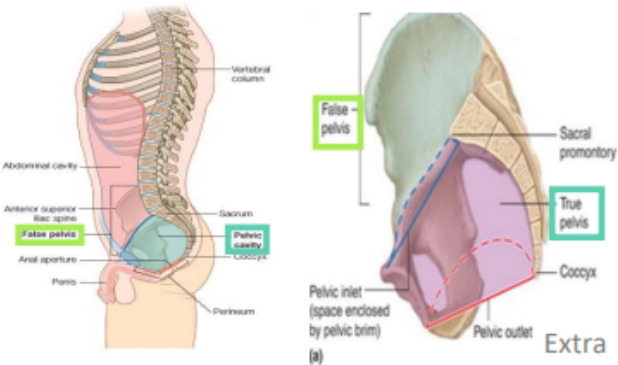
Its borders consist of:

- Anterior: inferior border of pubic symphysis
- Anterolateral ischiopubic ramus
- Posterolateral: Ischial tuberosities & sacrotuberous ligament
- Posterior: tip of coccyx



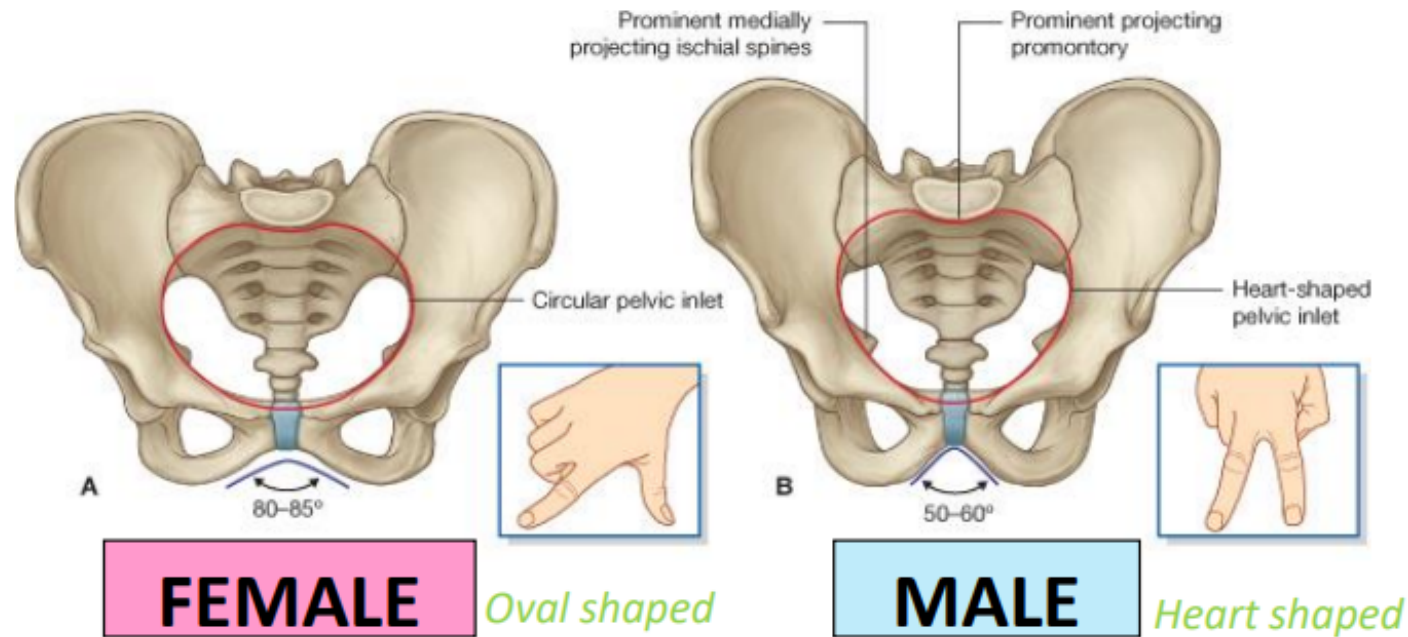
The pelvis is divided into two parts by the pelvic brim or inlet:

False or greater pelvis	True or lesser pelvis	
<u>Above</u> the brim (part of the abdominal cavity)	<u>Below</u> the brim. Has 3 parts:	
Bounded by: <u>Posteriorly:</u> Lumbar vertebrae. <u>Laterally:</u> Iliac fossae and the iliacus muscle. <u>Anteriorly:</u> Lower part of the anterior abdominal wall. It supports the abdominal contents	1. A Cavity: The cavity is a short, curved canal, with a shallow anterior wall and a deeper posterior wall. It lies between the inlet and the outlet.	
	2. Inlet (oval shape)	3. Outlet* (Diamond shape)
	<u>Anteriorly:</u> <u>Symphysis pubis</u> (upper border)	<u>Anteriorly:</u> <u>Symphysis pubis</u> (lower border)
	<u>Posteriorly:</u> <u>Promontory of sacrum</u> , <u>ala of sacrum</u> .	<u>Posteriorly :</u> Tip of <u>Coccyx</u> .
	<u>Laterally:</u> (both sides) <u>Ileopectineal (arcuate) lines</u> . (Ileo = illum \ pectineal = pubis)	<u>Anterolaterally:</u> <u>ischiopubic ramus</u> . <u>Posterolaterally:</u> <u>Sacro</u> tuberous ligament & <u>ischial</u> tuberosities.



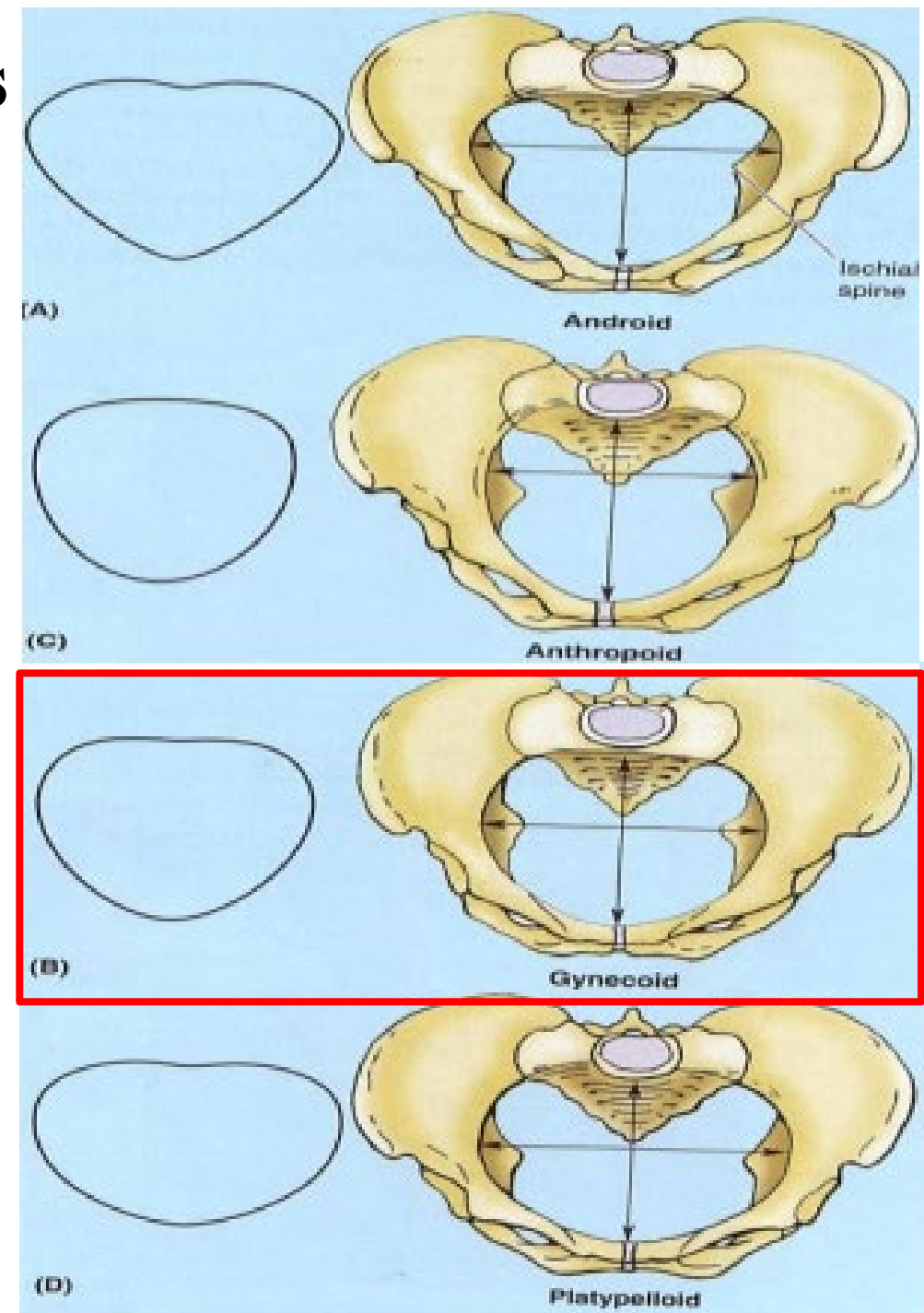
*recall these are the same boundaries as the perineum

Differences between male & female pelvis



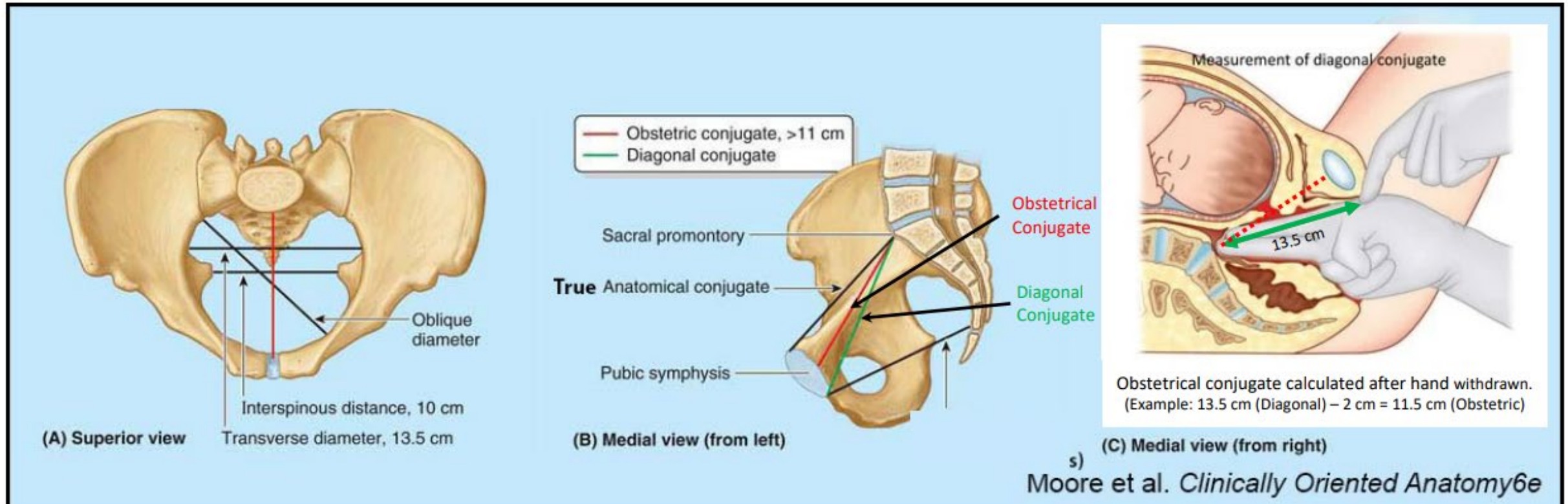
What is the main differences between male and female pelvis?

1. In female the **Sacrum** is usually wider and shorter, while in male is long and curved.
2. Also, the **Angle of the pubic arch** is wider.
3. The **promontory** and the **ischial spines** are less projecting, while the male is inverted.



Pelvic Measurements

- The diagonal conjugate diameter is the distance between the sacral promontory (palpated by the middle finger) and the anterior surface of the pubic symphysis' inferior margin.
- This distance can be determined during a pelvic examination, which is then used to calculate the obstetrical conjugate. Once the diagonal conjugate is measured, the examiner subtracts 1.5 to 2 cm from the diagonal conjugate measurement.

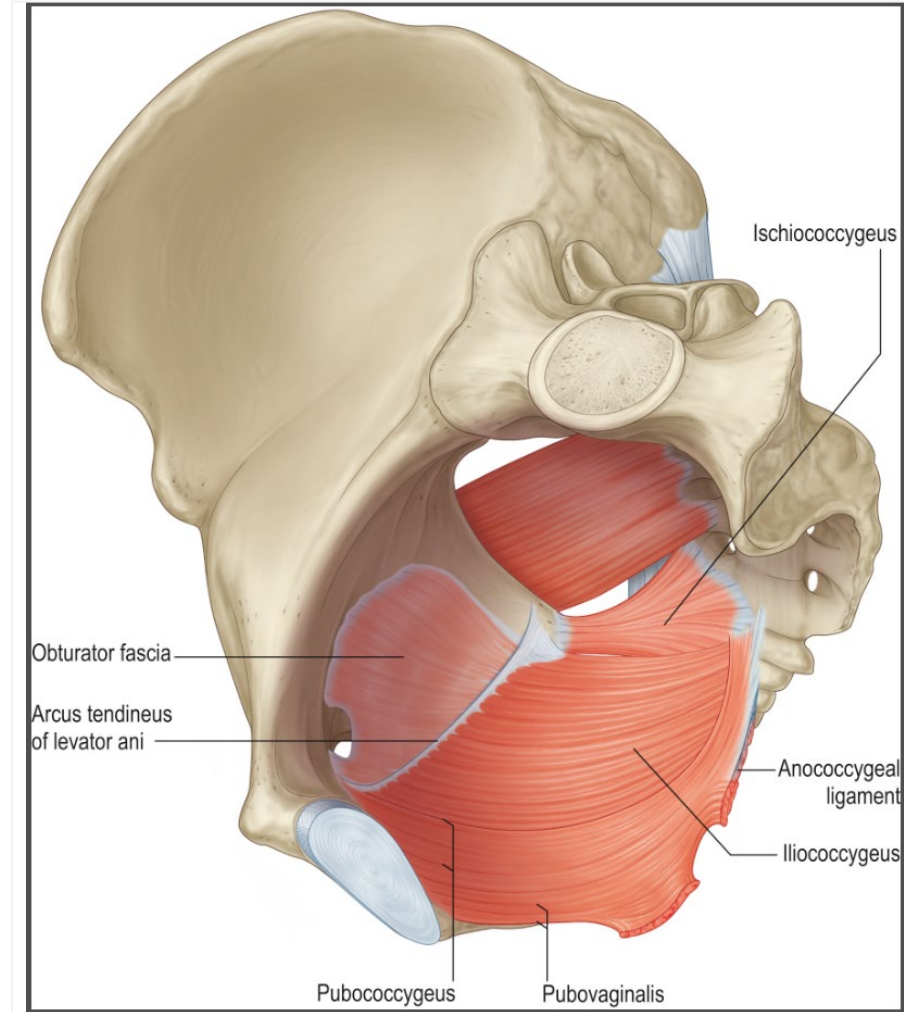


Muscles of the Pelvis

Lateral walls: **Obturator internus muscle** inserts on the internal surface of the obturator membrane and passes through the **lesser sciatic foramen** to insert on the greater trochanter of the femur. Within the pelvis, the obturator internus muscle is covered in a thick fascia called the **obturator fascia**.

Posterior walls: **Piriformis muscle**, its proximal attachment is on the anterior aspects of the S2-4 vertebral segments and passes through the **greater sciatic foramen** to its distal attachment on the greater trochanter of the femur.

Floor: **The pelvic floor (pelvic diaphragm)** is a funnel-shaped structure composed of the following muscles: **levator ani and coccygeus**.



Muscles of the Pelvis

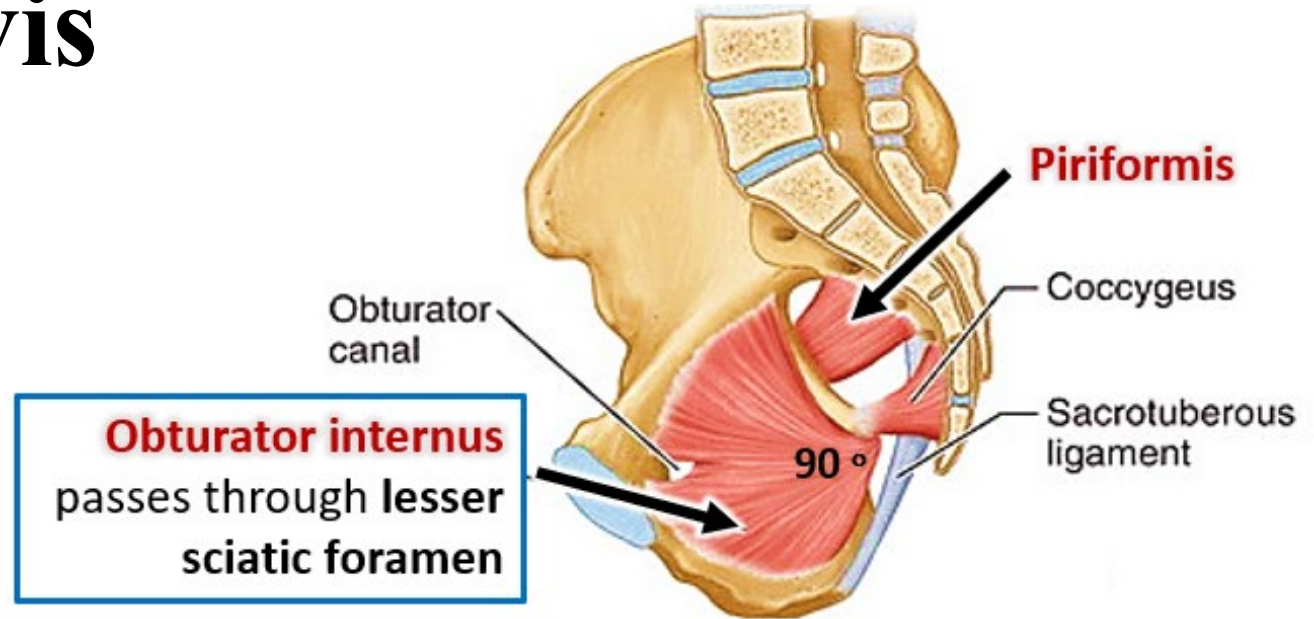
Piriformis:

O: anterior sacrum

I: greater trochanter

N: nerve to Piriformis

A: Lat. Rotation of femur



(A) Medial view of hemisected pelvis

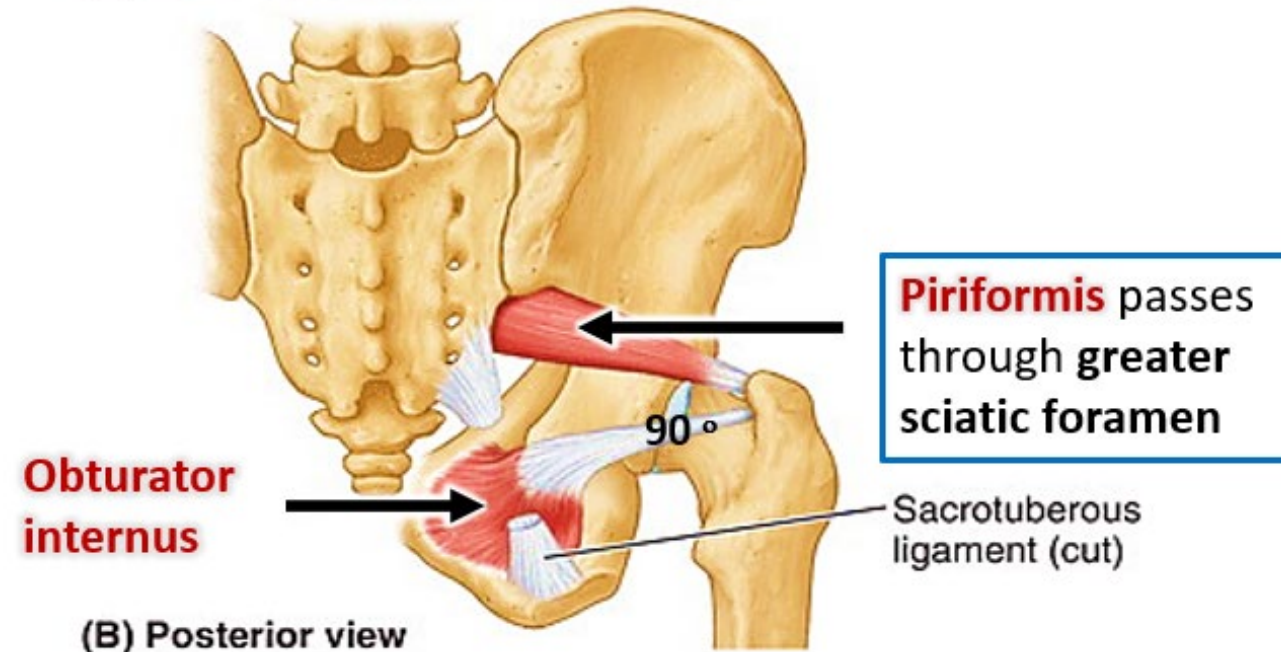
Obturator Internus:

O: inner surface of obturator foramen

I: greater trochanter

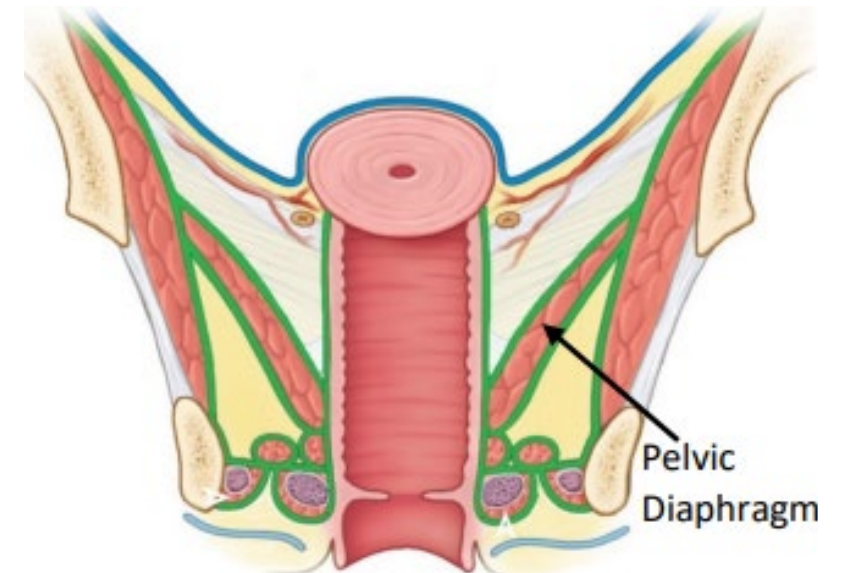
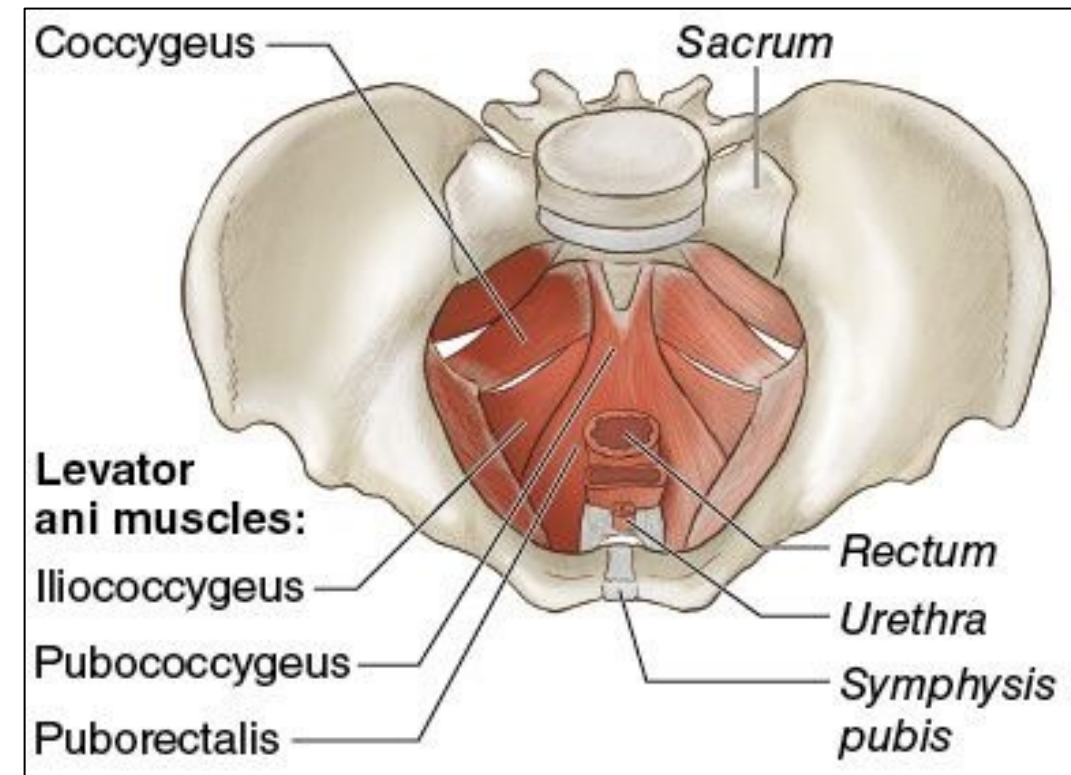
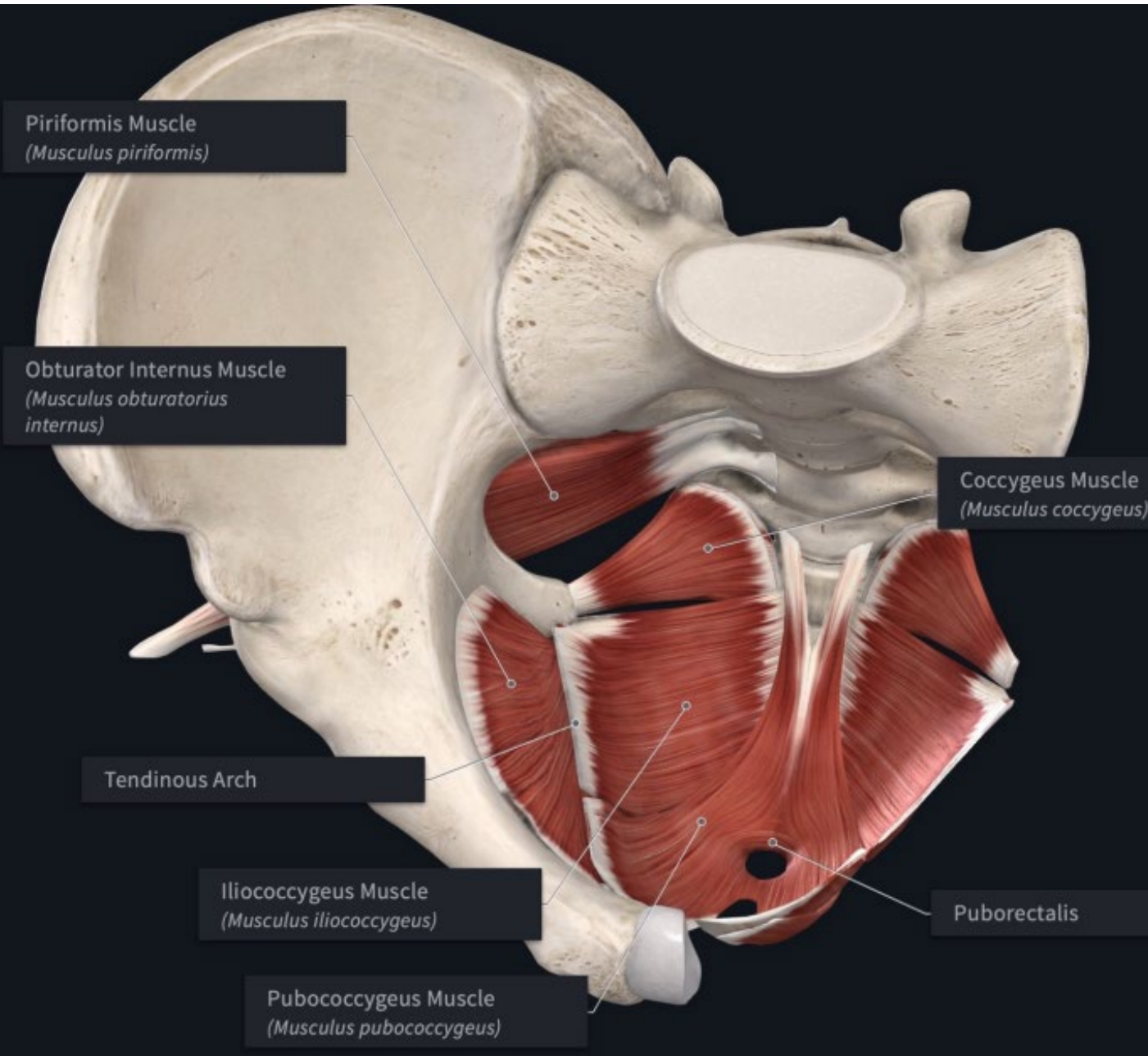
N: nerve to obturator internus

A: Lat. Rotation of femur



(B) Posterior view

Muscles of the Pelvis



Levator ani

- Its fibers are divided into 3 parts:

1. Pubococcygeus
2. Puborectalis
3. Iliococcygeus

1. Pubococcygeus

- Origin: originates from the posterior surface of the body of the pubis and passes downward & medially
- Insertion: inserted into the perineal body, anococcygeal body and coccyx.
- Action:
 - forms a sling around the prostate or the vagina*:

In males

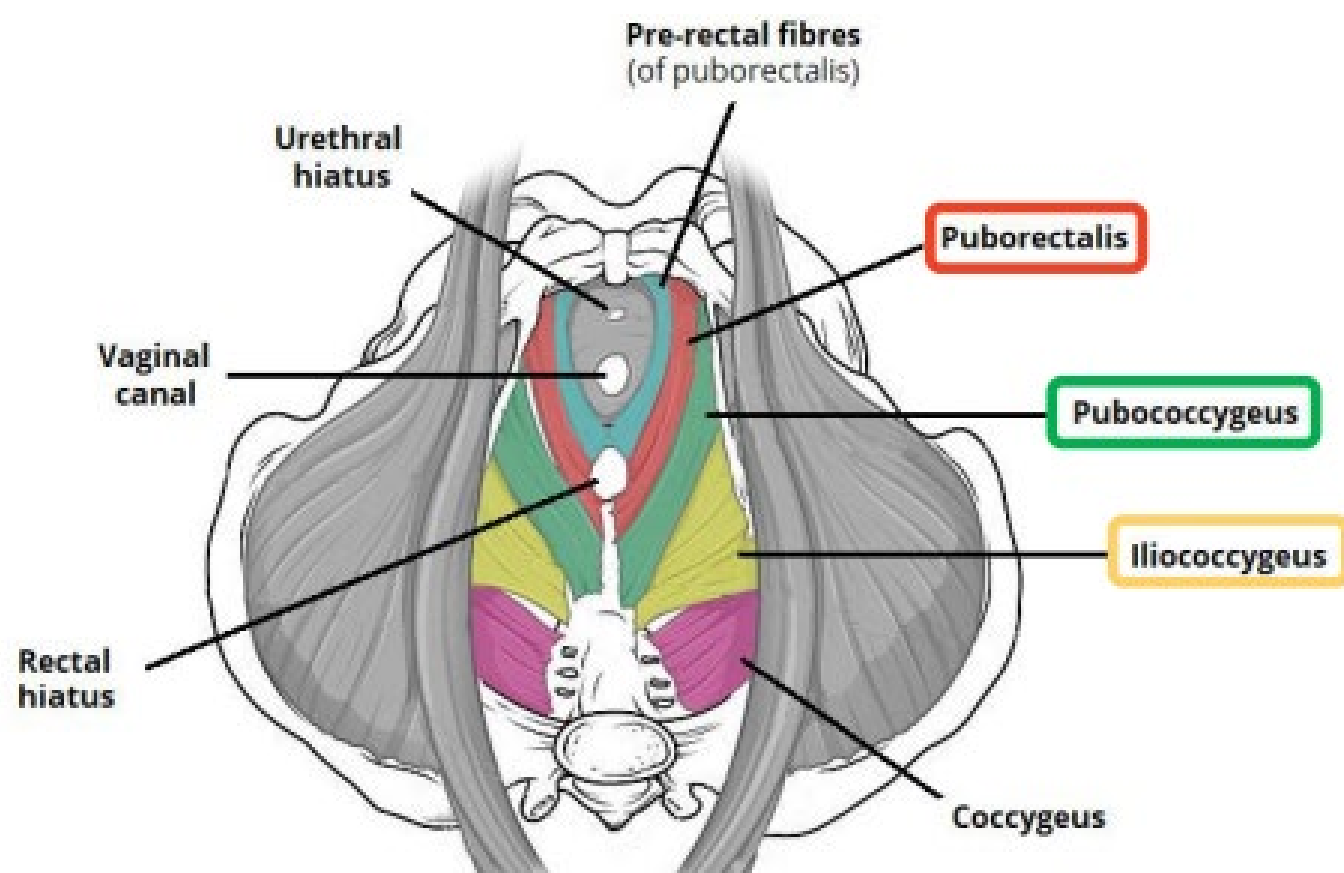
Levator prostate:

1. Supports prostate
2. Stabilizes perineal body

In females

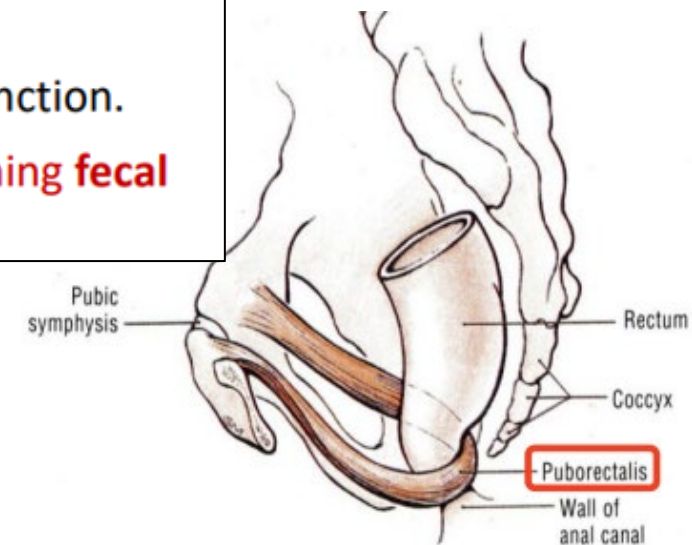
Sphincter vaginae:

1. Supports vagina
2. Stabilizes perineal body

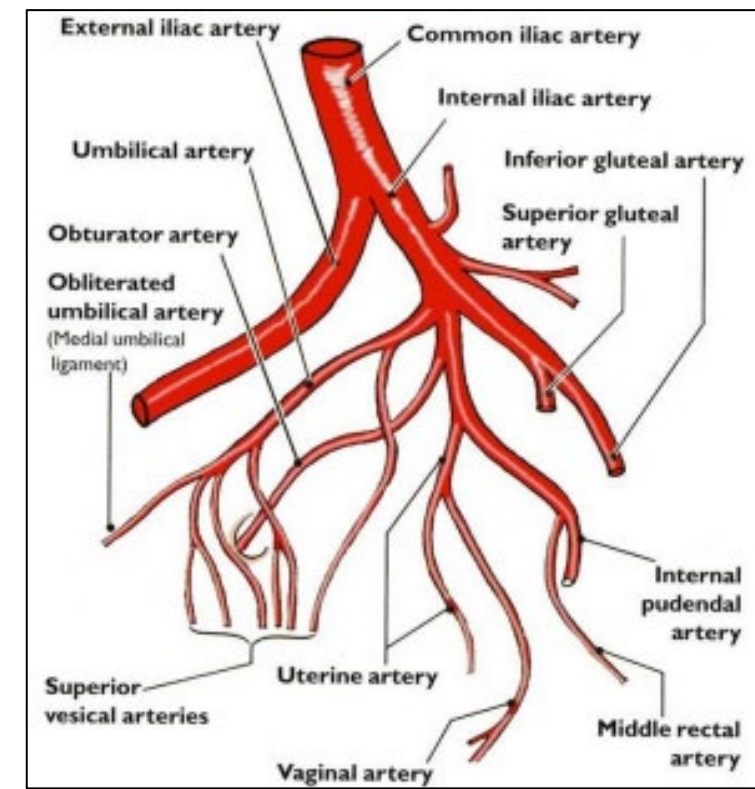
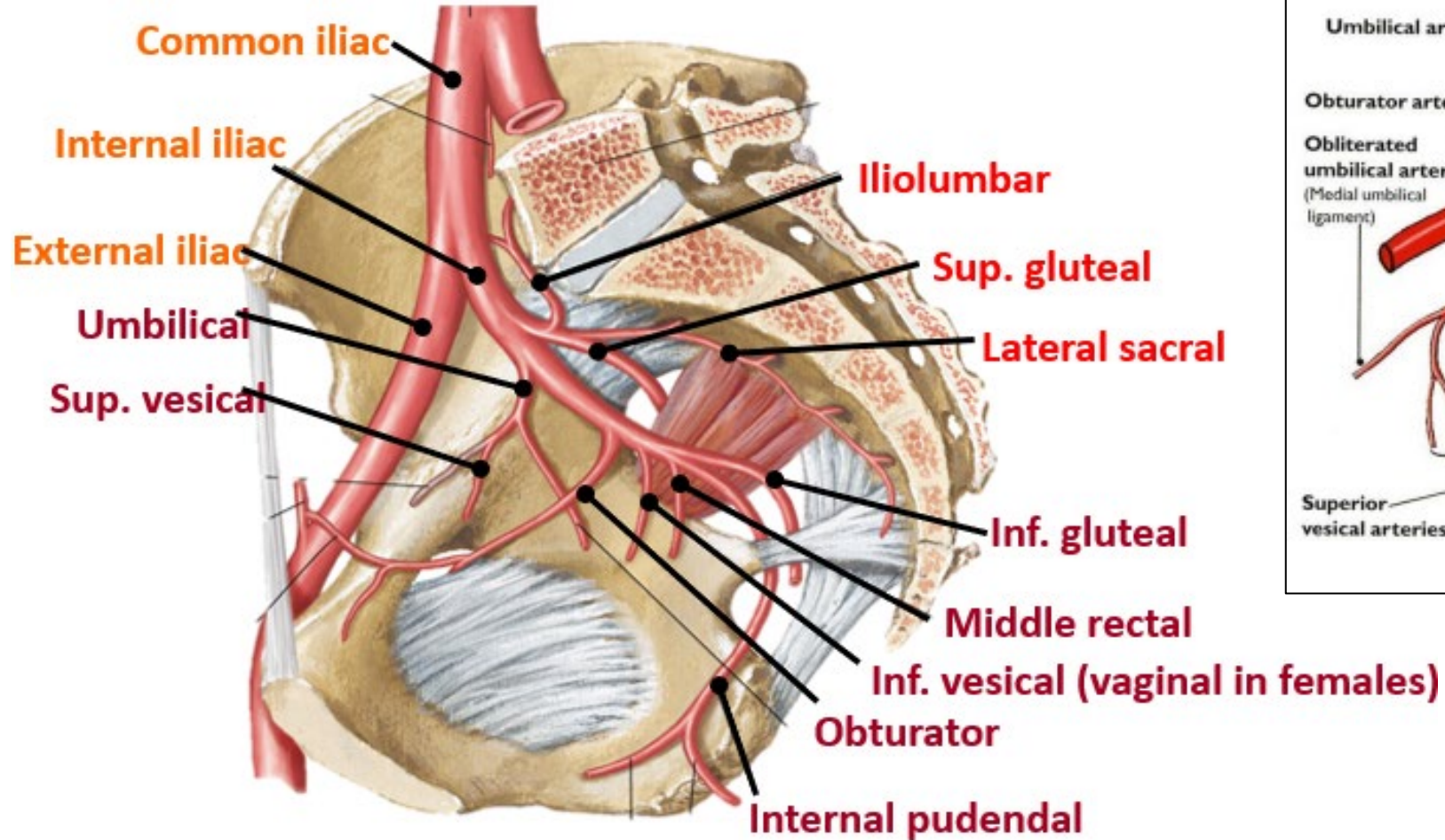


2. Puborectalis

- forms a sling around the recto-anal Junction.
- It has a very important role in maintaining **fecal continence**.

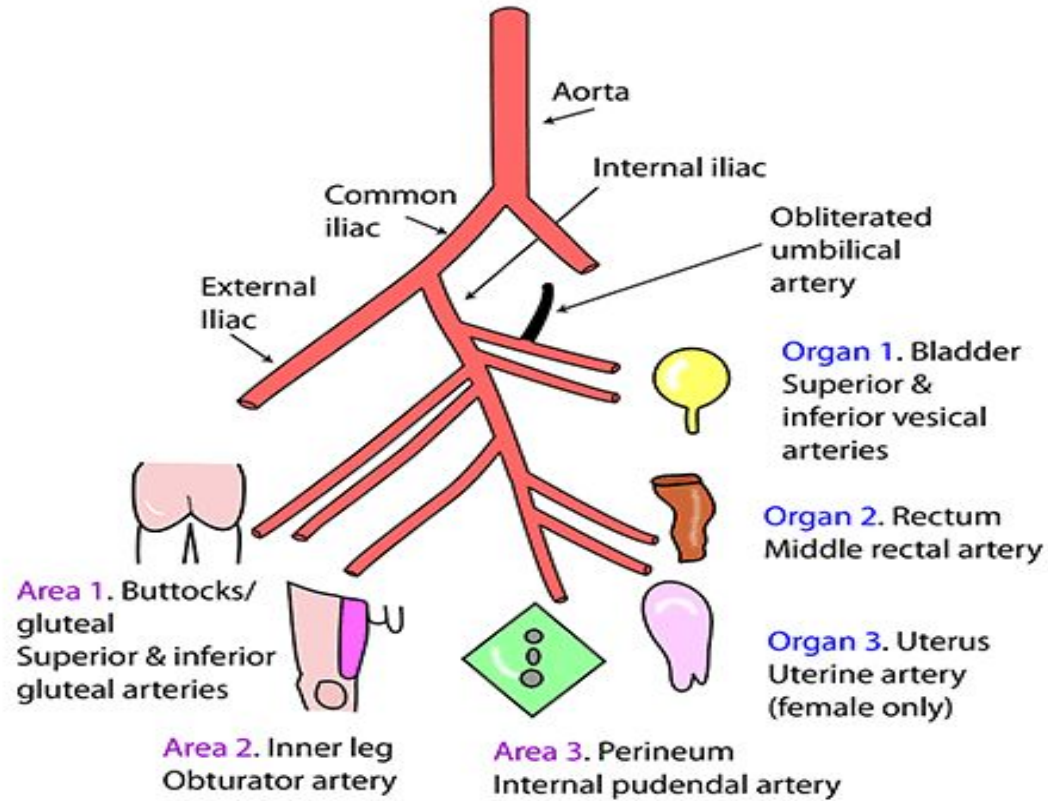


Internal iliac artery



Supplies the pelvis, perineum, gluteal region & medial thigh. Branches are **highly variable**, best identified and named for the structure they supply. Branches of the internal iliac a. are divided into **anterior & posterior trunks** for descriptive purposes. **Posterior branches: iliolumbar, lateral sacral, & superior gluteal arteries**; all other are from the anterior trunk.

BRANCHES OF THE INTERNAL ILIAC ARTERY



A simple representational diagram of the branches of the internal iliac artery. An easy way to remember them is to think of them in 2 groups, each with 3 branches. **Group 1** has 3 branches to organs (bladder, rectum and uterus) and **Group 2** has 3 branches to areas (buttocks/gluteal, adductor compartment and perineum).

ILIOLUMBAR: Passes laterally, behind obturator nerve & psoas. Lumbar branch to psoas, quadratus lumborum & spine. Iliac branch to iliacus, iliac bone, anastomosis at anterior superior iliac spine

LATERAL SACRAL: Passes inferiorly, lateral to anterior sacral foramina & anterior to nerve roots & piriformis

ARTERY TO VAS: Usually off superior vesical (or inferior vesical)

UTERINE ARTERY: In female it largely replaces middle rectal (or inferior vesical)

Parietal Branches

From posterior division

1. Iliolumbar artery.
2. Lateral sacral arteries (2 branches.)
3. Superior gluteal artery.

From anterior division:

4. Obturator artery.
5. Inferior Gluteal artery.

Visceral Branches

(all from anterior division only)

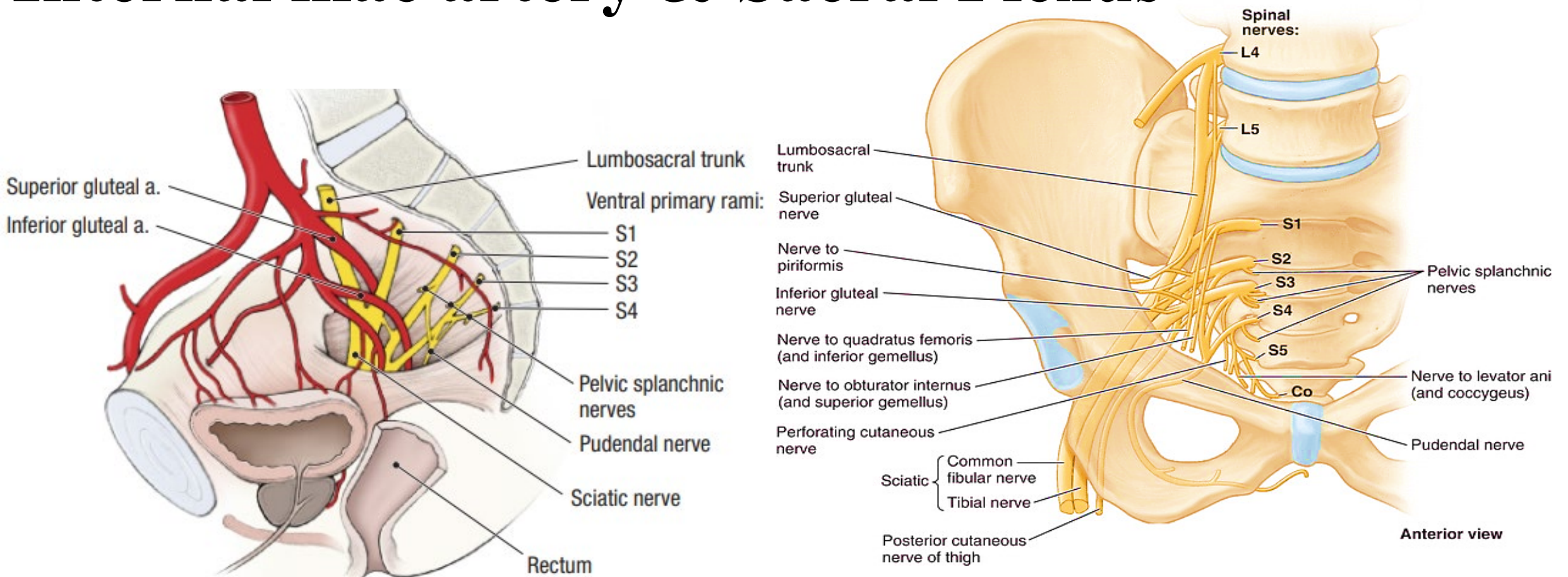
1. Umbilical artery : Gives the superior vesical* artery:
The distal part of this artery fibrosed and forms the **Medial Umbilical Ligament**.
2. **Inferior Vesical** artery in male or **vaginal** in female:
In the male it supplies, the Prostate and the Seminal Vesicles. It also gives the artery of the Vas Deferens.
3. **Middle rectal** artery
4. Internal pudendal artery
It is the main arterial supply to the perineum.

Visceral Branches (In Female)

1. Vaginal artery:
Replaces the inferior vesical artery.
2. Uterine artery: has tortuous* course
Crosses the Ureter superiorly and supplies the uterus & uterine tubes.
N.B. The ureter may be wrongly ligated (cut) in **hysterectomy** (removal of uterus)

**Water
under the
bridge**

Internal iliac artery & Sacral Plexus

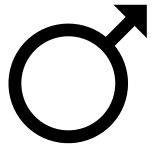


- **Superior gluteal a.** (between lumbosacral trunk & S1).
- **Inferior gluteal a.** (between S1 & S2) OR (between S2 & S3).
- **Internal pudendal a.** (leaves the pelvis through the greater sciatic notch below the piriformis to cross the ischial spine and reach the perineum).

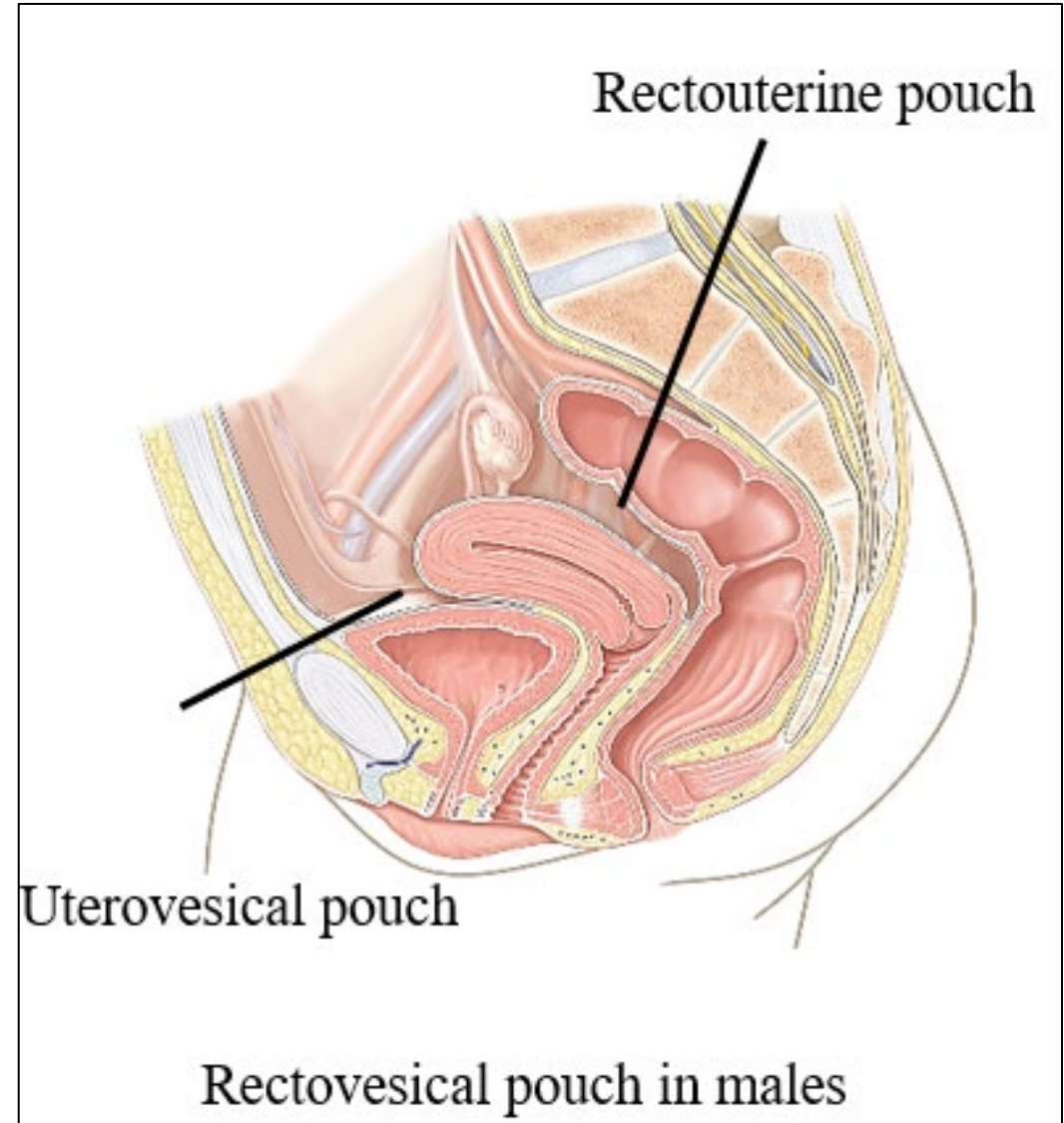
Female & Male Pelvic Peritoneum



- Open sac
- Two pouches



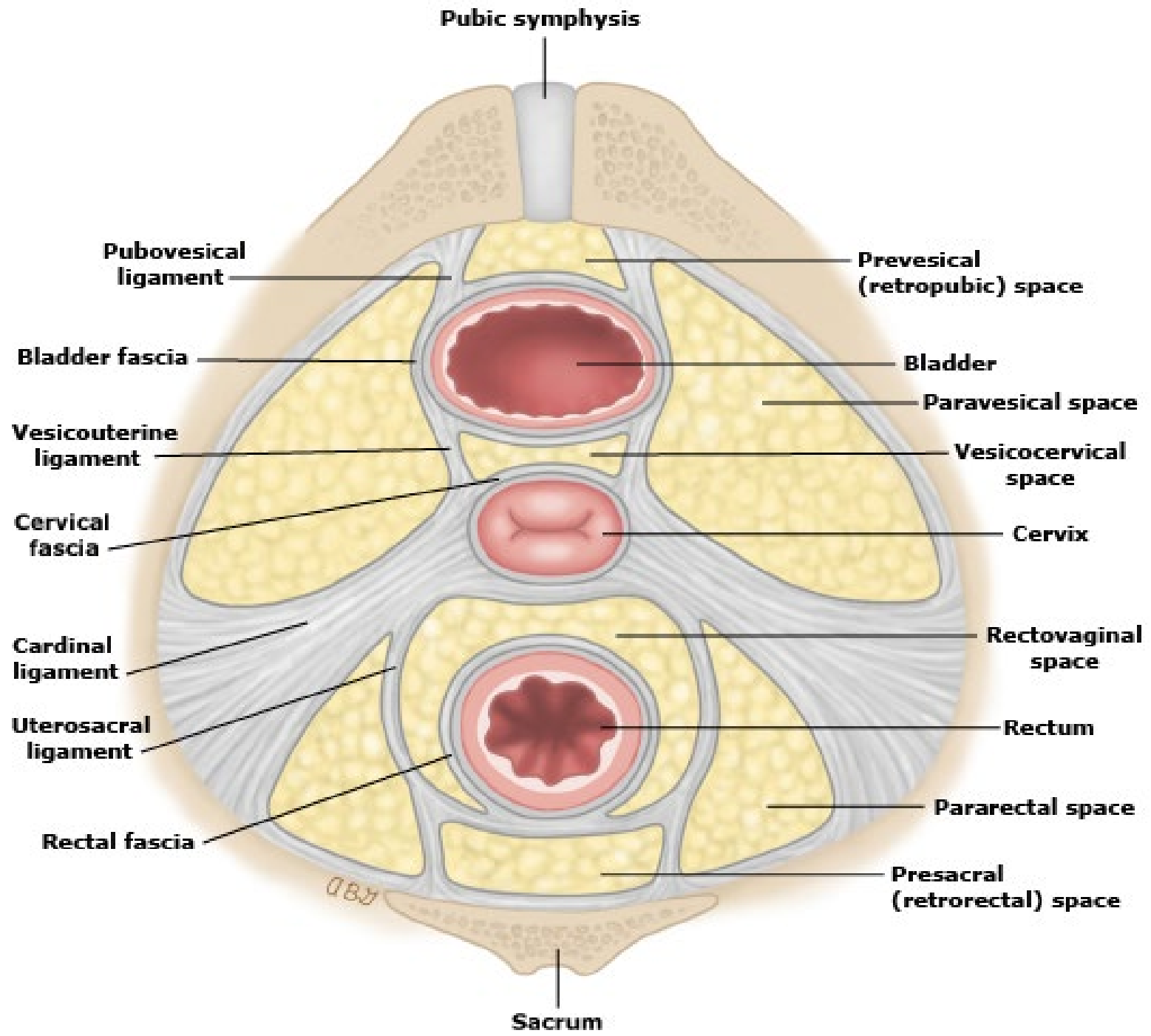
- Closed sac
- One pouch



Pelvic Spaces

Pelvic connective tissue divides the subperitoneal pelvic area into different spaces that are filled with fatty or loose areolar connective tissue.

These are potential spaces that are particularly important for pelvic surgeries.





Questions

Question

A 16-year-old woman is brought to the emergency department with severe abdominal pain and fever. Laboratory examination is remarkable for an elevated white blood cell count and a positive test for pregnancy. Culdocentesis is performed to ascertain the presence of blood in the pelvis from a ruptured ectopic pregnancy. Through which of the following structures does the needle need to be inserted to perform culdocentesis?

- A. Through the perineal body into the vesicouterine space
- B. Through the posterior fornix of the vagina into the rectouterine pouch
- C. Through the anterior fornix into the endocervical canal
- D. Through the introitus into the vestibular gland
- E. Through the perineal membrane into the urogenital diaphragm

Question

A 16-year-old woman is brought to the emergency department with severe abdominal pain and fever. Laboratory examination is remarkable for an elevated white blood cell count and a positive test for pregnancy. Colpocentesis is performed to ascertain the presence of blood in the pelvis from a ruptured ectopic pregnancy. Through which of the following structures does the needle need to be inserted to perform Colpocentesis?

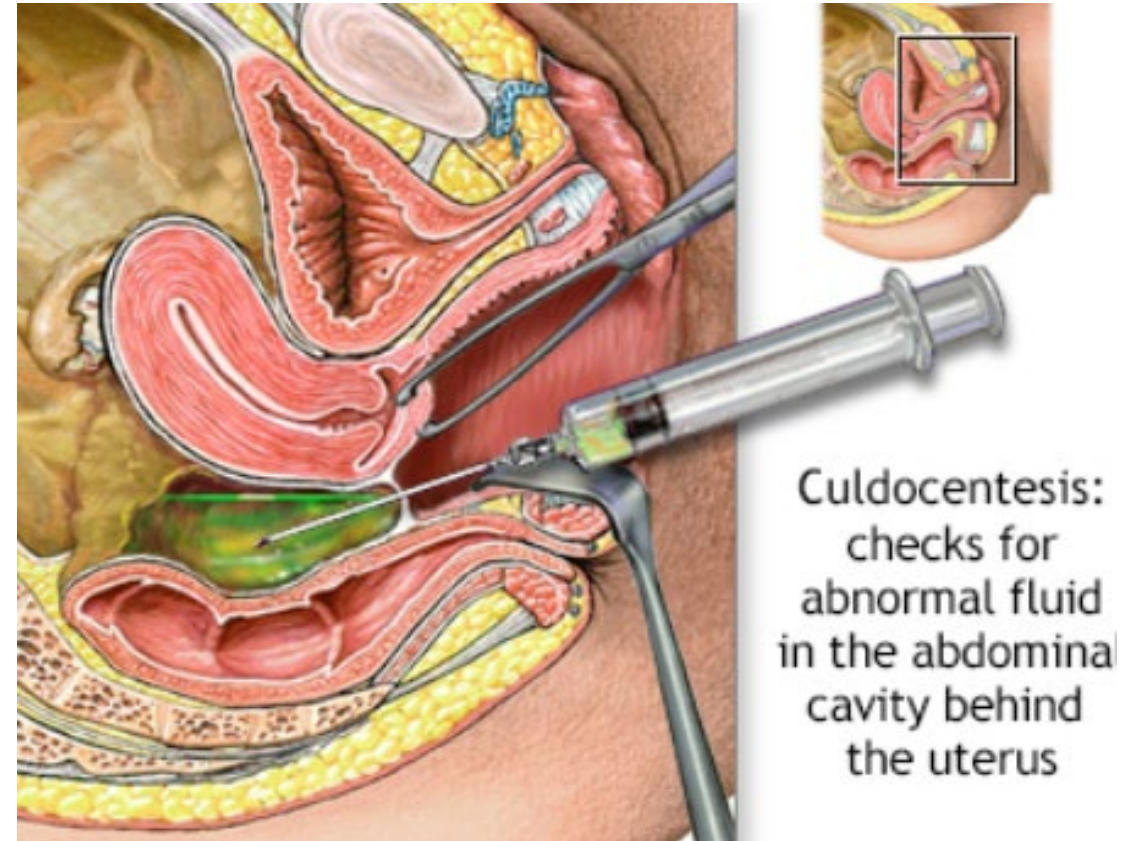
B

- A. Through the perineal body into the vesicouterine space
- B. Through the posterior fornix of the vagina into the rectouterine pouch
- C. Through the anterior fornix into the endocervical canal
- D. Through the introitus into the vestibular gland
- E. Through the perineal membrane into the urogenital diaphragm

Colpocentesis (or Culdocentesis)

Performed to aspirate fluid from the rectouterine pouch, usually to diagnose hemoperitoneum, such as from a ruptured ectopic pregnancy.

The posterior fornix is the deepest part of the vaginal canal located behind the cervix, providing direct access to the rectouterine pouch.



Question

A 17-year-old boy presents to the emergency department with a stab wound from a knife that entered the pelvis through the greater sciatic notch superior to the piriformis muscle. Following the injury, the patient develops weakness in hip abduction and pelvic instability.

Which of the following structures is most likely to be damaged?

- A. Sciatic nerve
- B. Internal pudendal artery & pudendal nerve
- C. Superior gluteal nerve & vessels
- D. Inferior gluteal artery & vessels
- E. Posterior femoral cutaneous nerve

Question

A 17-year-old boy presents to the emergency department with a stab wound from a knife that entered the pelvis through the greater sciatic notch superior to the piriformis muscle. Following the injury, the patient develops weakness in hip abduction and pelvic instability.

Which of the following structures is most likely to be damaged?

- A. Sciatic nerve
- B. Internal pudendal artery & pudendal nerve
- C. Superior gluteal nerve & vessels
- D. Inferior gluteal artery & vessels
- E. Posterior femoral cutaneous nerve

C

Question

A 62-year-old man is admitted to the emergency department due to increasing difficulty in urinating over a period of several months. Physical examination reveals prostatic hypertrophy. After several unsuccessful attempts to catheterize the penile urethra, the urologist orders drainage of the urinary bladder by the least invasive procedure, avoiding entry into the peritoneal cavity or the injury of any major vessels or organs. Which of the following spaces needs to be traversed by the needle to reach the bladder?

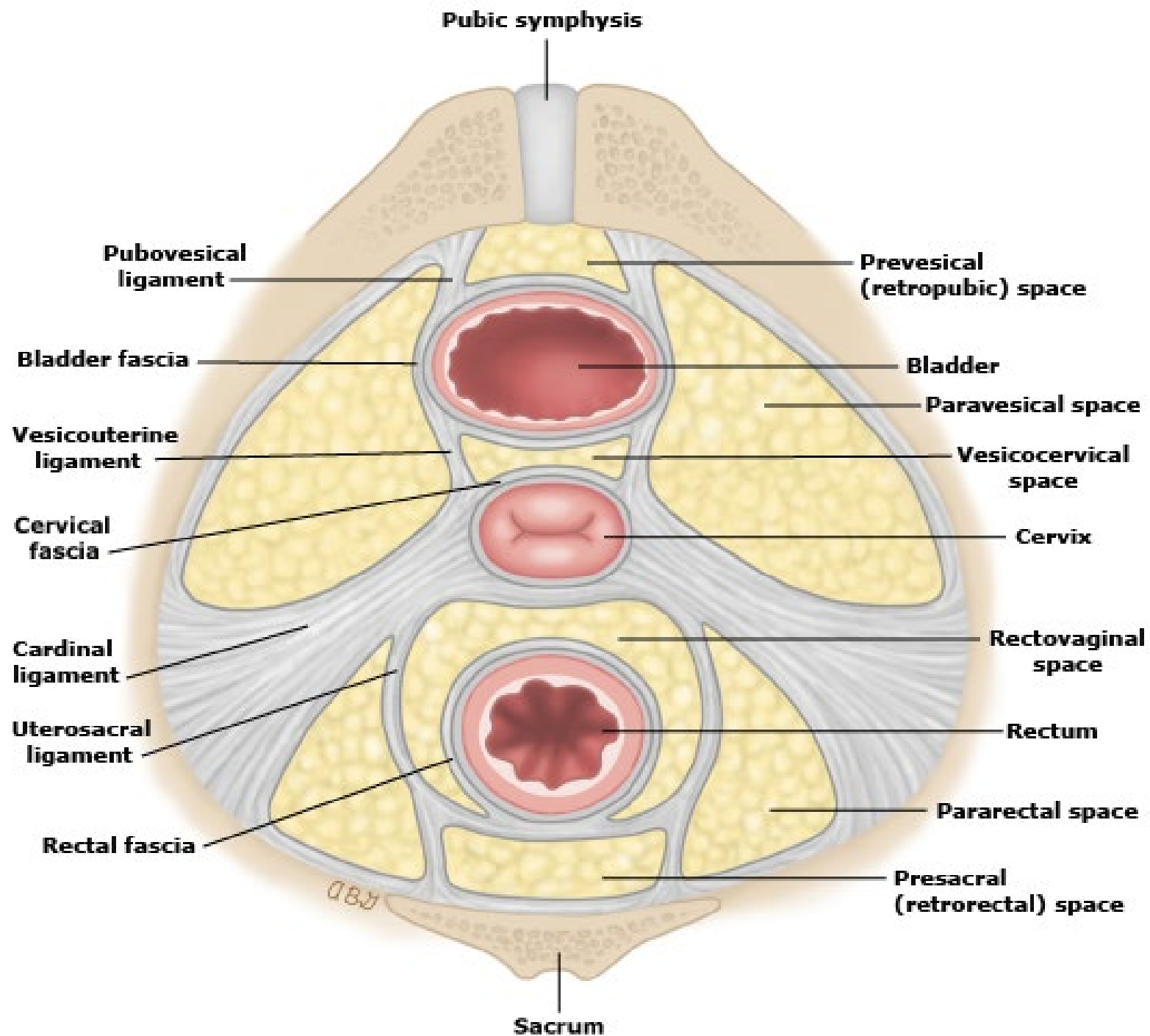
- A. Ischioanal fossa
- B. Perineal body
- C. Retropubic space (of Retzius)
- D. Superficial perineal pouch
- E. Deep perineal pouch

Question

A 62-year-old man is admitted to the emergency department due to increasing difficulty in urinating over a period of several months. Physical examination reveals prostatic hypertrophy. After several unsuccessful attempts to catheterize the penile urethra, the urologist orders drainage of the urinary bladder by the least invasive procedure, avoiding entry into the peritoneal cavity or the injury of any major vessels or organs. Which of the following spaces needs to be traversed by the needle to reach the bladder?

- A. Ischioanal fossa
- B. Perineal body
- C. Retropubic space (of Retzius)
- D. Superficial perineal pouch
- E. Deep perineal pouch

C



Question

A 34-year-old woman is admitted to the hospital complaining of urinary incontinence. MRI examination reveals that one of the skeletal muscles of the pelvis has a significant tear. Which of the following muscles is the most significant in terms of maintaining continence?

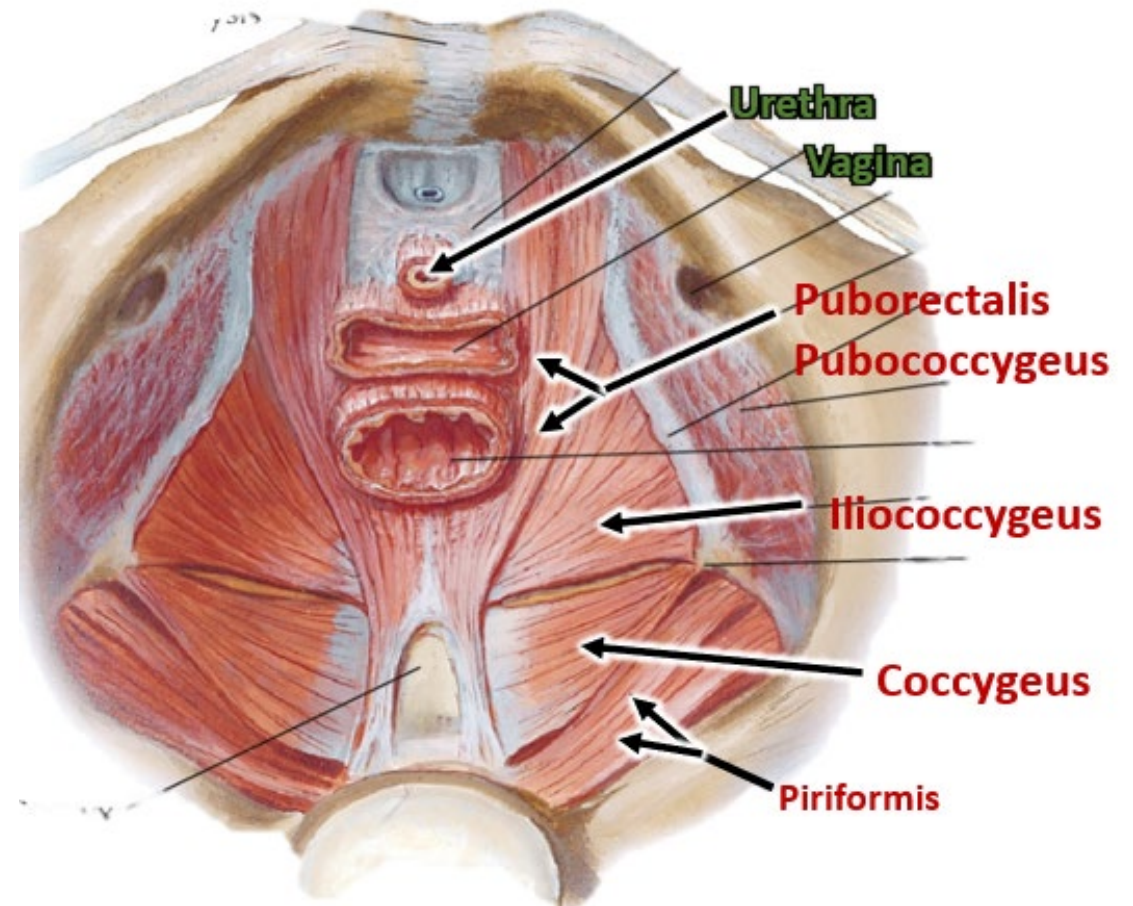
- A. Pubococcygeus
- B. Obturator internus
- C. Piriformis
- D. Coccygeus
- E. Iliococcygeus

Question

A 34-year-old woman is admitted to the hospital complaining of urinary incontinence. MRI examination reveals that one of the skeletal muscles of the pelvis has a significant tear. Which of the following muscles is the most significant in terms of maintaining continence?

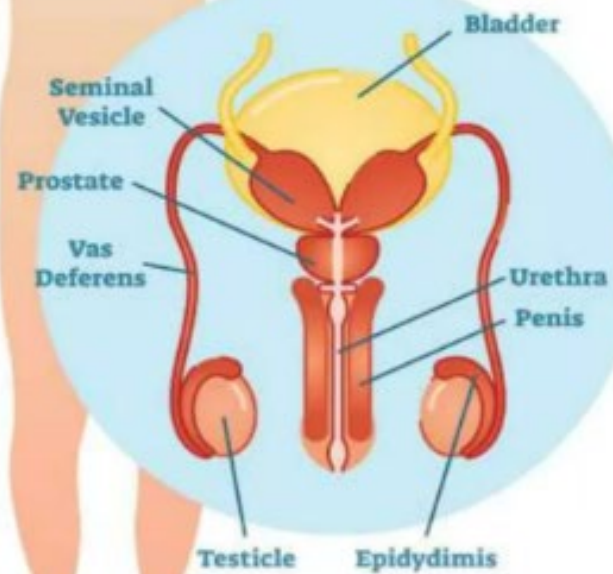
- A. Pubococcygeus
- B. Obturator internus
- C. Piriformis
- D. Coccygeus
- E. Iliococcygeus

A

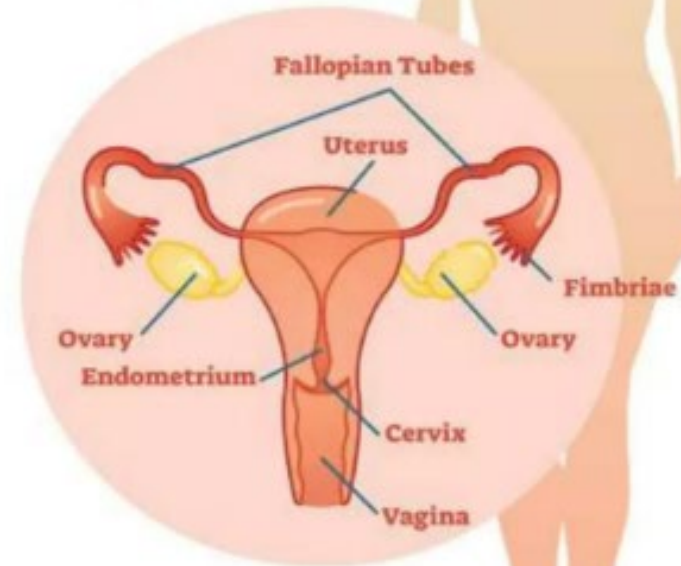


HUMAN REPRODUCTIVE SYSTEM

Male Organs



Female Organs



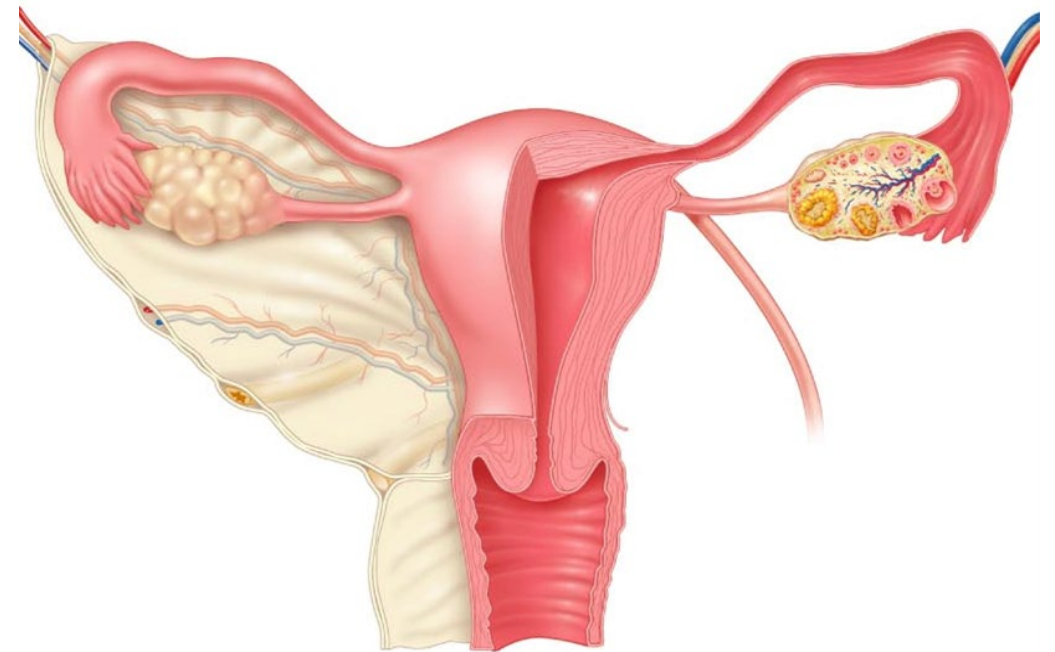
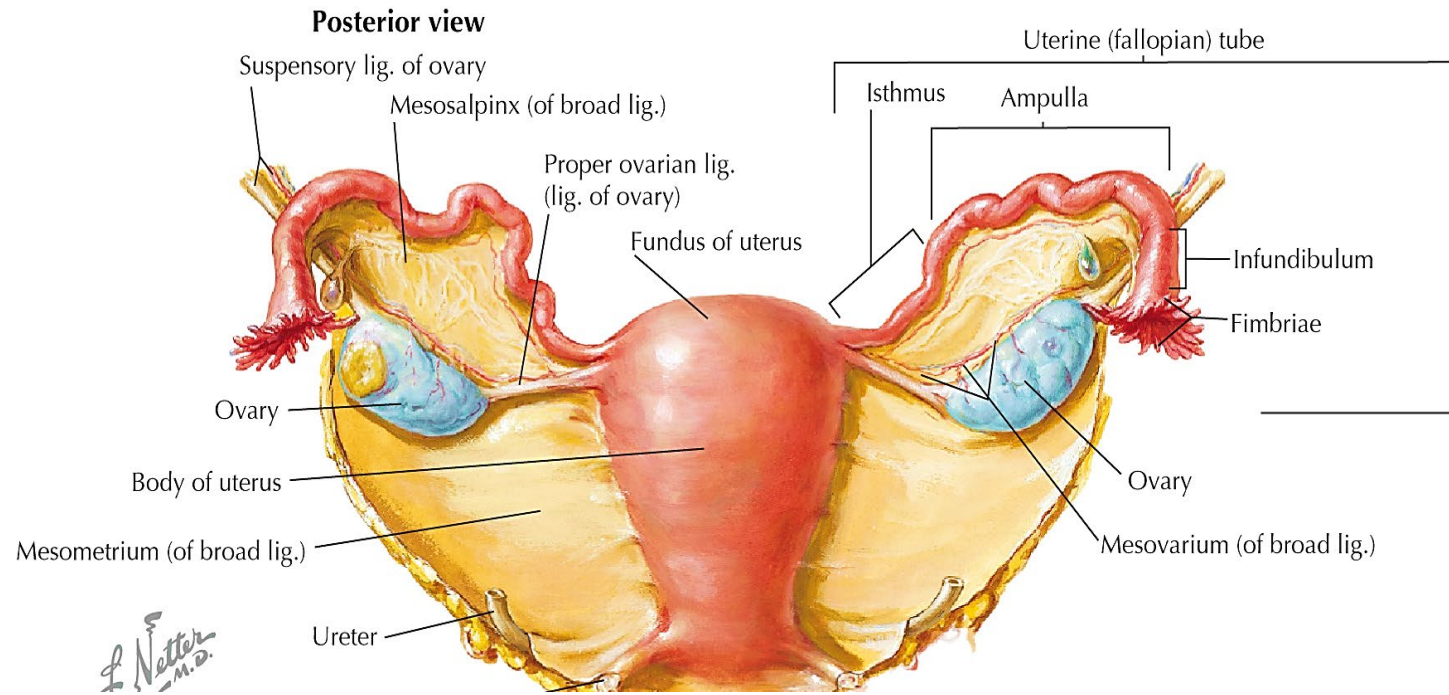
Uterus

Pear-shaped, 7-8 cm

- **Body** (fundus & isthmus)
- **Cervix** (neck): internal & external os, cervical canal
- Anteverted & anteflexed
- **Broad ligament** envelops ovaries, uterine tubes, uterus

Vagina: fibromuscular tube, 8-9 cm.

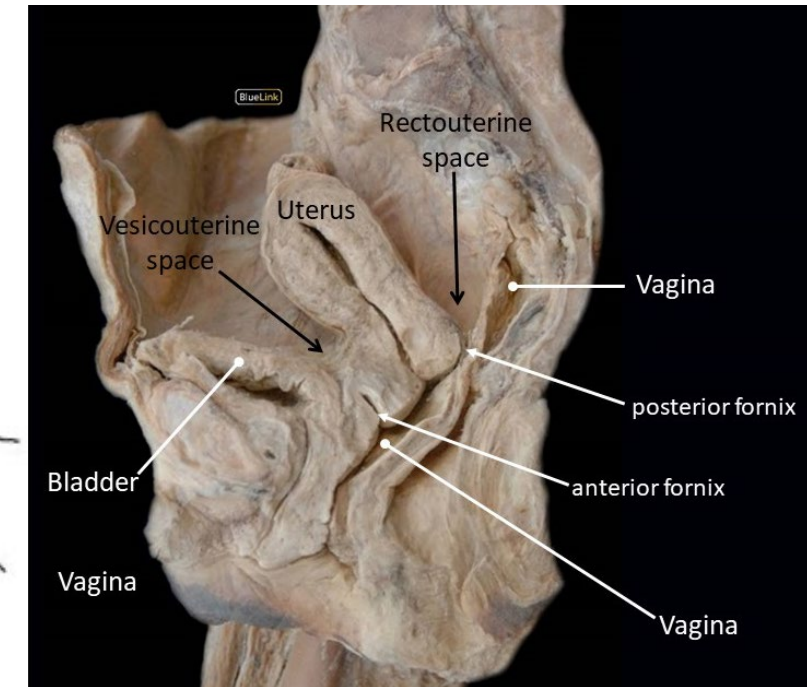
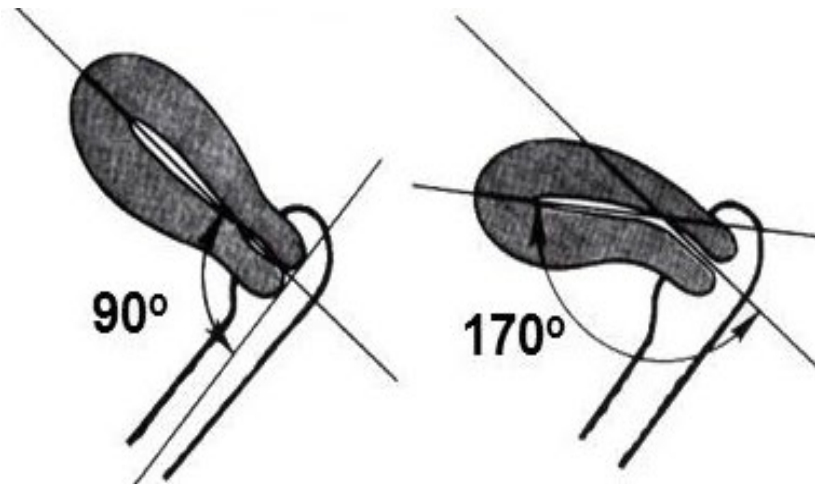
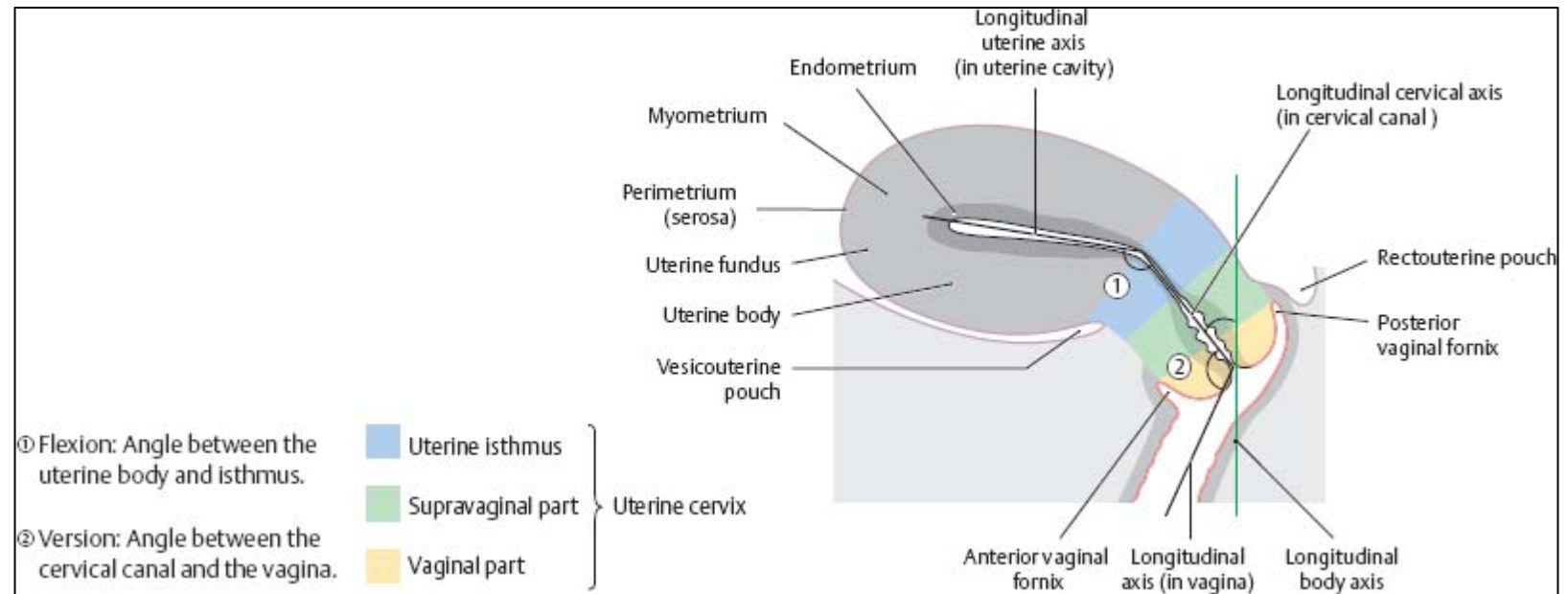
- **Fornix** (anterior, lateral, posterior)
- **Canal** (opens in the vestibule)



Uterus

Position

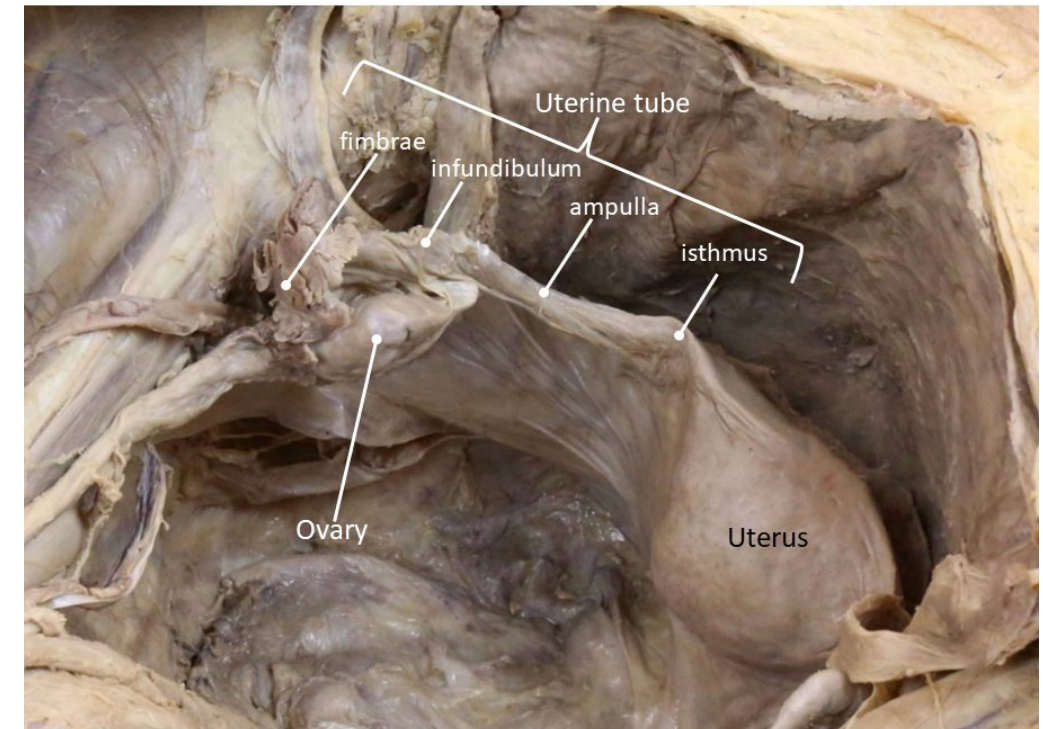
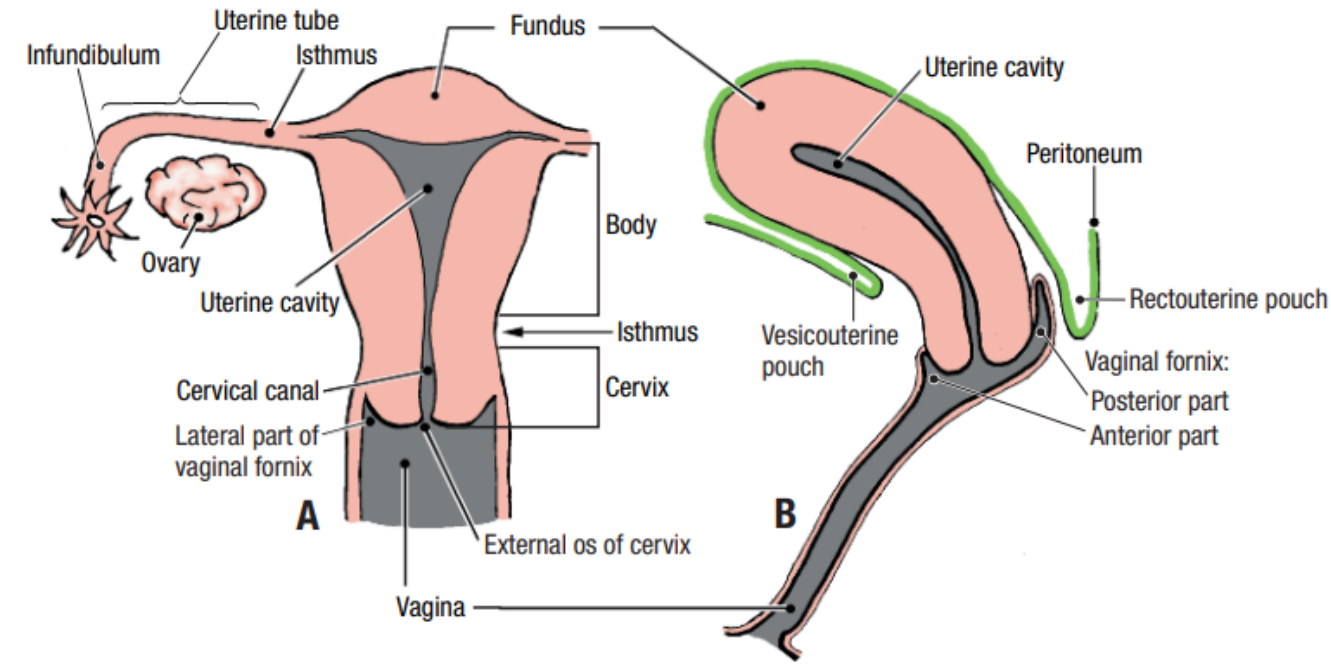
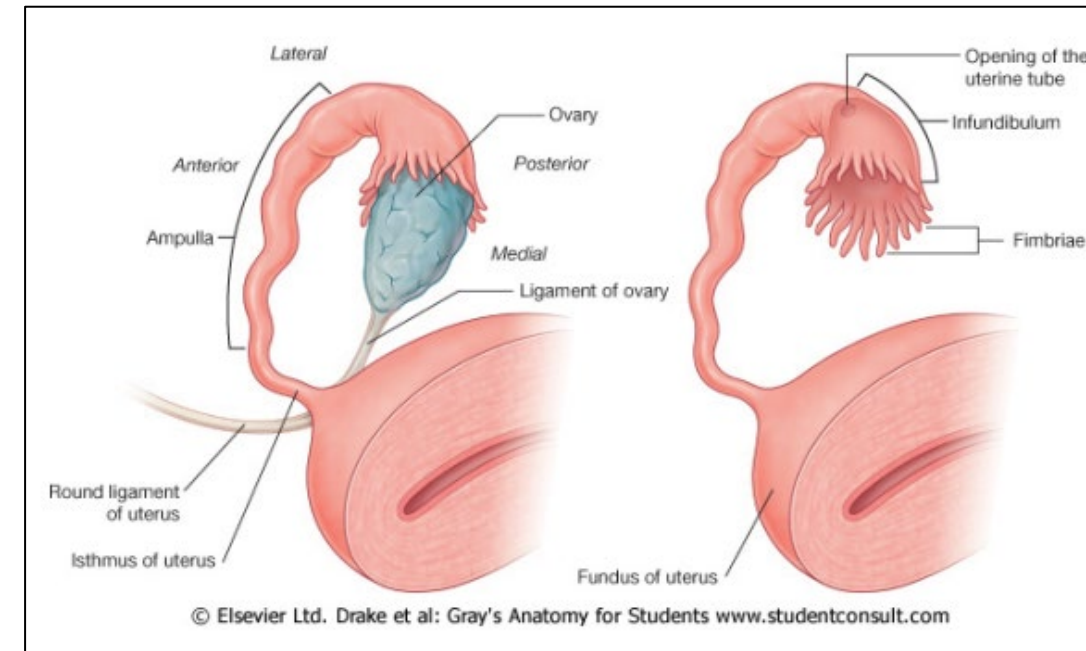
- Lies in the center of the pelvic cavity, between bladder, anteriorly, and rectum, posteriorly, usually it is anteversion and anteflexion
- **Anteversion**
 - axis of uterus at 90° with that of vagina
- **Anteflexion**
 - axis of body of uterus is bent forward with that of the cervix forming an angle of about 170° (body of uterus is bent slightly forward at isthmus)



Fallopian Tube

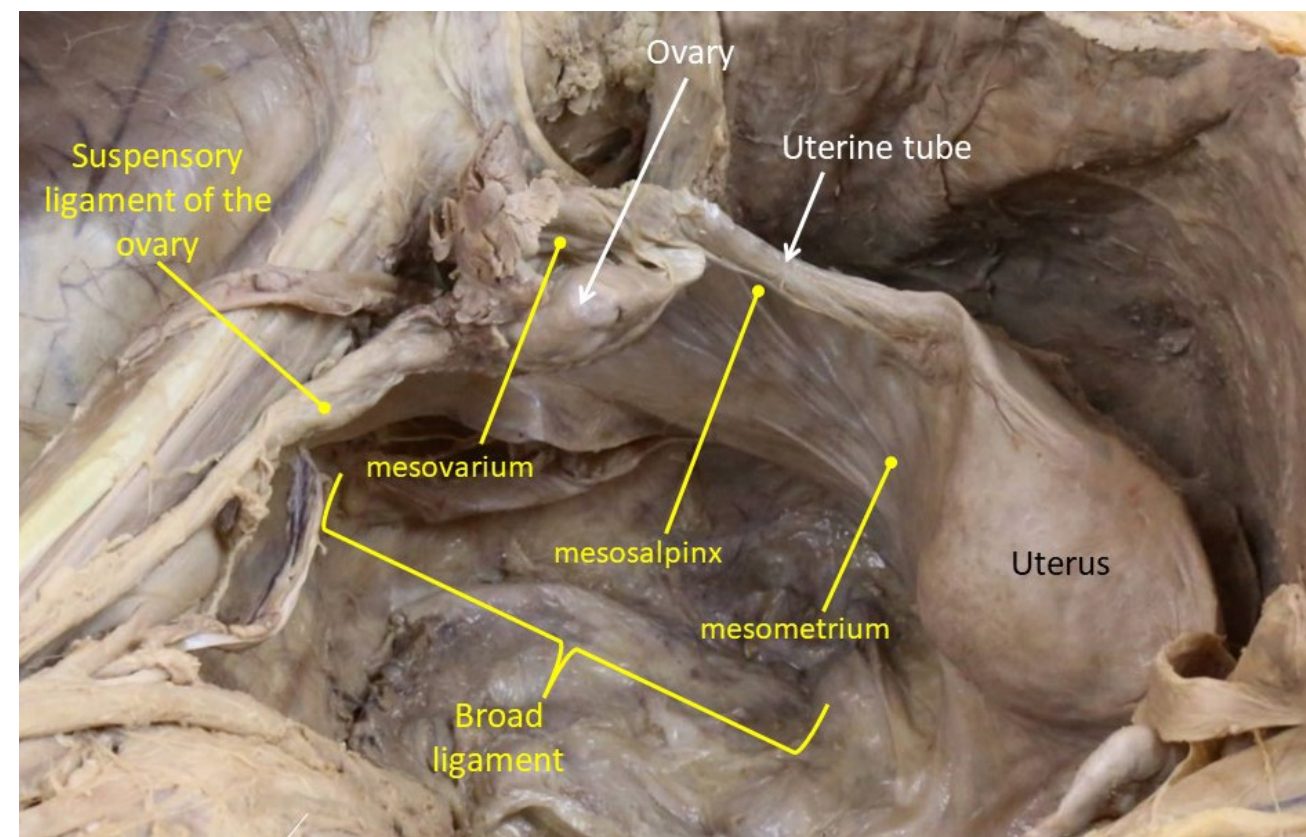
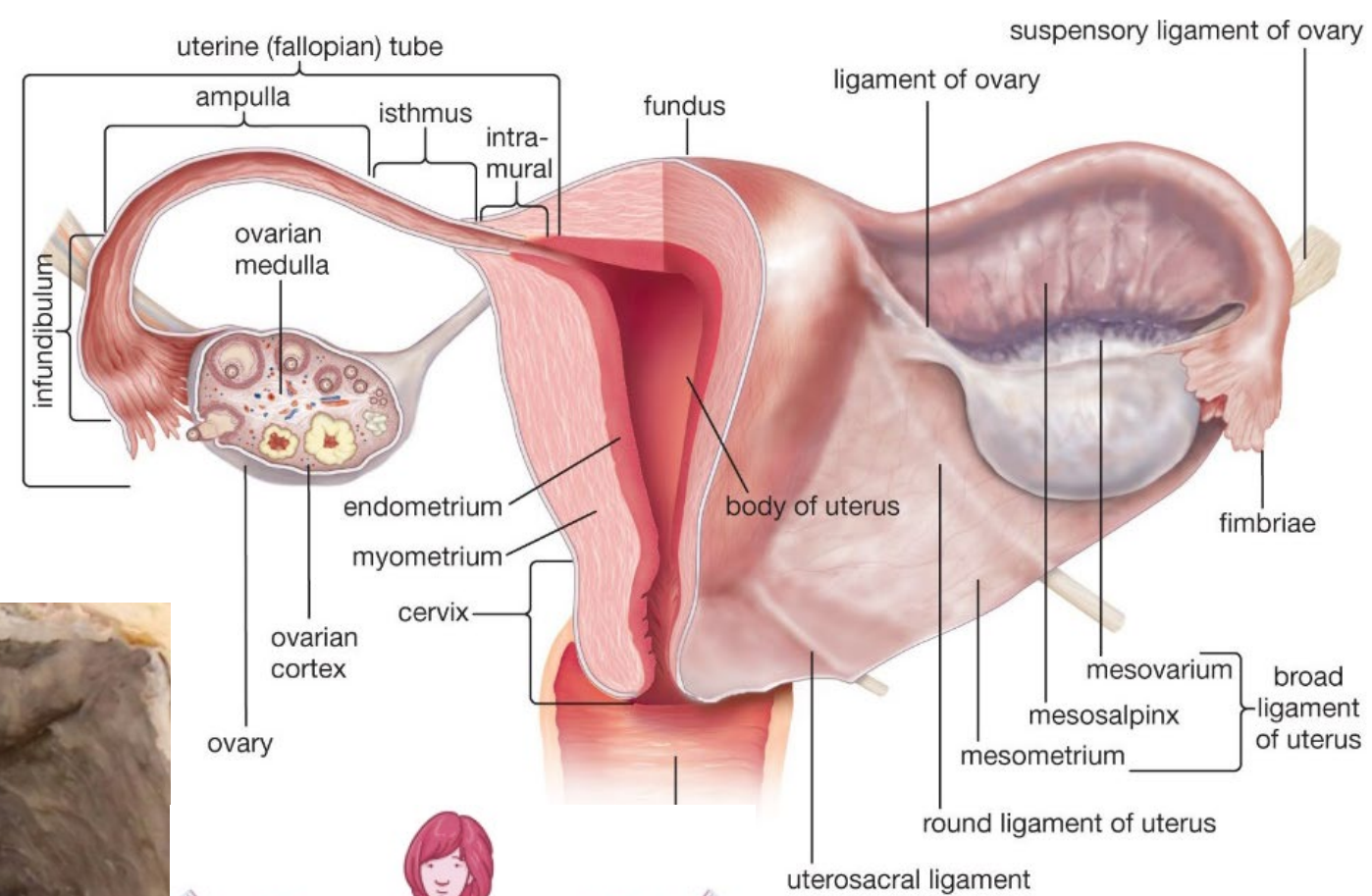
Uterine tube: 8-10 cm

- **Infundibulum:** fimbriated
- **Ampulla:** site of fertilization
- **Isthmus:** joins the uterus
- **Intramural portion:** in the uterine wall, opens into the cavity



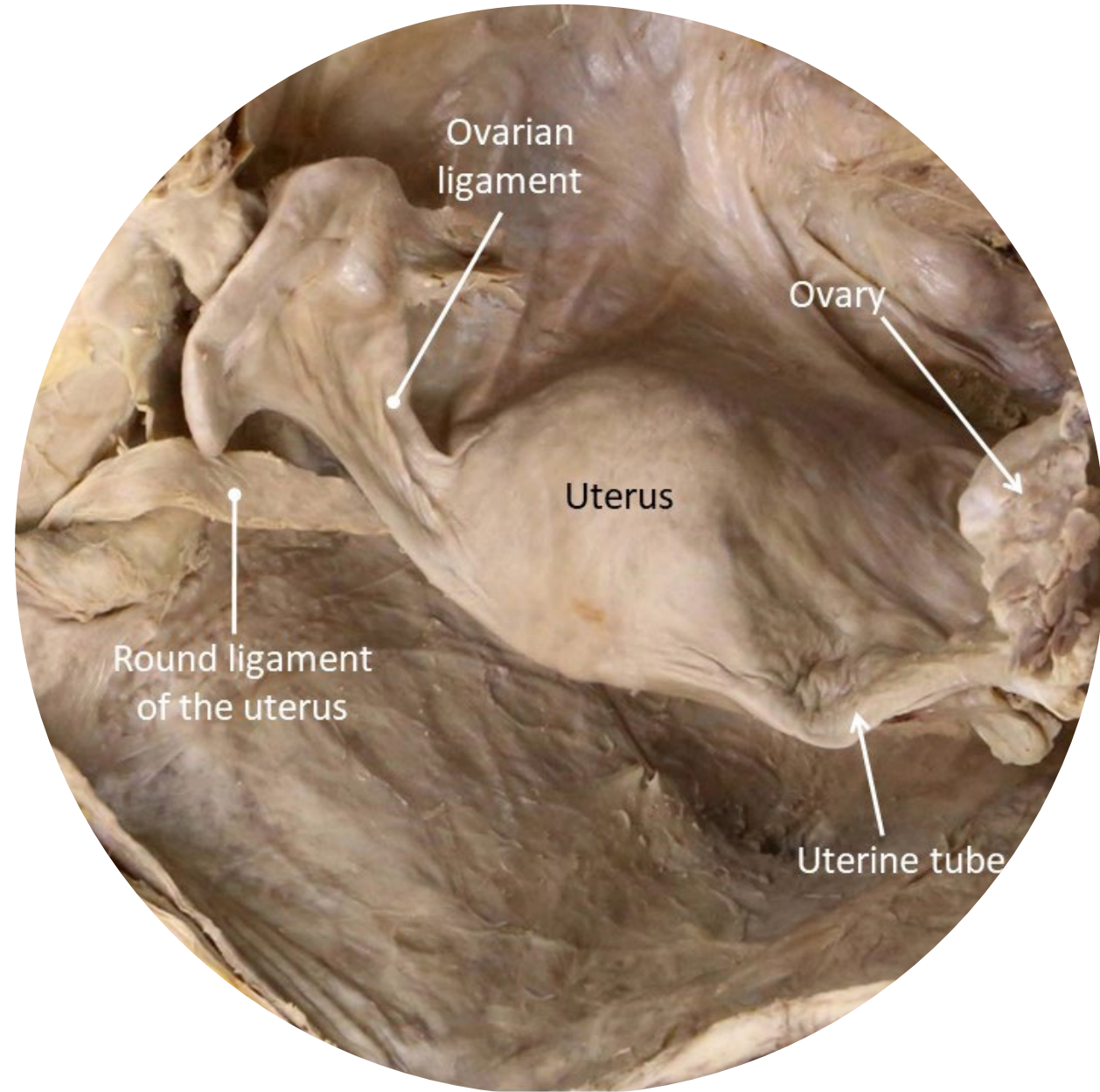
Broad Ligament

- **Mesovarium**
- **Mesosalpinx**
- **Myometrium**



Other Ligaments

- **Suspensory ligament of the ovary** contains the ovarian artery and vein.
- **(Proper) ovarian ligament**, which connects the uterus and ovary.
- **Round ligament (of the uterus)**, which goes through the deep inguinal ring to the inguinal canal to the labia majora.



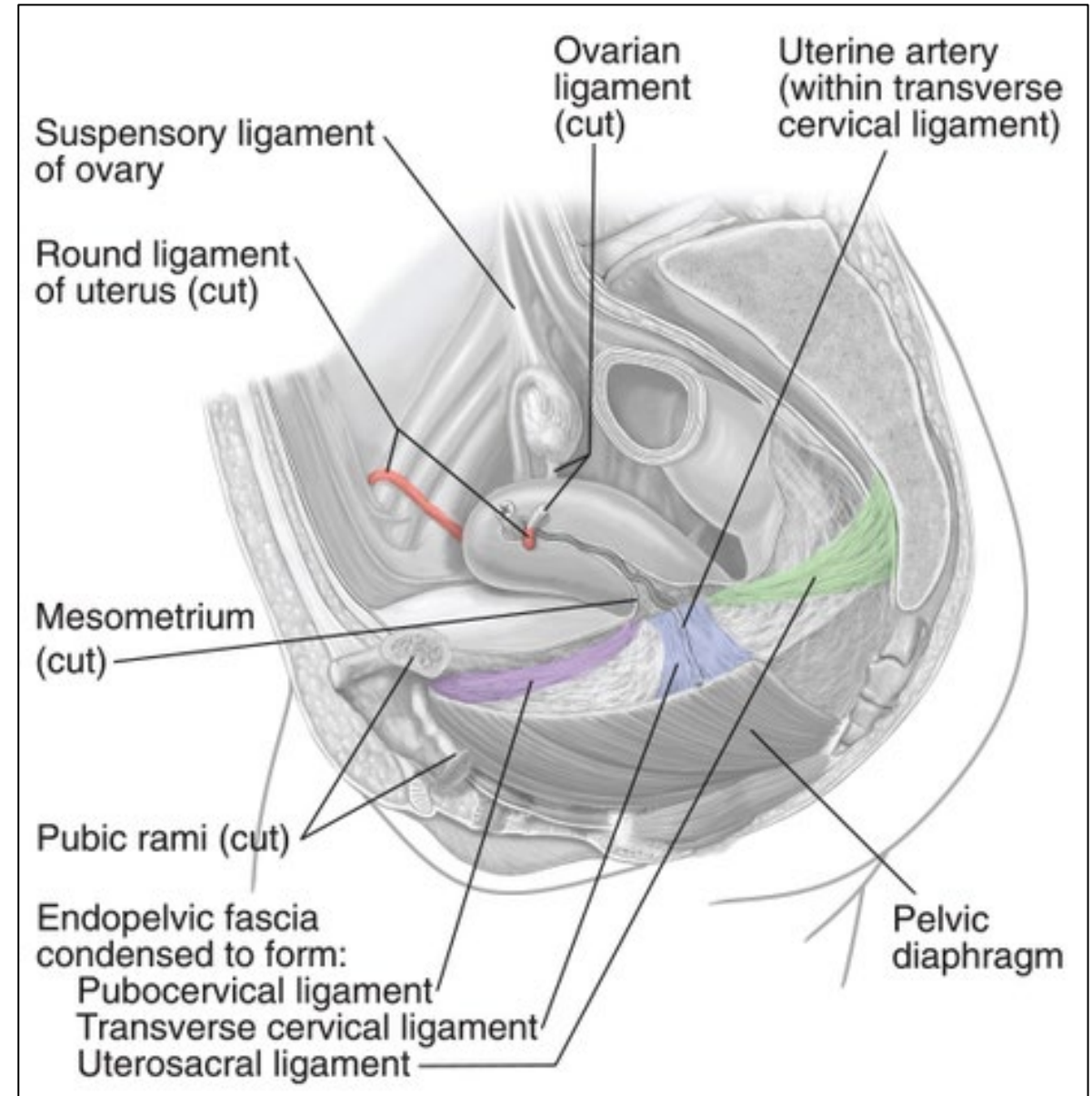
Endopelvic Fascia

It is the connective tissue that surrounds & supports the pelvic organs & connects them to the pelvic wall.

It provides passive support to the cervix & upper vagina.

Subdivided into:

- **Cardinal ligament** (transverse cervical ligament)
- **Uterosacral ligament**
- **Pubocervical ligament**



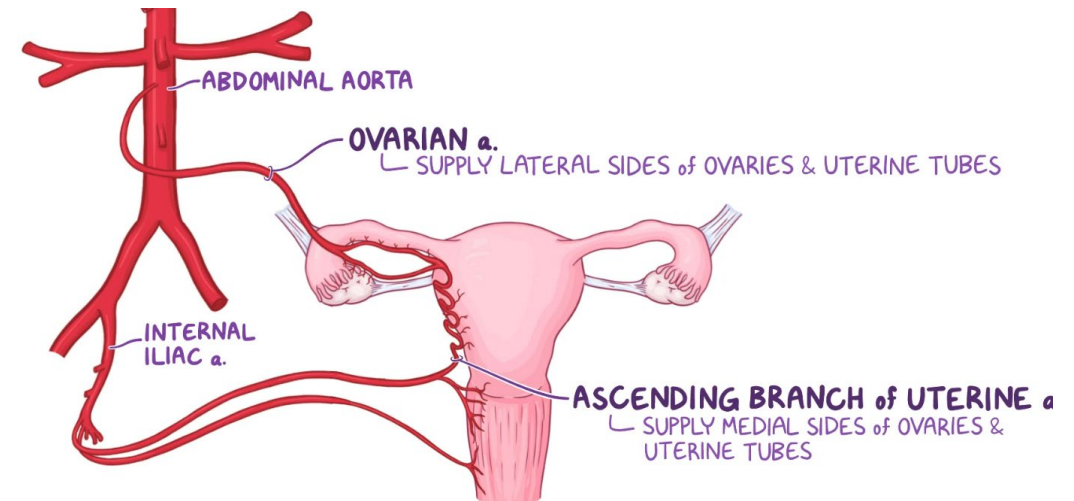
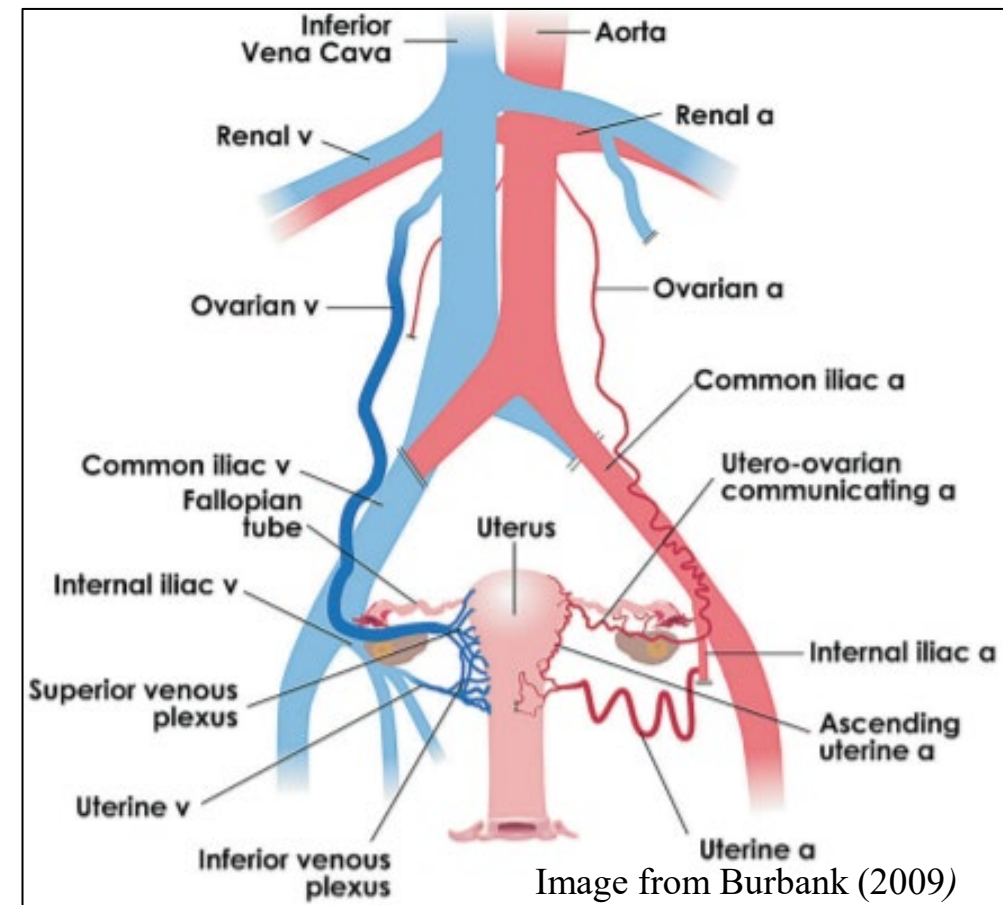
Blood Supply

Arterial supply:

- Ovarian a (from abdominal aorta)
- Uterine a. (from internal iliac a)
- Vaginal a. (from internal iliac a)

Venous drainage:

- Rt. ovarian v to inferior vena cava
- Lt. ovarian v to Lt. renal v.
- Uterine, vaginal vv to internal iliac v



Female Reproductive Summary

Ovary: almond -shaped, 3-4 cm

- **Suspensory ligament of ovary** (ovarian vessels, lymphatics, autonomic fibers)
- Ovarian Lig. attaches ovary to uterus

• **Uterine tube:** 8-10 cm

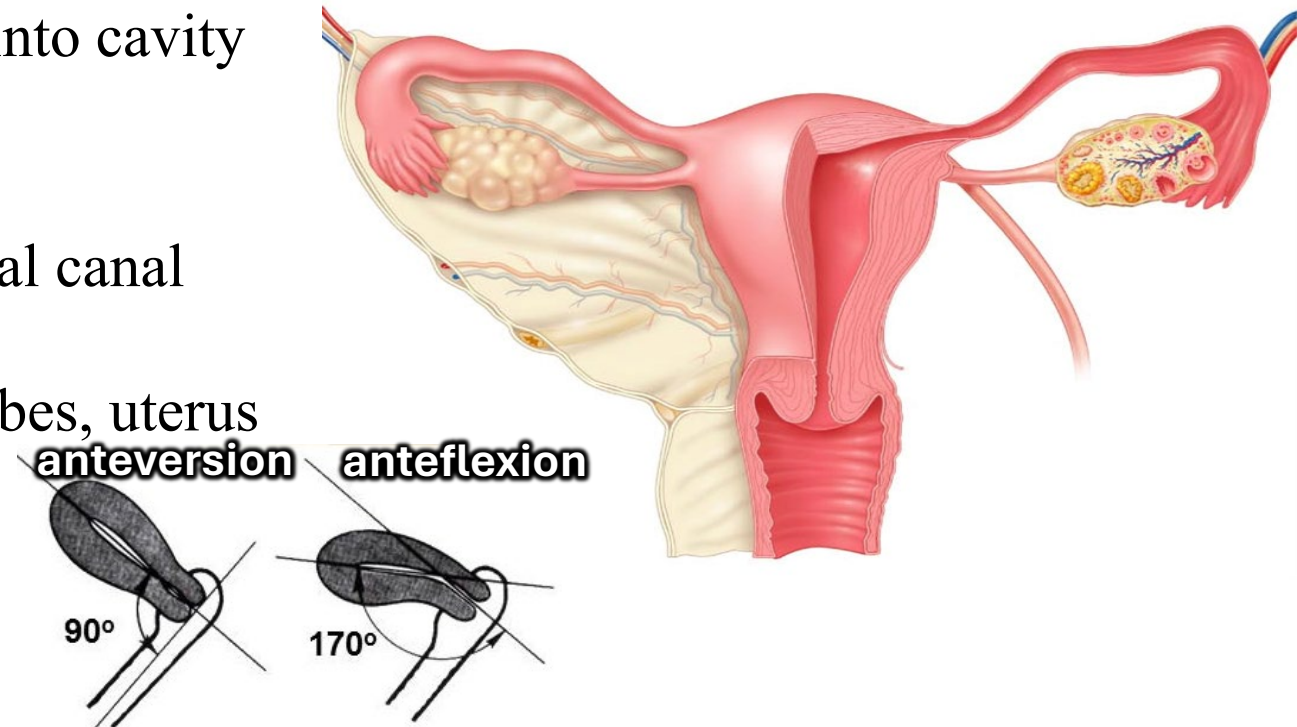
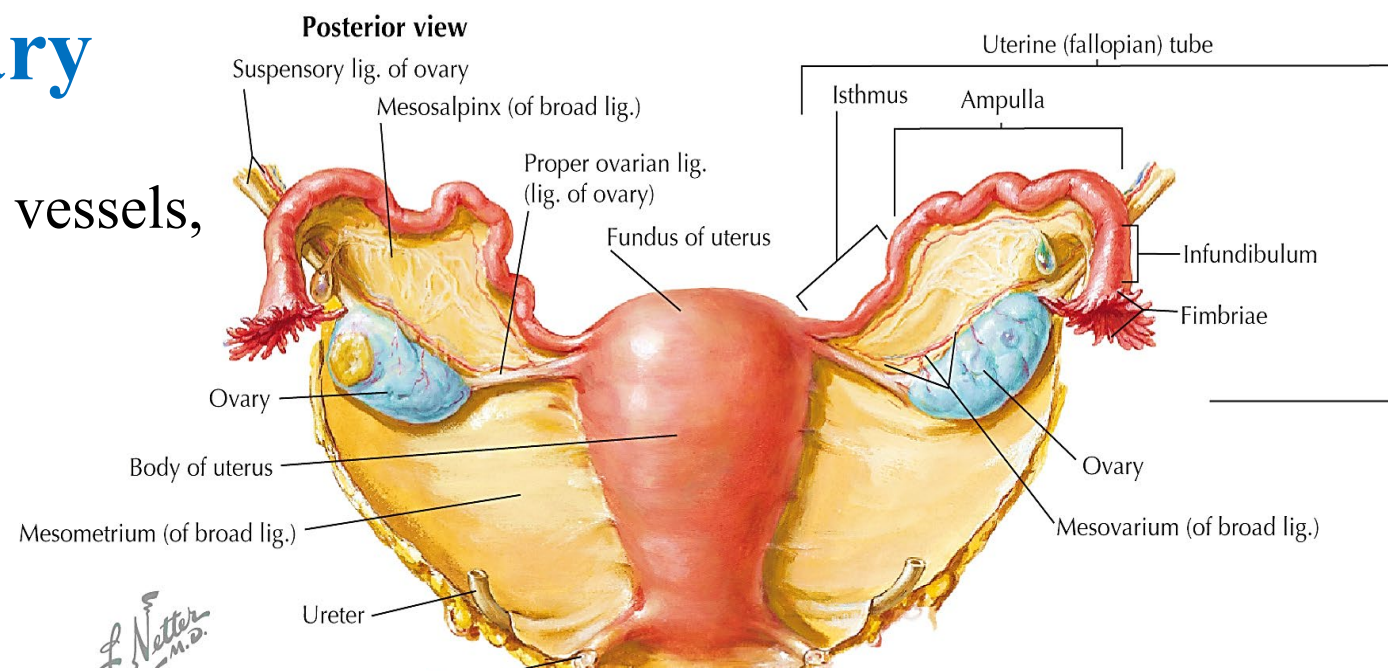
- **Infundibulum:** fimbriated
- **Ampulla:** site of fertilization
- **Isthmus:** joins uterus
- **Intramural portion:** in uterine wall, opens into cavity

• **Uterus:** pear shaped, 7-8 cm

- **Body** (fundus & isthmus)
- **Cervix** (neck): internal & external os, cervical canal
- Anteverted & anteflexed
- **Broad ligament** envelops ovaries, uterine tubes, uterus
- **Round ligament** enters inguinal rings

• **Vagina:** fibromuscular tube, 8-9 cm,

- **Fornix** (anterior, lateral, posterior)
- **Canal** (opens in vestibule)





Questions

Question

A 32-year-old woman is admitted to the hospital due to a palpable mass superior to the inguinal ligament. It is demonstrated by physical examination that she suffers from an indirect inguinal hernia. Due to the severity of the hernia, an open hernia repair is performed. Which of the following structures would the surgeon expect to find in the canal?

- A. Round ligament of uterus
- B. Urachus
- C. Suspensory ligament of the ovary
- D. Uterine tube
- E. Mesosalpinx



Question

A 32-year-old woman is admitted to the hospital due to a palpable mass superior to the inguinal ligament. It is demonstrated by physical examination that she suffers from an indirect inguinal hernia. Due to the severity of the hernia, an open hernia repair is performed. Which of the following structures would the surgeon expect to find in the canal?

- A. Round ligament of uterus
- B. Urachus
- C. Suspensory ligament of the ovary
- D. Uterine tube
- E. Mesosalpinx

A

Question

A 60-year-old woman suffers from pelvic pain that has been getting worse over the last 2 months. She reports that the pain is present every day and that its intensity does not vary based on her menstrual cycle. She has also noticed an increase in abdominal size despite a recent decrease in her appetite. After a series of imaging studies, a left-sided ovarian mass was found, and a surgical procedure was planned to excise the mass. Which of the following structures should be ligated in order to avoid excessive bleeding during the surgery?

- A. Round ligament
- B. Suspensory ligament
- C. Ovarian ligament
- D. Transverse cervical ligament
- E. Mesosalpinx

Question

A 60-year-old woman suffers from pelvic pain that has been getting worse over the last 2 months. She reports that the pain is present every day and that its intensity does not vary based on her menstrual cycle. She has also noticed an increase in abdominal size despite a recent decrease in her appetite. After a series of imaging studies, a left-sided ovarian mass was found, and a surgical procedure was planned to excise the mass. Which of the following structures should be ligated in order to avoid excessive bleeding during the surgery?

- A. Round ligament
- B. Suspensory ligament
- C. Ovarian ligament
- D. Transverse cervical ligament
- E. Mesosalpinx

B

Question

A 47-year-old woman with three children and a family history of breast cancer mutations in the BRCA1 and BRCA2 genes has made a decision to have an elective hysterectomy as a prophylactic treatment. During ligation of the uterine artery, which of the following adjacent structures is most susceptible to iatrogenic injury?

- A. Ureter
- B. Internal iliac artery
- C. Internal iliac lymph nodes
- D. Obturator nerve
- E. Lumbosacral trunk

Question

A 47-year-old woman with three children and a family history of breast cancer mutations in the BRCA1 and BRCA2 genes has made a decision to have an elective hysterectomy as a prophylactic treatment. During ligation of the uterine artery, which of the following adjacent structures is most susceptible to iatrogenic injury?

- A. Ureter
- B. Internal iliac artery
- C. Internal iliac lymph nodes
- D. Obturator nerve
- E. Lumbosacral trunk

A

Question

A 45-year-old woman is admitted to the hospital with lower abdominal and pelvic discomfort. Combined laparoscopic and MRI examinations reveal tears of the ligaments supporting the uterus with moderate uterine prolapse. Which of the following ligaments provides direct support to the uterus and thereby resists prolapse?

- A. Broad ligament
- B. Round ligament of the uterus
- C. Uterosacral Ligament
- D. Arcus tendineus fascia pelvis
- E. Levator ani muscle

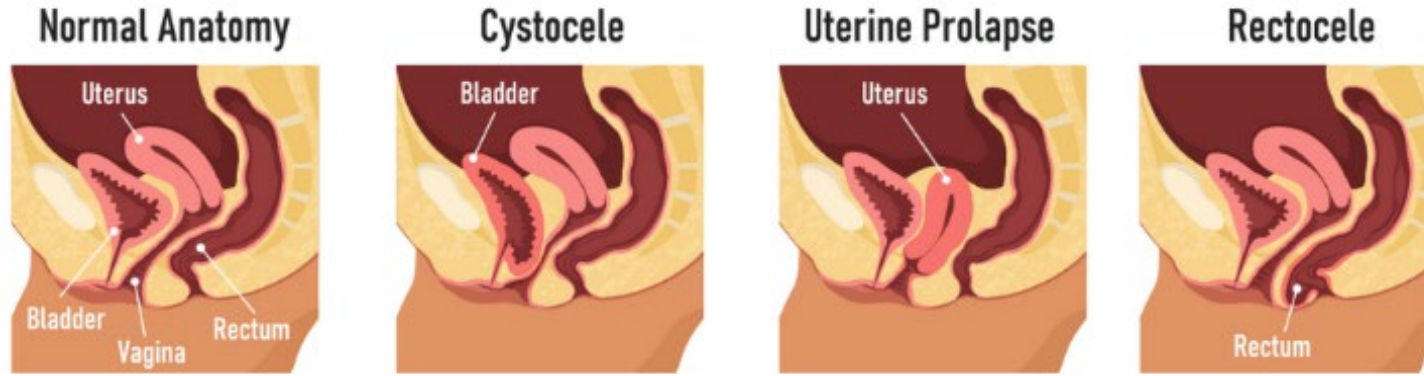
Question

A 45-year-old woman is admitted to the hospital with lower abdominal and pelvic discomfort. Combined laparoscopic and MRI examinations reveal tears of the ligaments supporting the uterus with moderate uterine prolapse. Which of the following ligaments provides direct support to the uterus and thereby resists prolapse?

- A. Broad ligament
- B. Round ligament of the uterus
- C. Uterosacral Ligament
- D. Arcus tendineus fascia pelvis
- E. Levator ani muscle

C

TYPES OF PELVIC ORGAN PROLAPSE



The uterosacral ligaments and the transverse cervical (cardinal) ligaments are the two main ligaments stabilizing the uterus. They help to inhibit the uterus from prolapsing into the vagina.

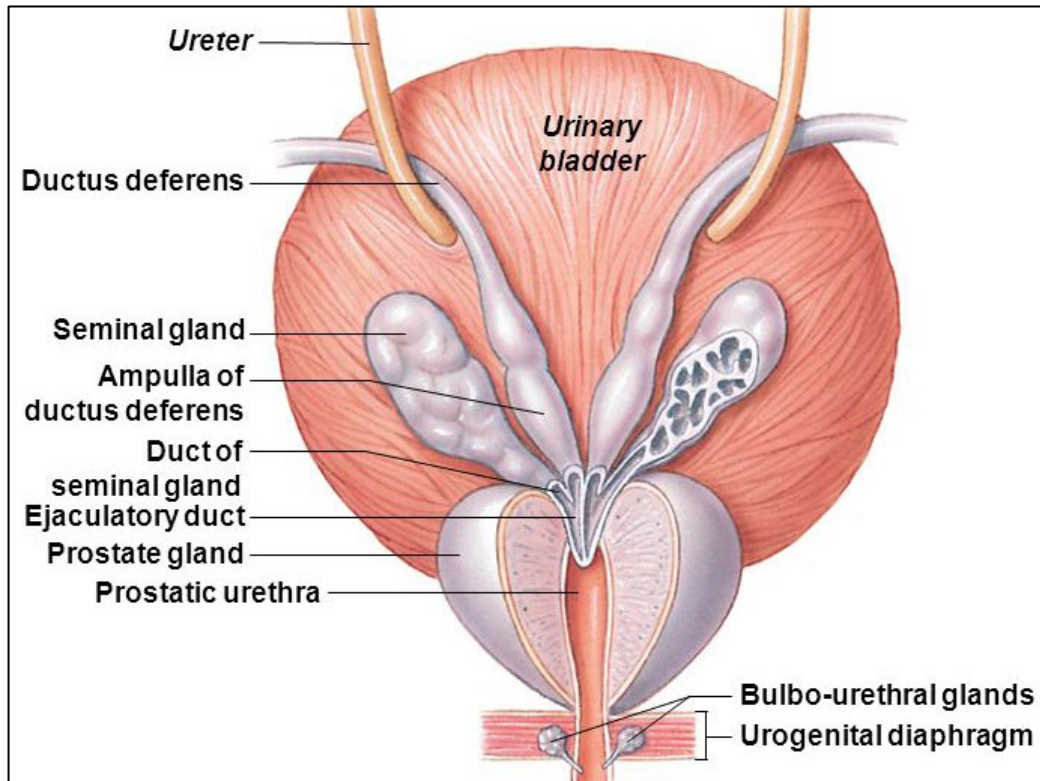
The round ligament of the uterus is related to the descent of the ovaries in embryologic development and continues into the inguinal canal. This ligament is the remnant of the gubernaculum.

The broad ligament is the peritoneal covering over the uterine tubes, uterus, and vessels.

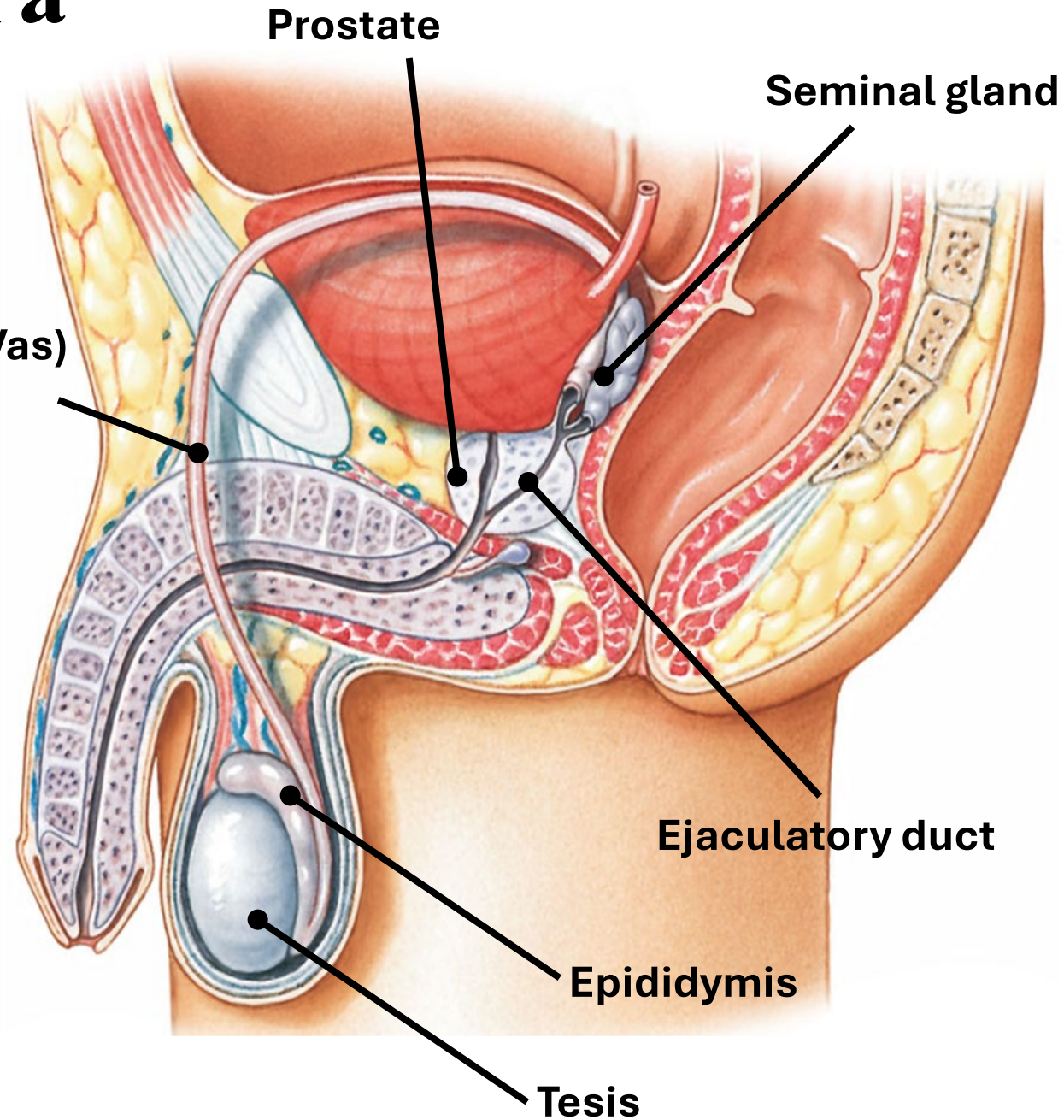
The arcus tendineus fascia pelvis joins the muscle fascia of the levator ani to the pubocervical fascia on the vagina and is not directly associated with the uterus or its ligaments. The levator ani muscles contribute to the floor of the pelvis and support all of the pelvic viscera indirectly; it does not, however, prevent prolapse of the uterus into the vagina.

Male Reproductive Viscera

Prostate
seminal vesicle
ejaculatory duct
Testis, epididymis, Vas deferens



Ductus (Vas) deferens

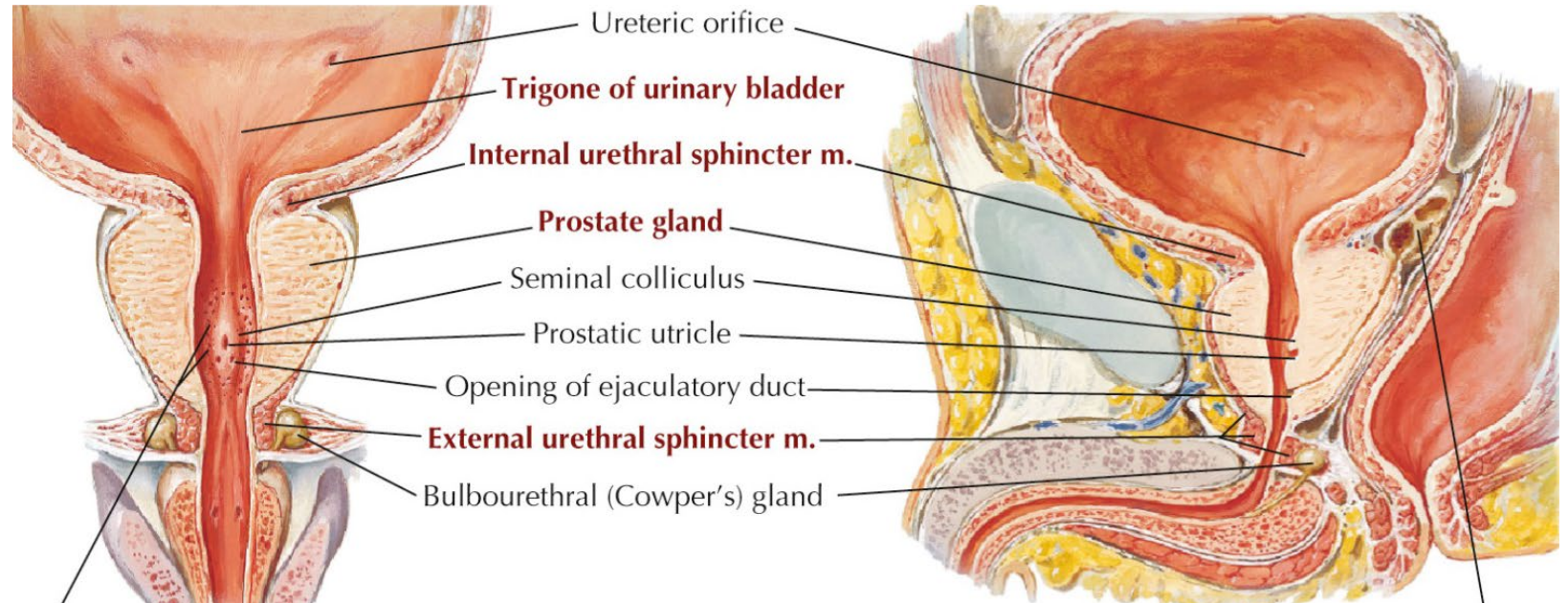
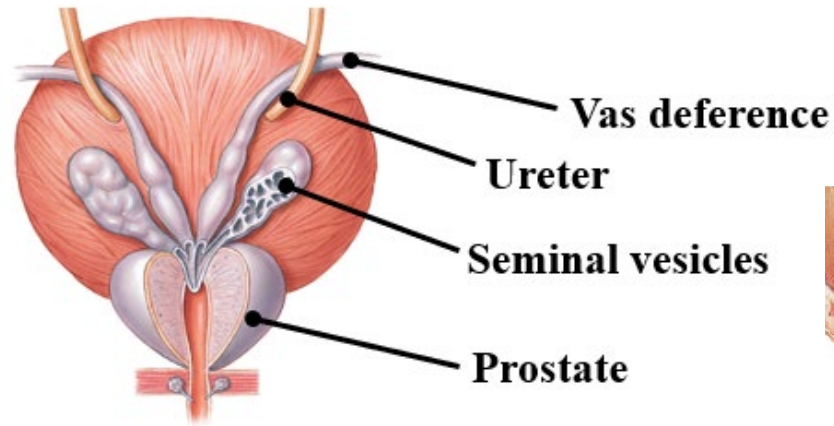


Male Reproductive Viscera

Seminal Vesicle :

- Contributes 70% of ejaculate
- Produce viscous & alkaline fluid, nourishes & protects spermatozoa from acidic vagina.

Ejaculatory duct: joins the prostatic urethra

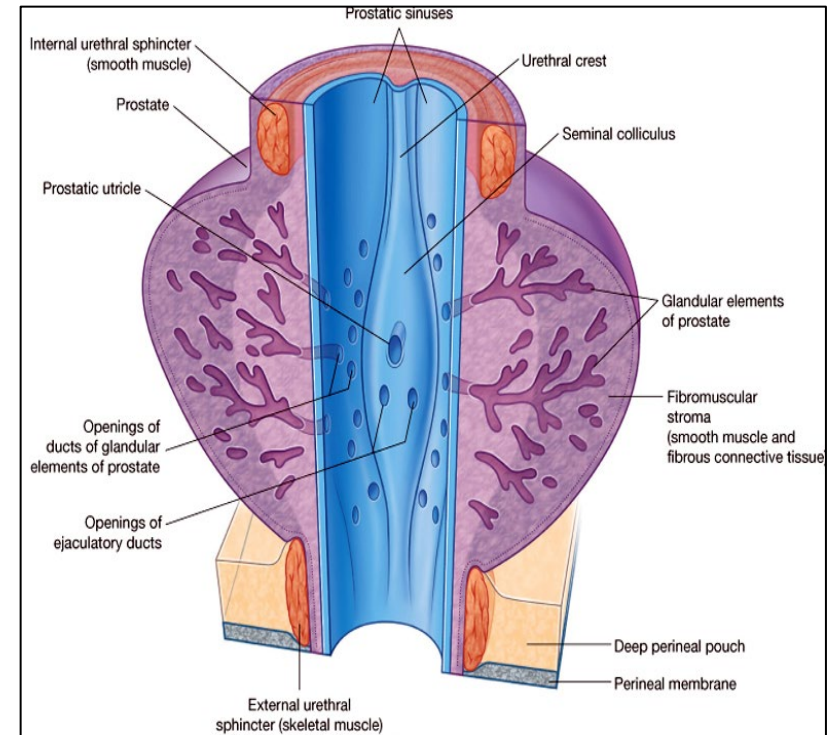
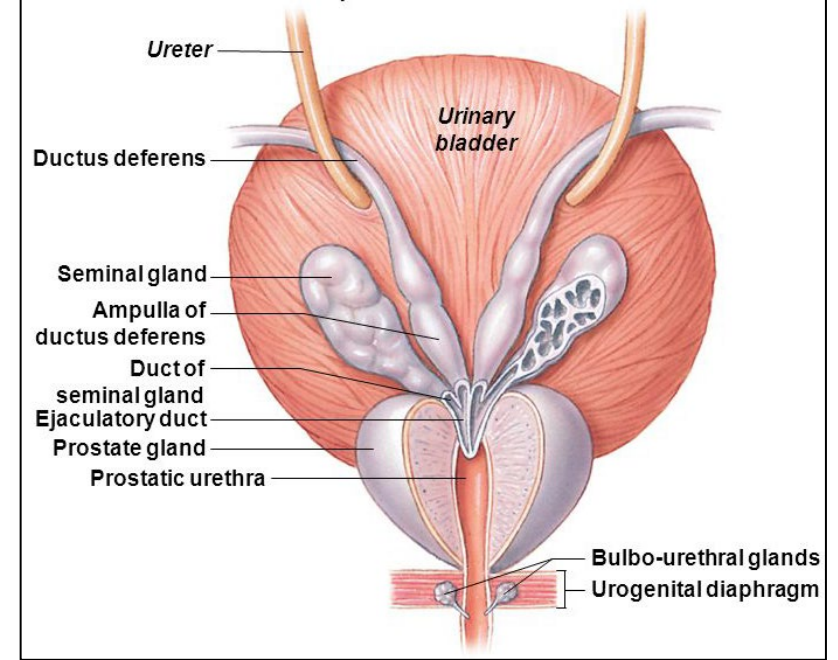


Prostate

- Walnut-sized; contributes 20% of ejaculate
- Produces thin, alkaline fluid that liquefies coagulated semen in the vagina; contains citric acid, proteolytic enzymes, sugars & phosphate.
- About 3 to 5 ml of semen and 100 million sperm/ml are present in each ejaculation. Ejaculate pH7 - pH8.

IUS: Involuntary, smooth muscle, located at the bladder neck, critical for preventing retrograde ejaculation by sealing the bladder outlet during emission.

EUS: Voluntary, skeletal muscle, located at the membranous urethra, mainly for urinary control; assists in urethral closure during ejaculation but does not have the primary retrograde flow barrier role.

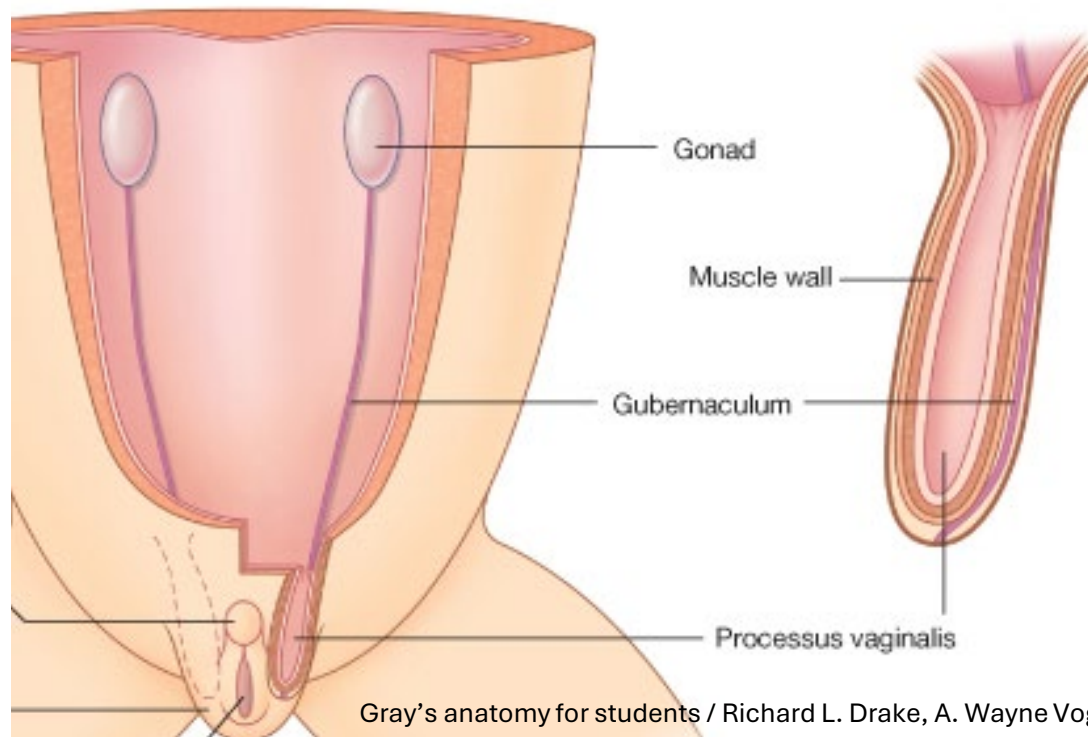


Inguinal Canal Story

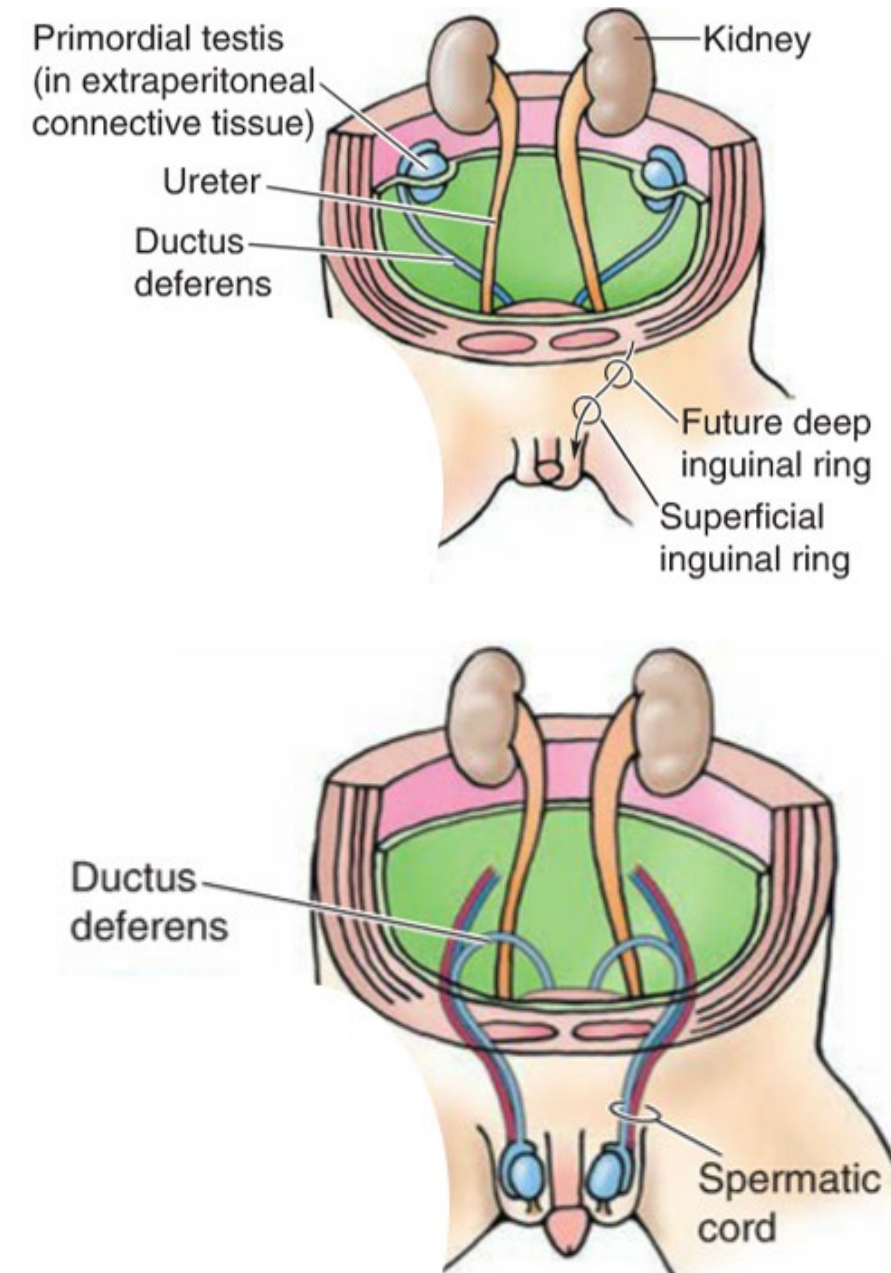
Inguinal canal is formed in relation to the relocation of the gonads during the fetal development.

Gubernaculum: is a fibrous band extend from the caudal part of testis. It plays a role in the decent of the testis.

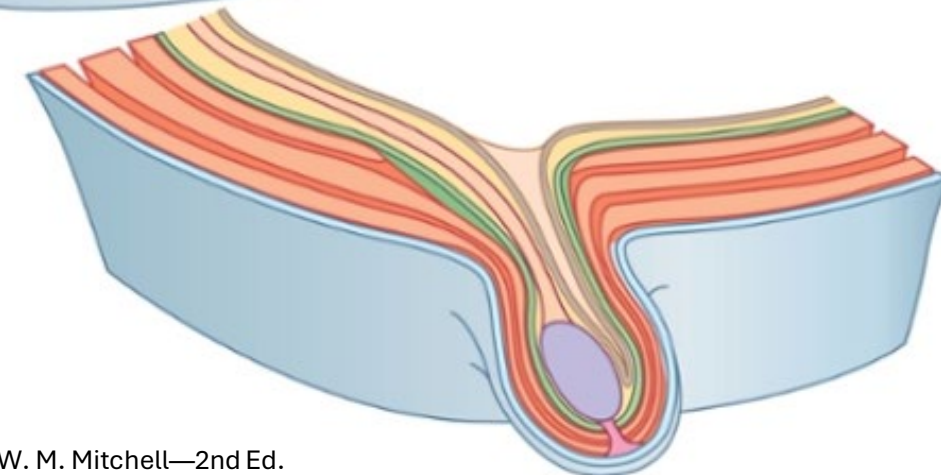
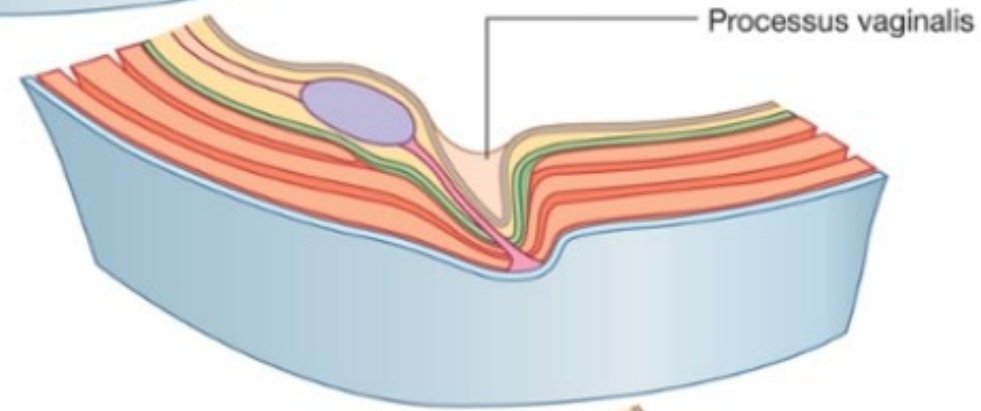
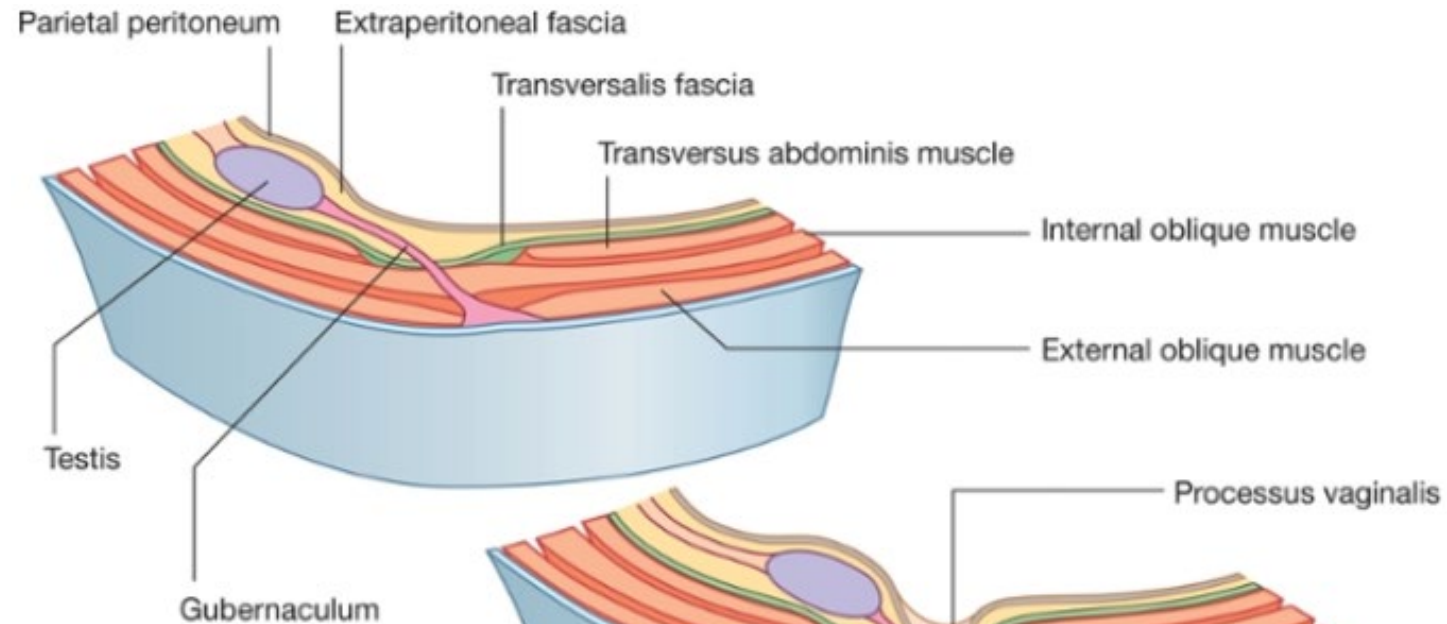
Processus vaginalis: is a diverticulum or out pouching of the peritoneum.



Gray's anatomy for students / Richard L. Drake, A. Wayne Vogl, & Adam W. M. Mitchell—2nd Ed.



Descent of Testes



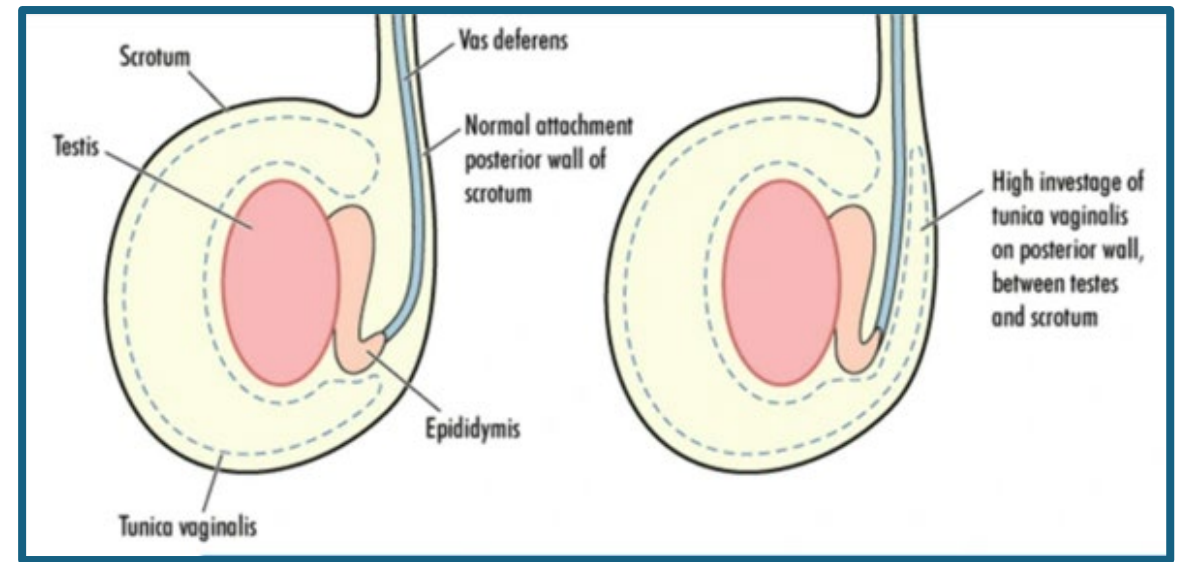
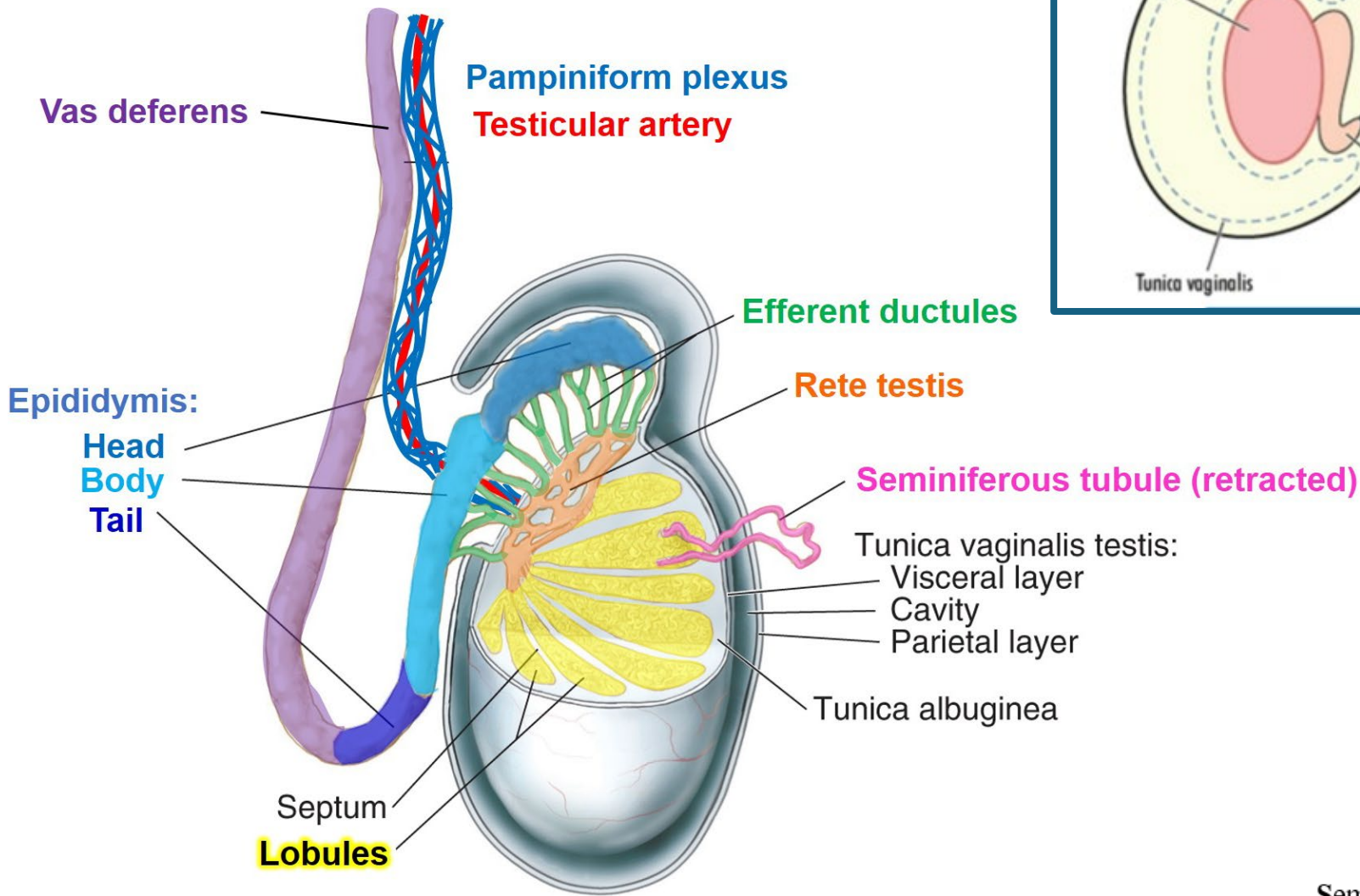
Scrotum Layers

Skin
Dartos fascia & dartos muscle
External spermatic fascia
Cremaster muscle
Cremasteric fascia
Internal spermatic fascia
Tunica vaginalis
• Parietal layer
• Visceral layer

Abdominal Wall Derivative

Skin
Subcutaneous tissue
External oblique muscle
Internal Oblique muscle
Fascia of internal oblique muscle
Transverse abdominal muscle
Transversalis fascia
Extraperitoneal fat
Peritoneum

Testis

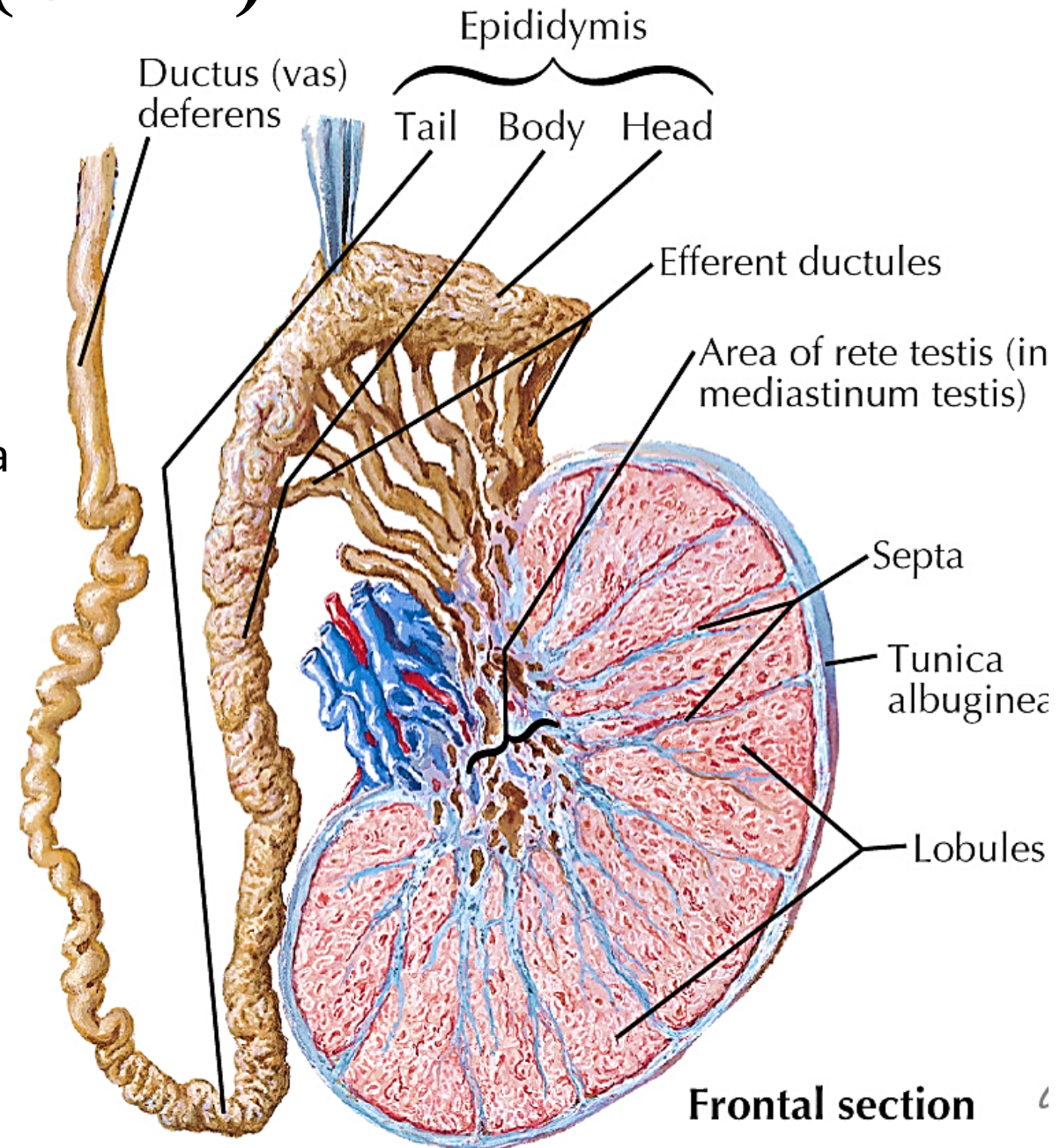


Seven Up

Seminiferous tubules, Epididymis, Vas (ductus) deferens, Ejaculatory ducts, (Nothing), Urethra, Penis

Male Reproductive Viscera (Cont.)

- **Testis:** paired gonads
 - Lobules of seminiferous tubules
 - Tunica albuginea: collagenous capsule
 - Tunica vaginalis: peritoneum (visceral & parietal layers).
- **Rete Testis (straight tubules):** drain spermatozoa from seminiferous tubules.
- **Epididymis:** long coiled tube (6 m)
 - Sperm maturation & storage site
 - Efferent ductules.
- **Ductus (vas) deferens:** muscular tube (40 cm), ascends in the spermatic cord, enters the pelvic cavity, joins the ejaculatory duct



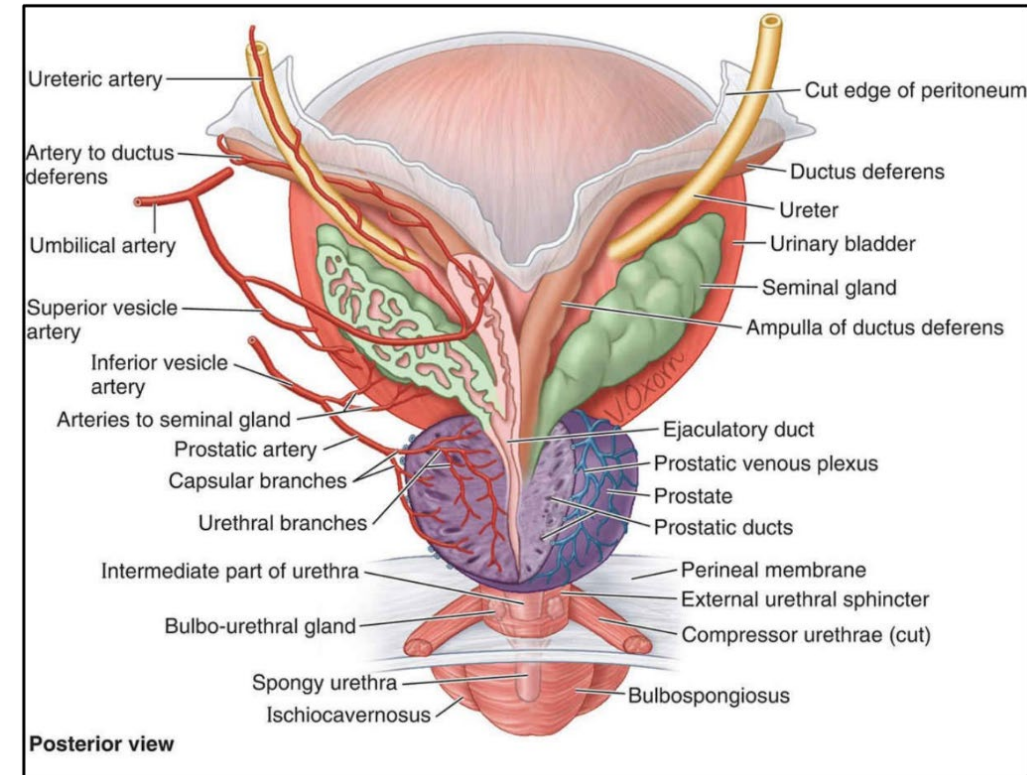
Blood Supply of Male Reproductive System

Arterial supply:

- Testicular a (from aorta)
- Branches of internal iliac aa:
 - ✓ superior vesical a – gives off a. to the ductus deferens to testes and ejaculatory ducts
 - ✓ inferior vesical a – branches supply seminal vesicles; prostatic branches to prostate; occasionally will give off a to the ductus deferens
 - ✓ middle rectal a – branches supply seminal vesicles, prostate

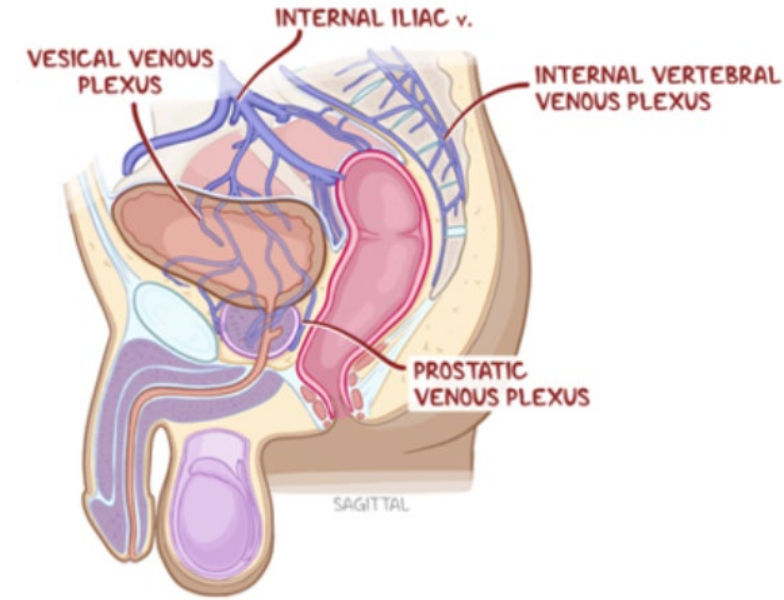
Venous drainage:

- From testis and epididymis: pampiniform venous plexus to testicular v; Rt testicular v. drains to IVC; Lt testicular v. drains to Left renal Vein.
- From prostate: prostatic venous plexus between the fibrous capsule of the prostate and the prostatic sheath drains into internal iliac veins.
- Other structures (seminal vesicles, ejaculatory ducts): veins join the prostatic and vesical venous plexuses

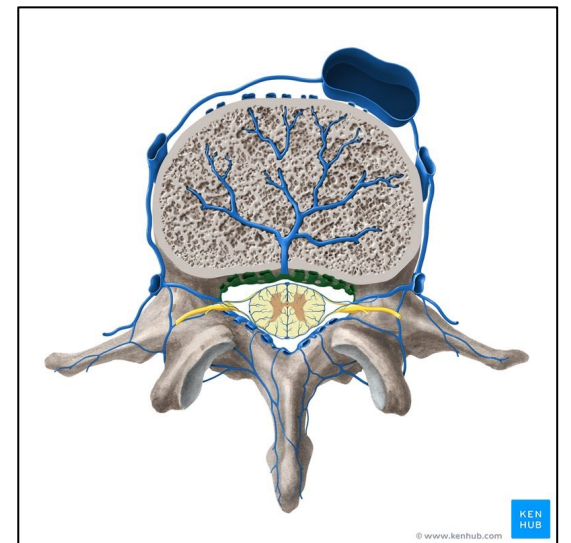


The Vertebral Venous Plexus (Batson's plexus)

A key pathway for the spread of cancer from the pelvic region, including the uterus and bladder, to the spine. This network of valveless veins allows cancer cells to bypass the normal circulatory filtering systems, such as the lungs, and directly access the vertebral column.

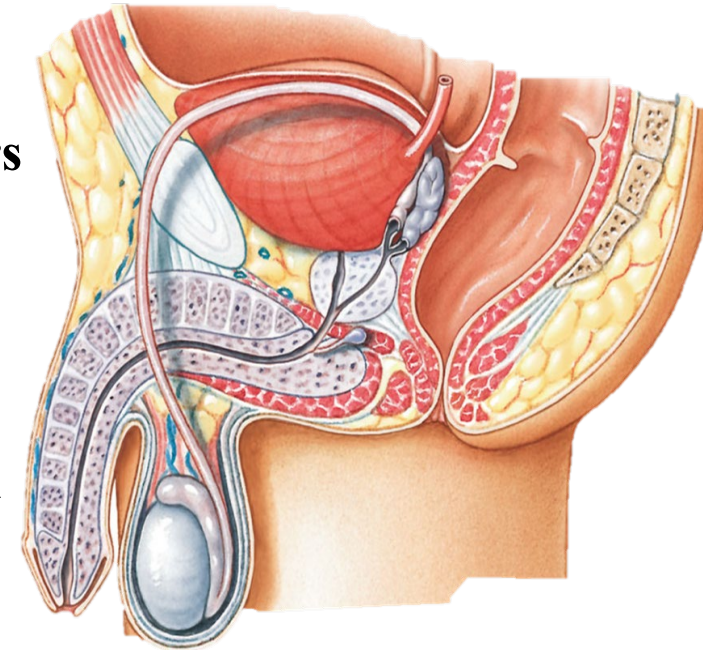
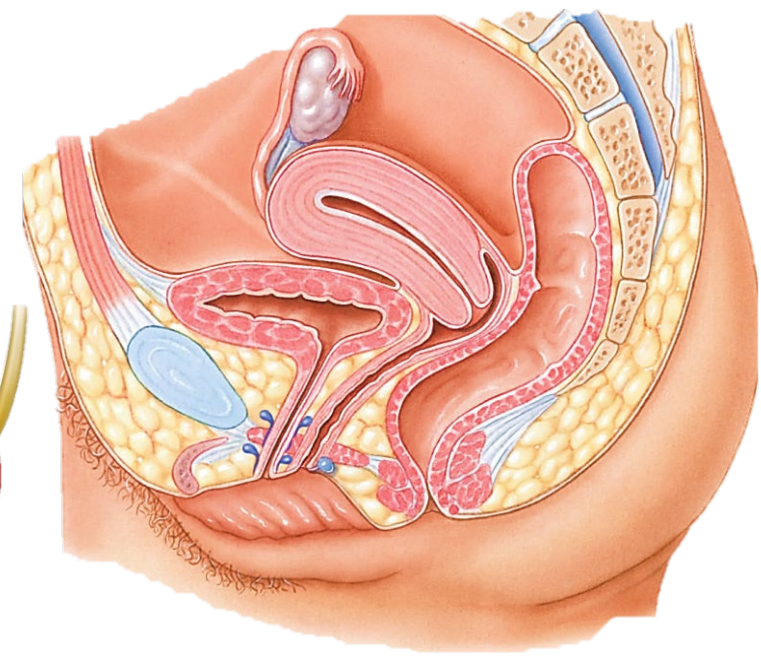
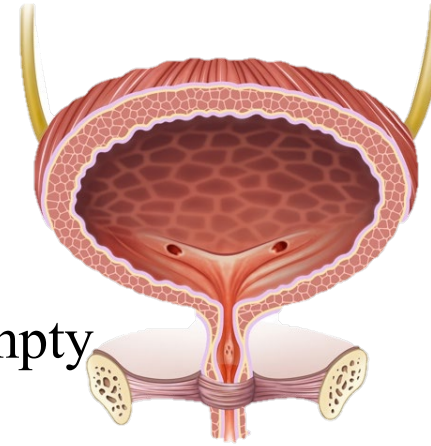


- **Valveless design:** Veins have no valves, allowing blood and cancer cells to move in both directions.
- **Pressure changes:** Straining or coughing raises belly pressure, pushing prostate blood straight into the spine.
- **Direct pathway:** Pelvic veins link directly into the Batson venous plexus surrounding the spinal column.

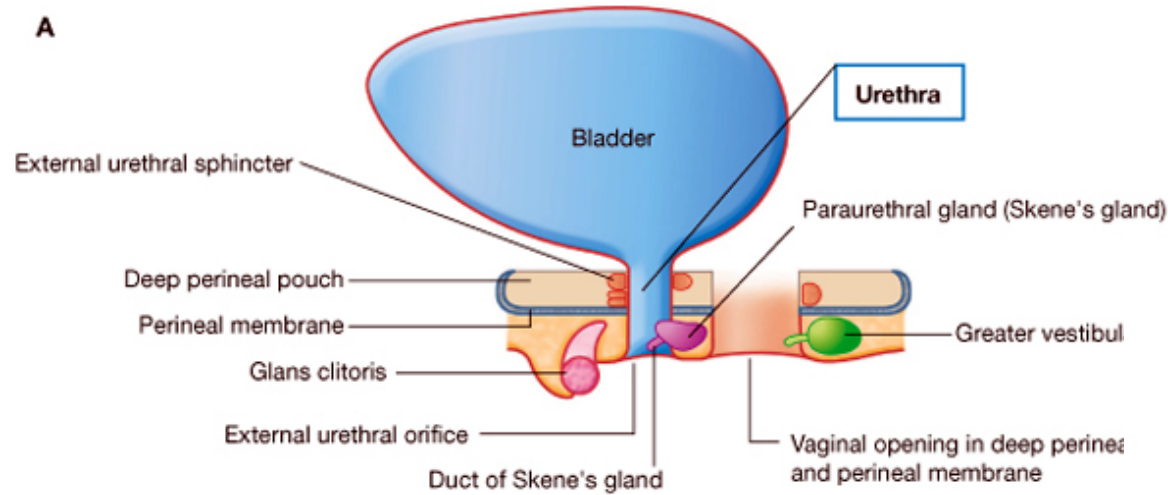


Pelvic Viscera: Distal Urinary Tract

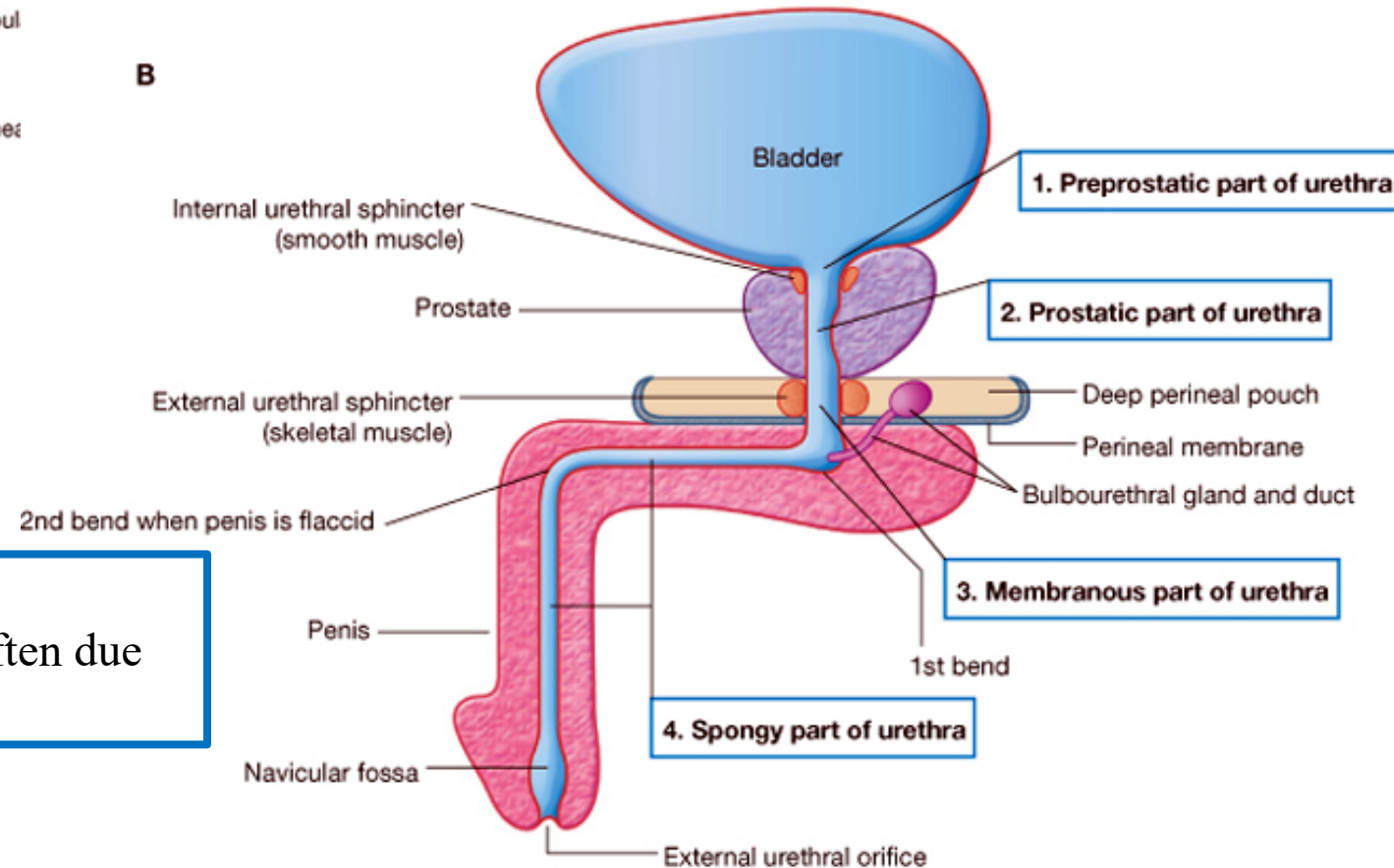
- **Ureters:** enter posterolateral muscular wall of bladder in oblique fashion, provides sphincter-like action.
- **Urinary Bladder:** posterior to pubic symphysis
 - Holds ~ 500 mL of urine
 - **Trigone of bladder**, mesonephric origin
 - **Detrusor muscle**
 - Sensitive to distention, signals brain the need to empty
- **Urethra:**
 - **Female** (3-4 cm)
 - **External sphincter:** skeletal muscle, voluntary, innervated by somatic fibers in the pudendal nerve (S2-S4).
 - **Male** (18-20 cm): prostatic (3cm), membranous(2cm), & spongy.
 - **Internal sphincter:** smooth muscle, involuntary, @ ejaculation, contracts to prevent semen from entering the bladder.
 - **External sphincter:** skeletal muscle, voluntary, innervated by somatic fibers in the pudendal nerve (S2-S4).



Male vs Female Urethra



B

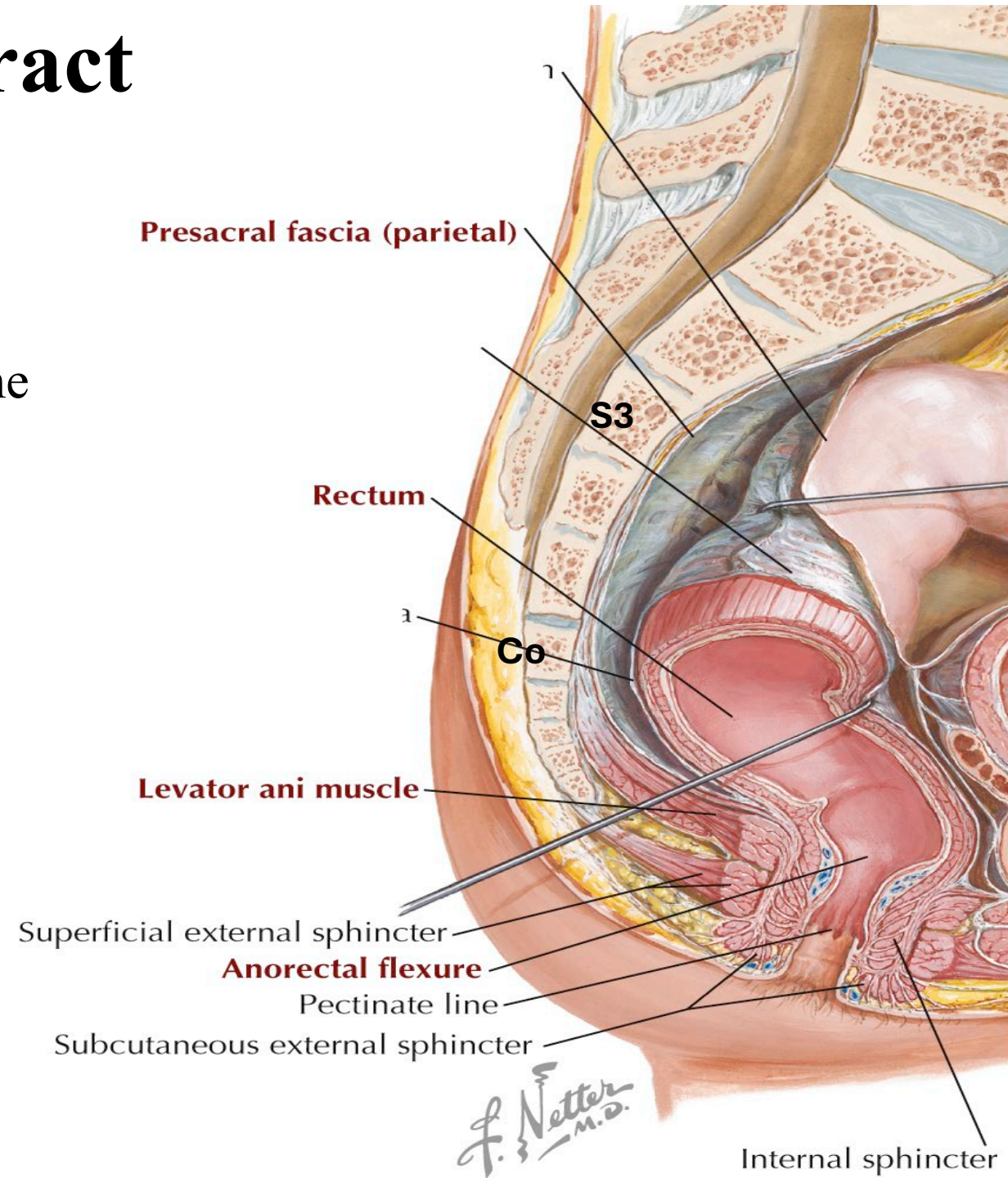


Clinical Focus

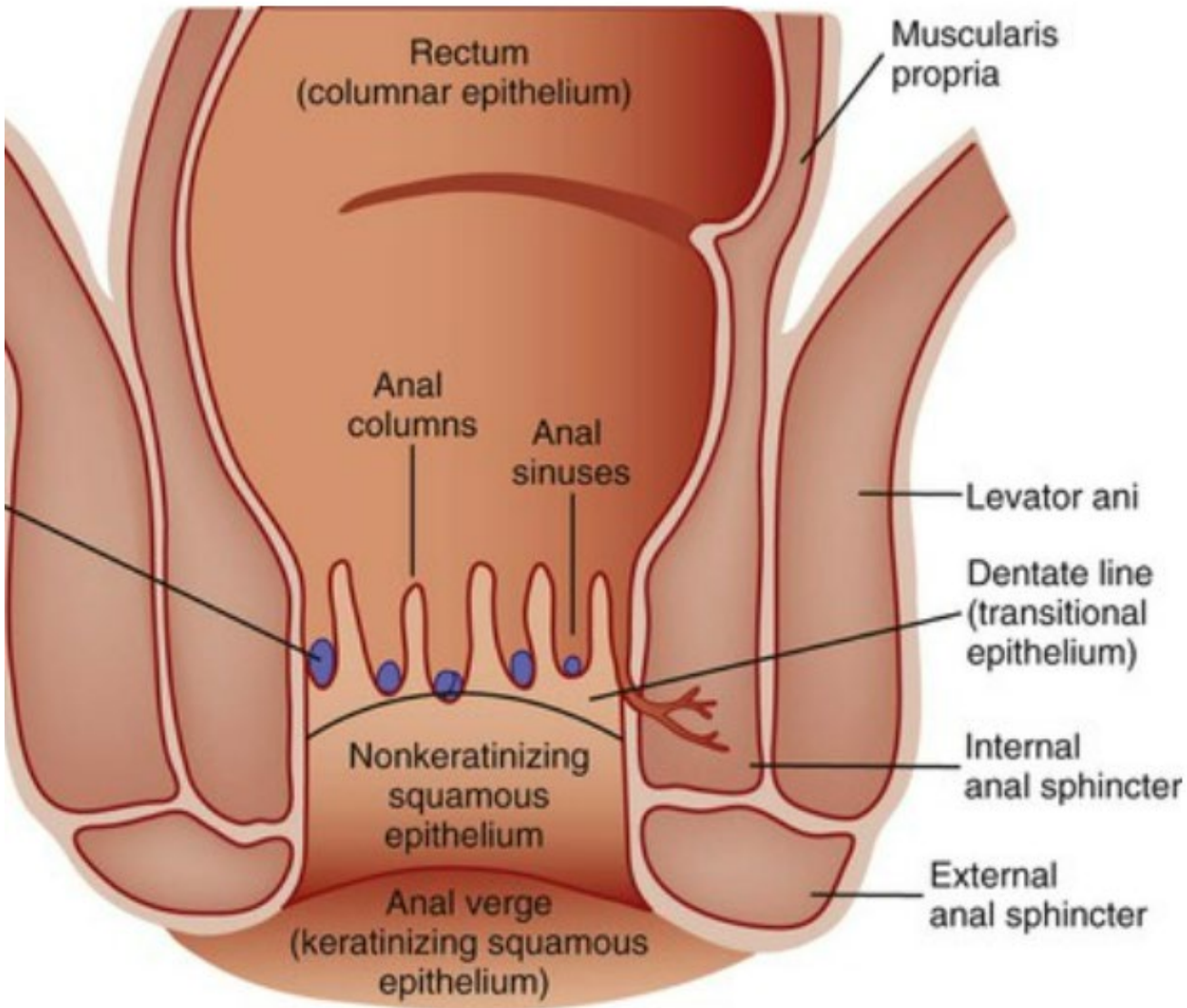
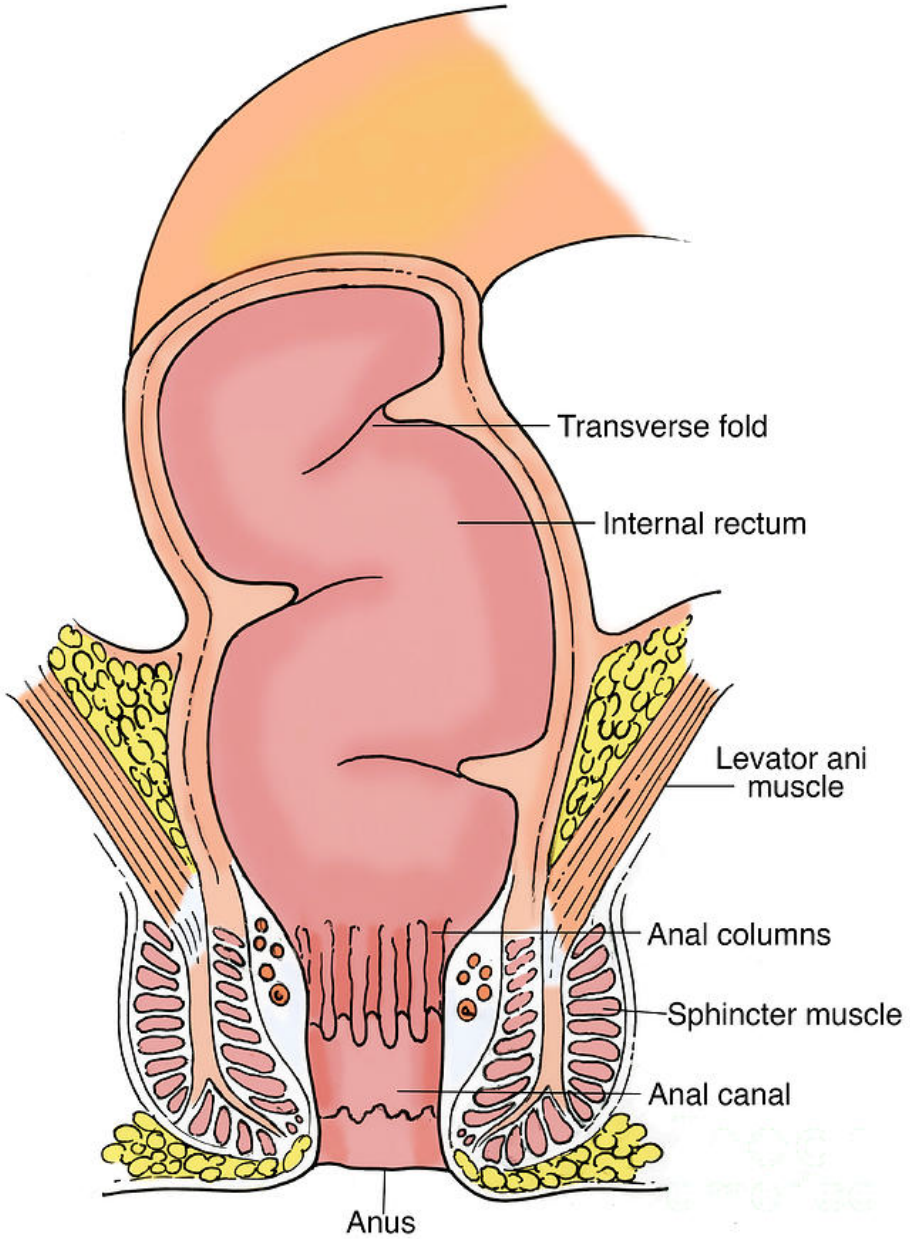
UTI: more common in females, shorter urethra. Most often due to *E. coli* infection

Pelvic Viscera: Distal GI Tract

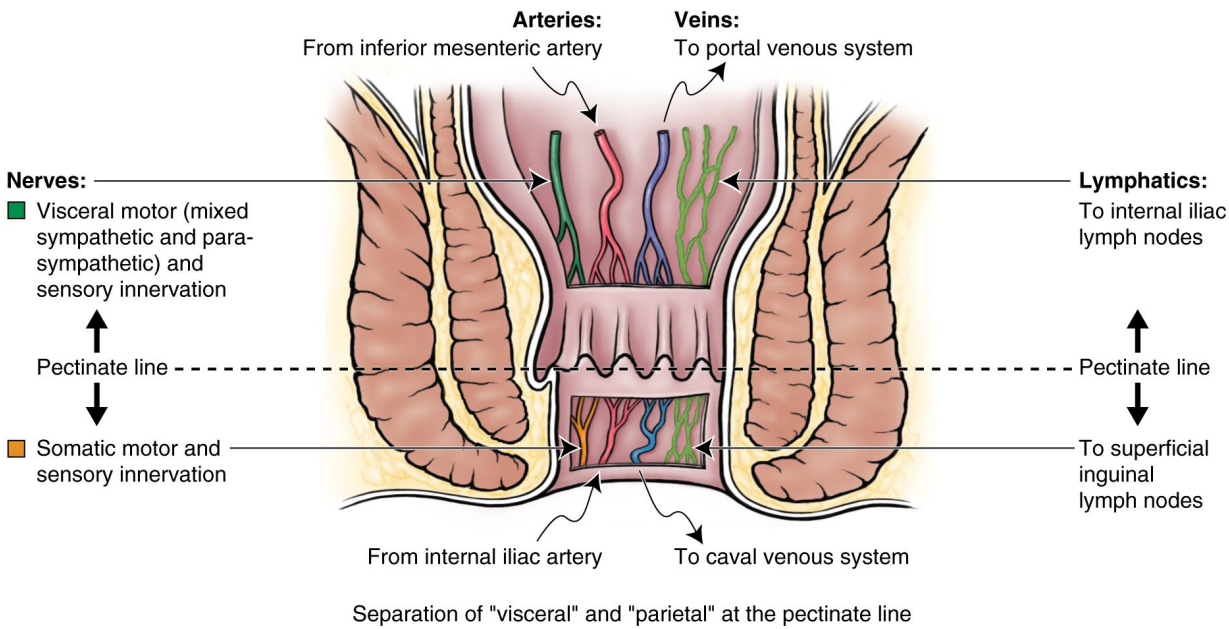
- **Rectum:** terminal part of the large intestine
 - Recto-sigmoid junction @ S3
- **Anal Canal:** below coccyx
 - **Anorectal flexure:** anal canal perforates the pelvic diaphragm (levator ani) maintains fecal continence through muscle tone of puborectalis, relaxes during defecation; flexure straightens
 - **Internal anal sphincter:** smooth muscle, involuntary
 - **External anal sphincter:** skeletal muscle (deep, superficial, and subcutaneous)
 - **Pectinate line:** Demarcates changes in visceral (above) from somatic (below) portions



Rectum & Anal Canal



The Pectinate Line



- Above the line → Endoderm derived
- Portal system (sup. Rectal V. to Inferior mesenteric V.)
- **Superior rectal a from IMA**
- Columnar epithelium
- Painless (GVA → Visceral innervation)

PECTINATE LINE

- below the line → Ectoderm derived
- Caval system (inf. rectal V. to internal pudendal V.)
- **Inferior rectal a from Internal Pudendal a**
- Stratified squamous
- Painful (GSA → somatic innervation)



Questions



Question

A 25-year-old man is admitted to the hospital with testicular pain. Physical examination reveals a swollen and inflamed right testis. A computed tomography (CT) scan examination reveals abnormal accumulation of fluid in the cavity of the tunica vaginalis. Which of the following is the most likely diagnosis for this patient?

- A. Varicocele
- B. Rectocele
- C. Cystocele
- D. Hydrocele
- E. Hypospadias

Question

A 25-year-old man is admitted to the hospital with testicular pain. Physical examination reveals a swollen and inflamed right testis. A computed tomography (CT) scan examination reveals abnormal accumulation of fluid in the cavity of the tunica vaginalis. Which of the following is the most likely diagnosis for this patient?

- A. Varicocele
- B. Rectocele
- C. Cystocele
- D. Hydrocele
- E. Hypospadias

D

It can be:

- congenital (due to a patent processus vaginalis)
- Acquired (from infection, trauma, inflammation)

Question

A 35-year-old man is admitted to the hospital after being kicked in the groin while playing football. During physical examination, it is noted that the left testicle of the patient is swollen. An MRI examination reveals coagulation of blood in the veins draining the testis. Into which of the following veins would a thrombus most likely pass first from the injured area?

- A. Inferior vena cava
- B. Left renal vein
- C. Left inferior epigastric
- D. Left internal pudendal
- E. Left iliac vein

Question

A 35-year-old man is admitted to the hospital after being kicked in the groin while playing football. During physical examination, it is noted that the left testicle of the patient is swollen. An MRI examination reveals coagulation of blood in the veins draining the testis. Into which of the following veins would a thrombus most likely pass first from the injured area?

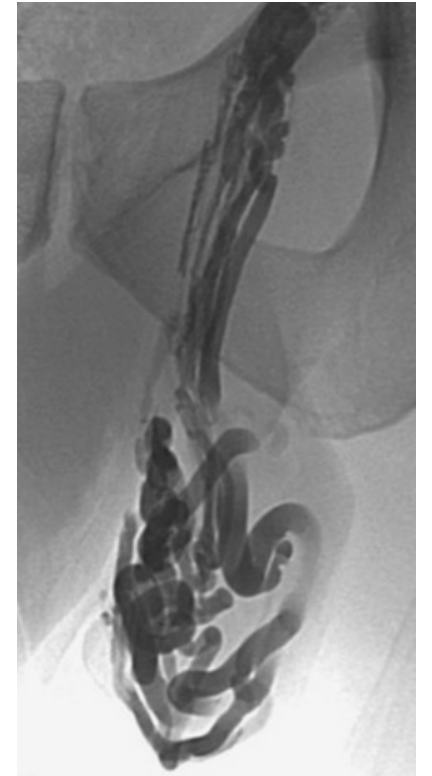
- A. Inferior vena cava
- B. Left renal vein
- C. Left inferior epigastric
- D. Left internal pudendal
- E. Left iliac vein

B

Question

A 54-year-old man is admitted to the hospital with severe back pain. Upon radiographic examination, the scrotum resembles a “bag of worms.” Which of the following conditions will be most likely associated with this radiographic picture?

- A. Varicocele
- B. Rectocele
- C. Cystocele
- D. Hydrocele
- E. Hypospadias

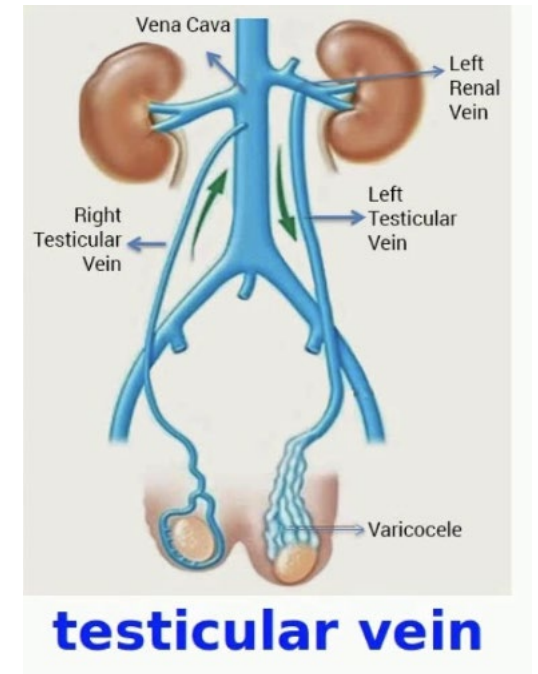


Question

A 54-year-old man is admitted to the hospital with severe back pain. Upon radiographic examination, the scrotum resembles a “bag of worms.” Which of the following conditions will be most likely associated with this radiographic picture?

- A. Varicocele
- B. Rectocele
- C. Cystocele
- D. Hydrocele
- E. Hypospadias

A



Question

A 72-year-old man with cancer of the prostate gland presents with loss of consciousness and seizures. A CT scan is performed and a brain tumor is diagnosed. The tumor spread to the brain from the pelvis via the internal vertebral venous plexus (of Batson). What feature of the plexus allows this to happen?

- A. The internal venous plexus contains the longest veins in the body.
- B. The internal venous plexus has valves that ensure one-way movement of blood.
- C. The internal venous plexus is located in the subarachnoid space.
- D. The internal venous plexus is, in general, valveless.
- E. The internal venous plexus is located in the subdural space.



Question

A 72-year-old man with cancer of the prostate gland presents with loss of consciousness and seizures. A CT scan is performed and a brain tumor is diagnosed. The tumor spread to the brain from the pelvis via the internal vertebral venous plexus (of Batson). What feature of the plexus allows this to happen?

- A. The internal venous plexus contains the longest veins in the body.
- B. The internal venous plexus has valves that ensure one-way movement of blood.
- C. The internal venous plexus is located in the subarachnoid space.
- D. The internal venous plexus is, in general, valveless.
- E. The internal venous plexus is located in the subdural space.

D

The background of the image is a close-up of a stone wall covered in ancient Egyptian hieroglyphs. The hieroglyphs are carved into the stone and include various symbols such as lotus flowers, birds, and geometric shapes. The lighting is warm, highlighting the texture of the stone and the relief of the carvings.

Thank You